Industrial Robot
SCARA Robot THE600-E Robot Manual

Rev. 4

November 30, 2022

Important

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Be sure to read this manual and that you fully understand its contents before operating the industrial robot.
- After reading this manual, keep it in an easily accessible place for future reference.

Original Instructions

Shibaura Machine

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Introduction

Thank you for purchasing SHIBAURA MACHINE's robot system.
This manual describes the basic specifications of the industrial robot, how to unpack and install the robot, how to connect wiring, how to install the hand, and how to maintain and inspect the robot.
Read this manual before actually unpacking the product.
Before performing the operations described in this manual, read through the Safety Manual so that you understand the safety measures.

This manual has the following chapters.

Specifications
Chapter 1: Specifications
This chapter describes the basic specifications and names of the robot parts.

Chapter 2: Functions
This chapter describes the key points for using the functions provided with the robot.

Installation
Chapter 3: Transportation
This chapter describes how to transport the robot in its packed state.

Chapter 4: Unpacking
This chapter describes how to unpack the robot.
It also describes the key points for temporarily storing the unpacked robot.

Chapter 5: Transfer
This chapter describes how to move the unpacked robot to the installation site.
It also provides the external dimensions and weight of the robot.

Chapter 6: Installation of the Robot
This chapter describes the installation environment, required space, and installation procedure for the robot.

Chapter 7: Hand and Other Settings
This chapter describes how to mount the hand to the robot.
Finally, this chapter provides the allowable payloads for the hand.

Chapter 8: User Wiring
This chapter describes how to route the wiring to the hand.

Chapter 9: Teaching
This chapter describes the key points when teaching the robot.
This also describes how to guide the robot manually.

Chapter 10: Operation
This chapter describes how to rescue someone who is trapped by the robot and operation when the robot is brought to an emergency stop.
It also describes the key points for preparing to repack an installed robot.

Chapter 11: Repacking
This chapter describes the key points for repacking an installed robot.
It also describes the key points for transporting a repackaged robot.

Maintenance Manual
Chapters 12 and 13: Maintenance
These chapters describe the maintenance procedures for the various robot sections.

Chapter 14: Origin Position Setting and Position Detector Error
This chapter describes the origin position setting procedure for the robot and the corrective action taken if a position detector error occurs.

Chapter 15: Service Parts
This chapter describes service replacement parts of the robot.

Chapter 16: Disposal
This chapter describes disposal of the robot and its parts.

Manufacturer

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If you have purchased this robot outside of Japan, please contact the company that you purchased it from.

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Scope of Use

This product is a horizontally articulated industrial robot where the manipulator has two horizontally rotating joint axes and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed for use in a wide range of automation tasks such as transport work including the movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product for the purpose of automating these tasks and for incorporating into automation equipment.

This product is not intended for use in tasks involving human contact (such as tasks for medical care and nursing care) or to be incorporated in equipment that sustains human life (such as life support devices and incubators), equipment on which people ride (including transportation equipment such as cars and aerospace equipment such as airplanes, and recreational equipment such as roller coasters), and equipment that protects human life or the human body (such as interlocking devices).
This product is not intended to be incorporated in equipment for controlling large heat quantities (such as nuclear power controllers and incinerator controllers).
This product is prohibited from being used for tasks or incorporated in equipment where a failure or malfunction could likely pose a direct threat to human life or harm to the human body. The customer is responsible for providing any warranties if this product is used.

Usage Procedures

This section describes the usage requirements for the THE600-E robot. Usage of the product in a state outside the required conditions could result in unexpected risks. Be sure to always use the robot in an environment that meets the requirements

1. Main Specifications of Product: See Specifications "1.4 Specifications Table" and Installation "6.1 Installation Environment."
2. Usage Restrictions: Tables 1 to 6 each provide the "Restricted Specifications Sheet" indicating the usage restrictions.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. [8] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power to robot, controller, and optional parts while their covers are removed - Operation in other installation state than the specified (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Entry into the hazard zone of the robot when the servo power is on - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Movable range	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: Maintenance: Disposal: 	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - The drive power shall be turned off whenever a person enters the robot hazard zone. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of "transfer and installation" (transportation, unpacking, transfer, installation, and repacking), "use" (teaching, operation, and maintenance), and "disposal."

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	-
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operation procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	-
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty period The warranty period of the product is one of the following, whichever is shorter.
   1) 18 months after shipping from our factory
   2) 12 months after installation at customer site

2. Warranty scope
   1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
   2) We will repair a faulty product free of charge during the warranty period only if this product has been operated and used in accordance with the instruction manual supplied with the product.
      A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimers The warranty does not include the following cases.
   a) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
   b) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables*1).
   c) Irregularities confined to sensory quality such as noise not recognized as affecting functions
   d) When any modification or disassembly without our consent was performed
   e) Any failure or damage arising from inadequate maintenance and inspection or improper repair
   f) Any failure or damage due to an act of God, fire, and other force majeure
   g) Internal data such as programs and points created or modified by customers
   h) When a product purchased in Japan is taken abroad

*1): Consumable parts refer to the replacement parts for maintenance as listed in instruction Manual "15.2 Service Parts List."

4. Safety Information
   1) SHIBAURA MACHINE shall provide no guarantees for the standard performance of this product when used outside the specifications.
   2) If the customer does not observe the safety information appearing for DANGER, WARNING, and CAUTION contained in this instruction manual, SHIBAURA MACHINE will not be liable for any resulting injuries, death, or other accidents, damage, product failure, or any other problems.
   3) The safety information appearing for DANGER, WARNING, CAUTION and other explanations in this instruction manual are within the scope of usage assumed by SHIBAURA MACHINE, and they do not foresee all possible incidents for all situations that could occur. Please be aware that the scope of this safety information is limited.

Standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO 12100 “Safety of machinery” specifies risk reduction measures for machinery and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.

Based on the above, the ISO/IEC has defined “Guide 51,” which classifies the various standards in a hierarchal structure.
Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.

- ISO 10218, ISO 10218-1

After risk assessment is carried out, provision of residual risk information to the user is standardized.

- IEC 82079-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

Safety Information

This instruction manual contains important information to ensure the safe and correct use of the robot and to prevent injury to the operator and others and damage to property.
Be sure that you fully understand the following information (signal words and symbols) before reading the main text, and always follow the provided safety instructions.

[Screen explanation]

Signal word and symbol	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1)] to operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2)] and advice about performing work.

1) Injury refers to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) Damage to property refers to extensive damage to property or materials.

⚠ DANGER

- In order to ensure robot safety from installation to disposal, be sure to carefully read the separate instruction manual “Safety Manual” before starting the actual work.
  Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Specifications

This section describes the basic specifications, names of parts, coordinate system, and external dimensions of the THE600-E robot.

1. Specifications

1.1 Names of Parts

Fig. 1.1 shows the name of the robot parts.

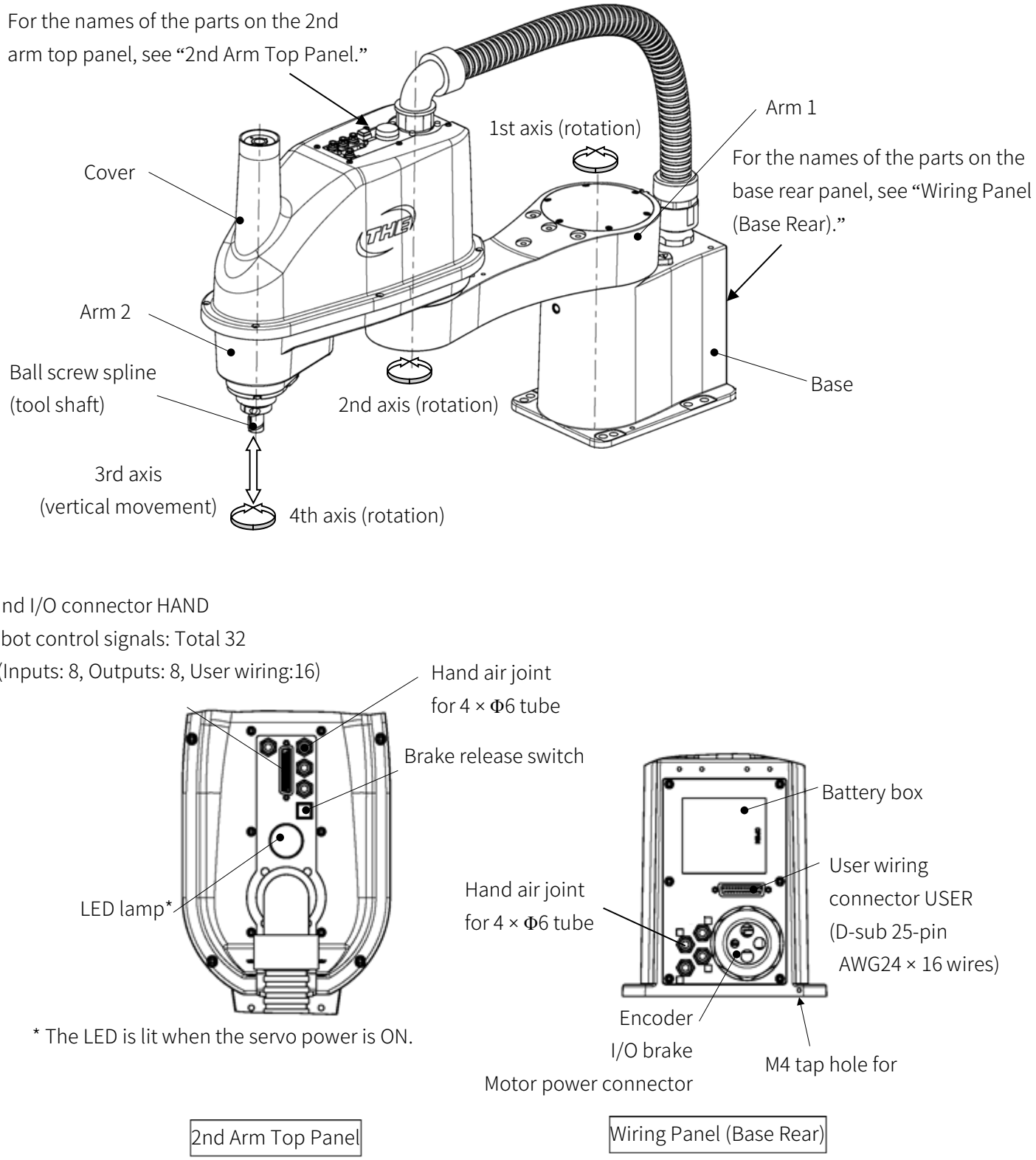


Fig. 1.1 Name of each part

1.2 Coordinate System

The robot's joint axis angle or stroke origin position (0° or 0 mm position) is factory-calibrated based on the reference planes.

Fig. 1.2 shows the base coordinate system and origin position of each axis joint angle.

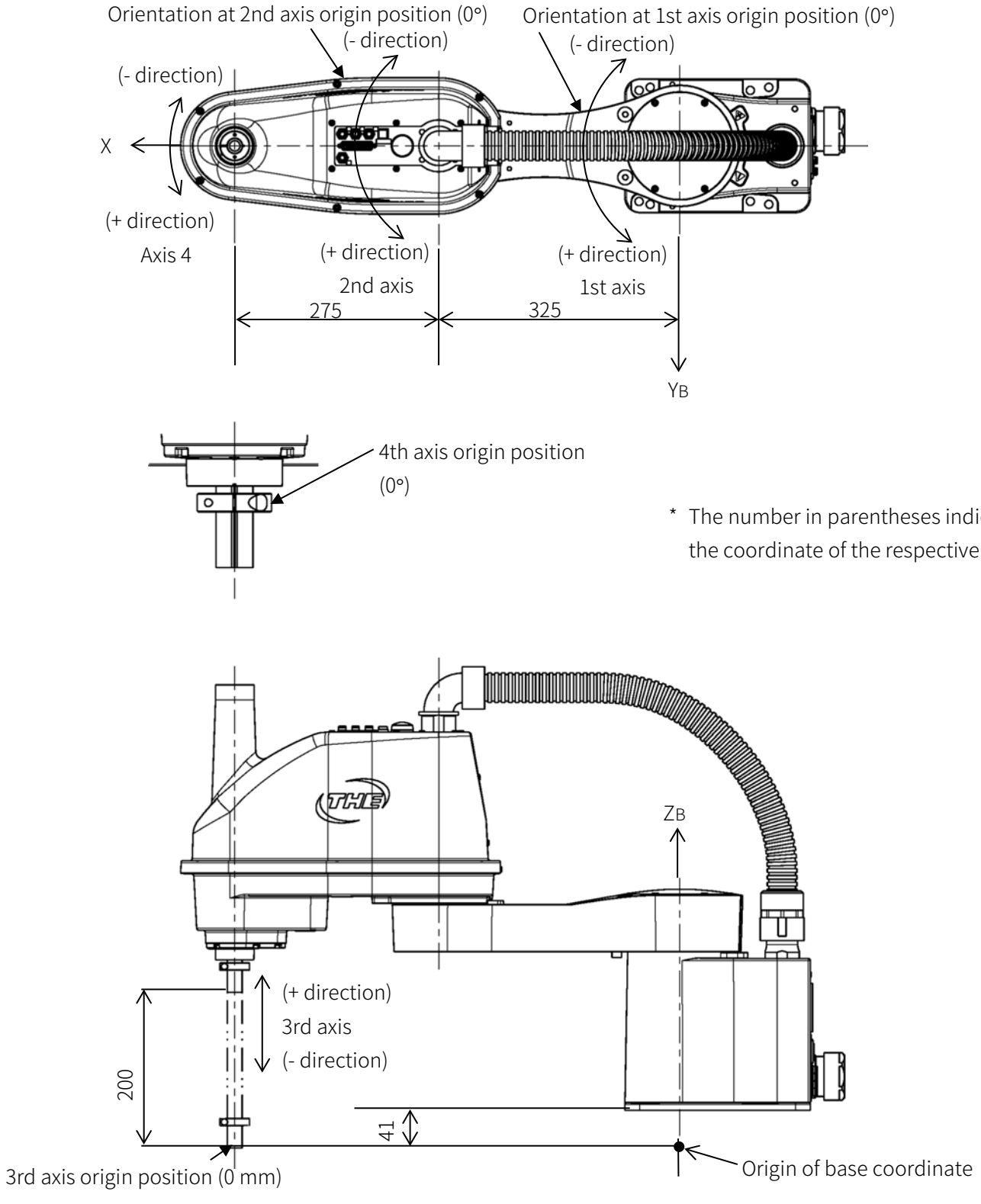


Fig. 1.2 Base coordinate system and joint axis angle origin position

1.3 External Dimensions

Fig. 1.3 refers to the external dimensions of the THE600-E.

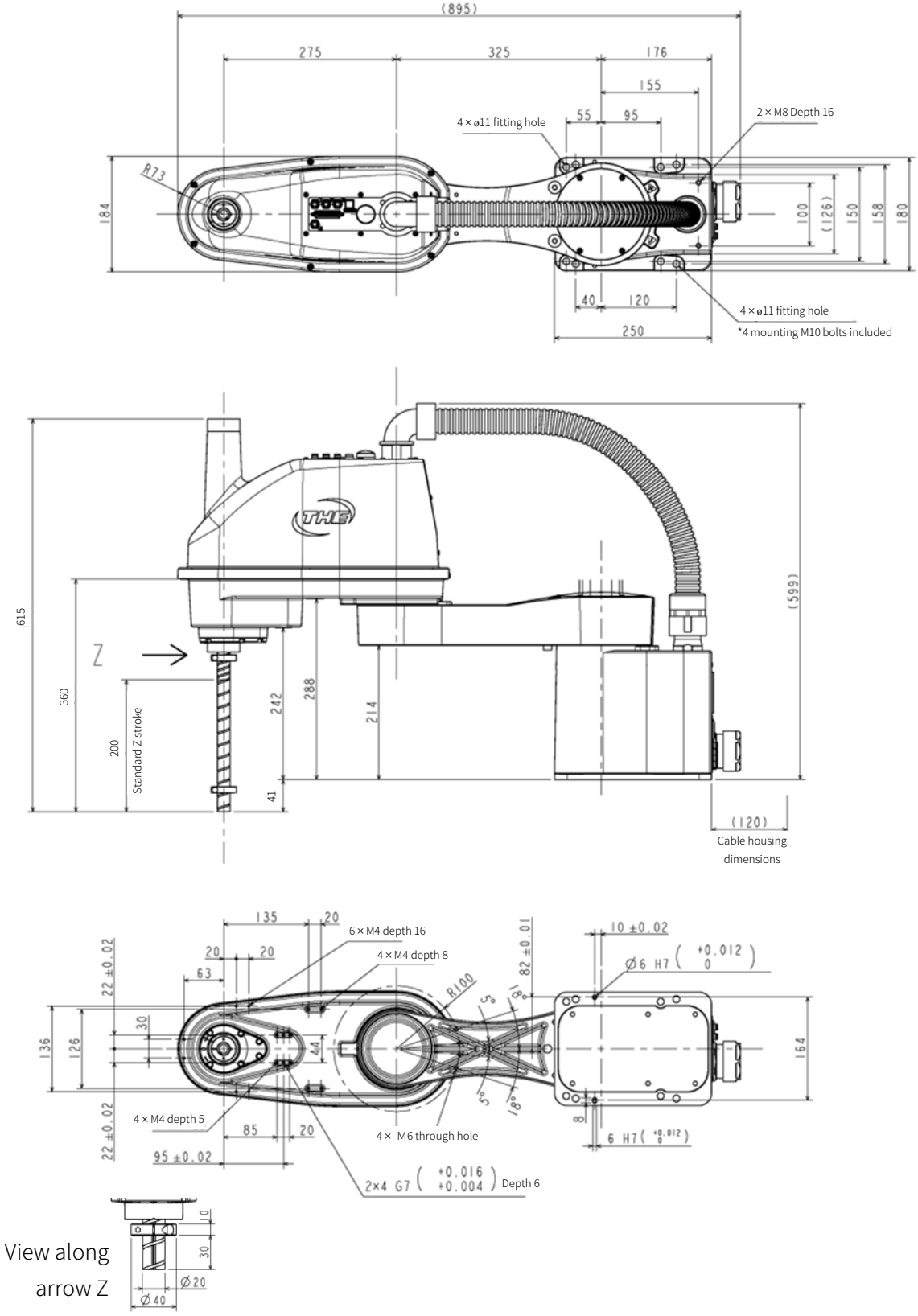


Fig. 1.3 External dimensions of the robot (THE600-E)

1.4 Specifications Table

Item		Specifications
Structure		Horizontally articulated SCARA robot
Type		THE600-E
Applicable controller		TS5000-EMS
Weight of robot body		31 kg
Number of controlled joint axes		Four (4)
Arm length		600 mm (275 mm + 325 mm)
Motor capacity/current limit	1st axis	1000 W / 28.2Ao-p
	2nd axis	400 W / 14.1Ao-p
	3rd axis	400 W / 18.4.Ao-p
	4th axis	400 W / 14.1Ao-p
Work envelope	1st axis	± 140°
	2nd axis	± 152°
	3rd axis	0 to 200 mm
	4th axis	± 360°
Maximum speed *1	1st axis	457°/s
	2nd axis	672°/s
	3rd axis	2000 mm/s
	4th axis	2359°/s
	1st and 2nd axes combined	8.0 m/s
Rated payload mass*1		2 kg
Maximum payload mass*1		12 kg
Permissible load inertia*1		0.25 kg·m^2
Maximum offset of payload center of gravity		100 mm
Repeatability*2	1st and 2nd axes combined	±0.01 mm
	3rd axis	±0.01 mm
	4th axis	±0.005°
Cycle time (when payload mass is 2 kg)*3		0.31 s
Drive system		AC servo motor
Position detection method		Absolute
Hand wiring		Inputs: 8, outputs: 8
Paint color*4 Material color*4		2nd Arm: Pantone 293C or equivalent Base and 1st arm: White AN-90 or equivalent Arm cover: White AN-90 or equivalent
Current amount (RMS)	1st axis	5.8A
	2nd axis	2.0A
	3rd axis	2.6A
	4th axis	0.8A

Maximum sound pressure level*5	73 dB (A)
Equivalent sound level (L_{Aeq}) *6	70.9 dB (A)
Power supply capacity*7	4.3 kVA

*1: The speed and acceleration are limited based on the operation pattern, payload weight, and the offset value.

*2: This is pose repeatability at a constant ambient temperature and machine temperature. It is not the absolute positioning accuracy.
It may exceed the specification values depending on the operation pattern, payload weight, and offset value.
X-Y and C represent values at the upper limit of Z. Path accuracy is not guaranteed.

*3: Continuous operation with a standard cycle operation pattern and that exceeds the effective payload ratio cannot be performed.
During rough positioning for back-and-forth motion by 300 mm in horizontal direction and 25 mm in vertical direction.

*4: The color may vary by the production lot, but this is not a problem in terms of product quality.

*5: The measurement conditions are as follows.
Operating conditions: Payload 2 kg, simultaneous movement of all joint axes, maximum speed, and maximum acceleration
Measurement position: Distance of 1 m from the 1st axis of the robot in five directions (front, rear, right, left, and up)
Maximum sound pressure position: 1 m directly above the 1st axis of the robot

*6: The measurement conditions are as follows.
Operating conditions: Payload 5 kg, simultaneous movement of all joint axes, maximum speed, and maximum acceleration
Measurement position: 1.6 m from the floor (860 mm from the installation surface) at a distance of -1 m in the X direction from the range of motion.

*7: The measurement conditions are as follows.
Operating conditions: Payload 2 kg, simultaneous movement of all joint axes, maximum speed, and maximum acceleration
Measuring instrument: Power analyzer (POWER ANALYZER 3390)

NOTICE

- Never use the robot in the high-temperature environment.
  Otherwise, the performance and lifespan of the alkaline batteries used for the position detector batteries will be reduced.
  - If the robot will be used in a high-temperature environment, please consult with the SHIBAURA MACHINE sales office.
- Never allow the lowered ball screw spline shaft to collide with an obstacle.
  Damage will occur before an alarm is triggered.
- When micro-vibrations occur, be sure to decelerate the robot.
  Otherwise, the performance and lifespan of the robot may be reduced. The robot will malfunction.
  - Use the ACCEL/DECEL command to adjust to an acceleration where the robot will not vibrate.
  - Micro-vibrations may occur on the robot tip depending on the orientation of the robot.
- To operate the 1st, 2nd, and 4th axes, be sure to raise the ball screw spline shaft as high as possible.
  If it is lowered, the shaft could be damaged prematurely.
  - To operate the 1st, 2nd, and 4th axes with the ball screw spline shaft lowered, use the SPEED command and ACCEL/DECEL command to adjust the operating speed and acceleration to prevent the ball screw spline shaft from vibrating.

2. Functions

2.1 Function for Specifying Pressing Force (N)

2.1.1 Usage Range

Figure 2.1 shows the range where the function for specifying the pressing force (N) can be used. Generally, use in the range indicated by “Usage allowed.” The range where the 3rd axis will drop is indicated by the “Drop” range. If the robot tip will be used with a light force, use it while being extremely careful about dropping. Do not use in the “Usage not allowed” range.
Table 2.1 shows the maximum margin of error when using this function. This data is intended to indicate the general trend, and so use it for reference only.

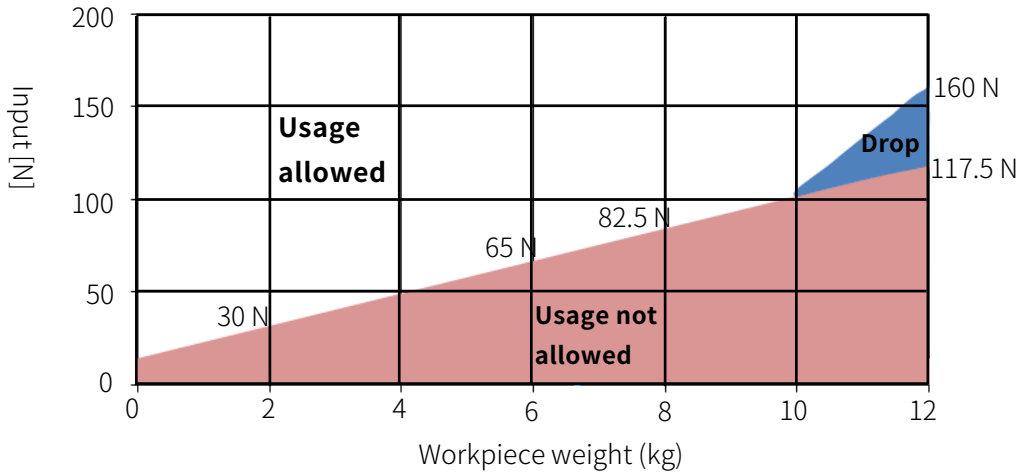


Figure 2.1 Usage range (THE600-E)

Table 2.1 Maximum margin of error in function for specifying pressing force

Model	Maximum margin of error
THE600-E	Approx. 30.0 N

2.1.2 Notes

NOTICE

- Be sure to always use the PAYLOAD command and specify a suitable setting value. If the PAYLOAD command is not set properly, a large difference could occur between the specified value and actual value.
- Use while grease is applied to the ball screw spline shaft. Usage without applying grease could lead to a large margin of error, resulting in damage to the pressed object or the ball screw spline shaft.
- If the robot was stopped for a half-day or longer, move the 3rd axis, and perform break-in operation for about 2 to 3 minutes. Usage without performing break-in operation could lead to a large margin of error, resulting in damage to the pressed object or the ball screw spline shaft.
- During the pressing operation, use with the SPEED command set to 10% or less. Although usage at conditions above 10% is possible, adjustment was performed at factory shipping so that a suitable value is obtained at 10%. If using under different conditions, a large margin of error could occur between the specified value and actual value, resulting in damage to the pressed object or the ball screw spline shaft.
- Adjustment was performed at factory shipping so that a suitable value is obtained when the override is 100%. Use at 100% for operation. However, in the operation check, operate at low speed to check that there are no problems, and then increase the override.

Installation

This section describes information you should know before unpacking, transferring, installing, and operating the robot.

3. Transportation

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with the arm	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

3.1 Transporting by Forklift

⚠ WARNING

- Never carry the packing boxes by hand when unloading them from a truck or other vehicle.
  A packing box could fall on a worker.
  - Use a forklift or similar equipment.
  - Do not allow people to approach the transportation route.

NOTICE

- Never carry the packing boxes by hand when unloading them from a truck or other vehicle.
  The worker may be unable to support the packing box and could fall over, resulting in damage to the robot.

The robot is packed and shipped in cardboard boxes. Unload the packing boxes stacked in the truck or other transportation equipment, and carry them to the unpacking location or other site. Transport carefully to ensure that the robot is not damaged.

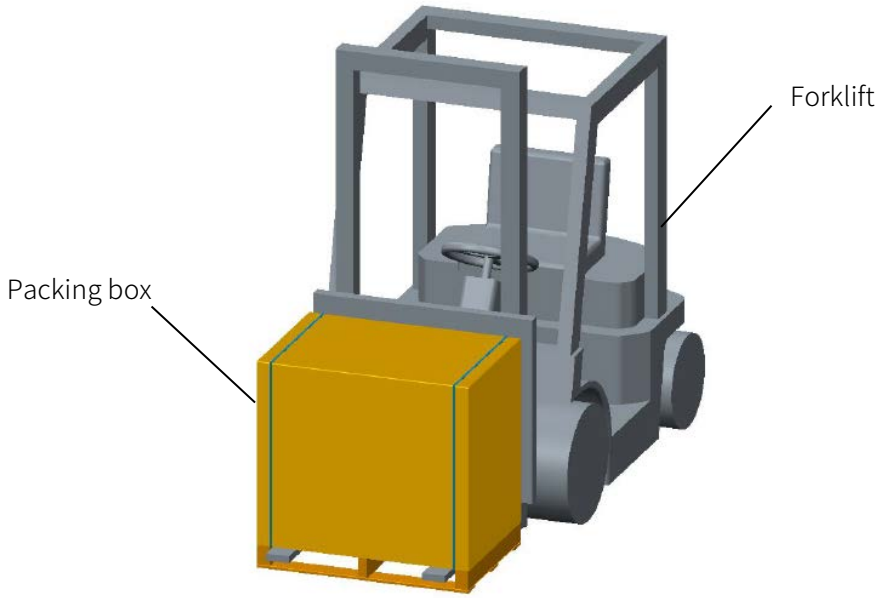


Fig. 3.2 Example of Transport Method for Packing Boxes

Table 3.1 Sizes and weights

Packing size	630 mm (W) × 900 mm (H) × 920 mm (D)
Packing weight	About 68 kg (not including options)

4. Unpacking

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

4.1 Unpacking

Fig. 4.1 shows the packing state. Unpack at a location suitable for later transfer and installation, and unpack while being careful not to damage the robot.

After unpacking, first, check that nothing has been damaged during transport and check the quantities of the accessories.

For robot and controller accessories, see the accessory list packed with the controller.

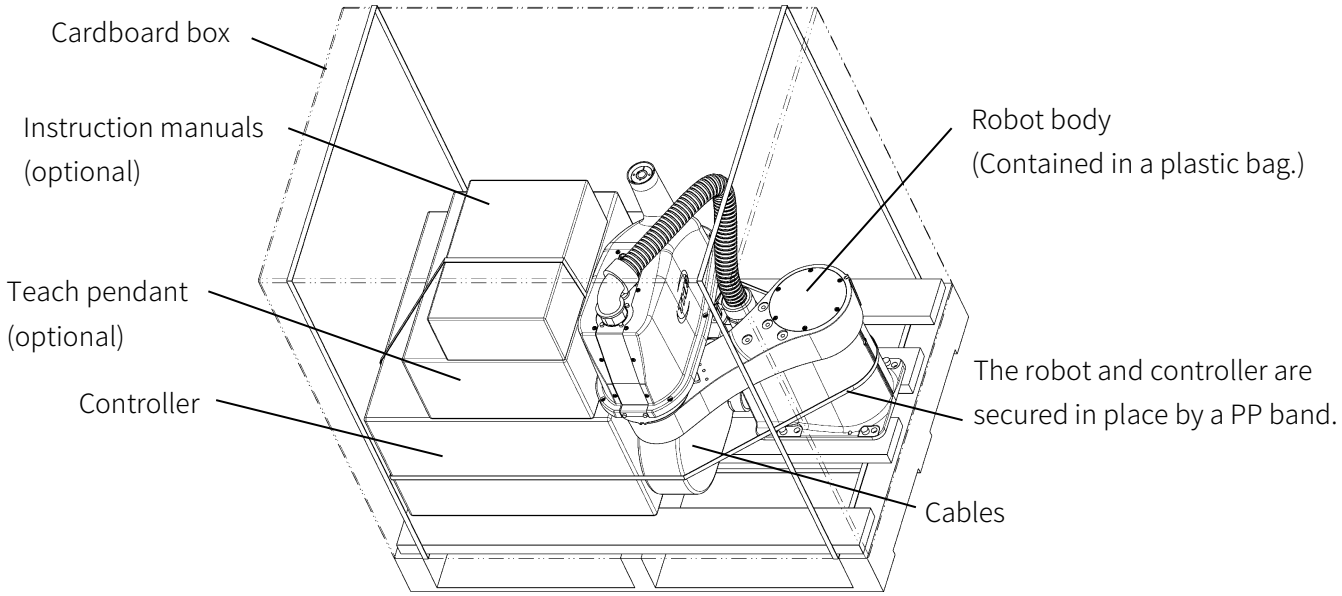


Fig. 4.1 Packaging state (THE600-E)

Table 4.1 Sizes and weights

Packing size	630 mm (W) × 900 mm (H) × 920 mm (D)
Packing weight	About 68 kg (not including options)

⚠ CAUTION

- When cutting the packing strap with the cutter, never point the blade edge towards a person. Otherwise, the person could get cut by the cutter.
  - Use a nippers or scissors.
  - Wear protective gloves.

⚠ DANGER

- After the work is completed, be sure to always check that the robot is not damaged.
  If the power is turned on for a malfunctioning robot, the robot could hit the worker due to an unexpected movement.
  - Check the appearance based on the external view shown in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or the parts are insufficient, please contact SHIBAURA MACHINE Sales Division.

NOTICE

- After the work is completed, be sure to always check that the robot is not damaged.
  A damaged robot could malfunction.
  - Check the appearance based on the external view shown in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or the parts are insufficient, please contact SHIBAURA MACHINE Sales Division.
- Be sure to dispose of any unneeded packing materials in accordance with the customer's company regulations.
  When storing the packing materials, be sure to manage them to prevent them from being used for other purposes than packing the robot.

MEMO

When storing the robot after unpacking or repacking for transportation, the customer should obtain a new packing strap (PP band), and secure the product in place as shown in Fig. 4.1.

⚠ WARNING

- Be sure to secure the robot to be stored in a packing box as it was delivered.
  Otherwise, the robot could fall over.
  - The customer must obtain new packing straps (PP bands).
  - If no packing box is available, use the mounting holes in the base to secure in place.

MEMO

- *When taking out the robot, refer to 5 Transfer.*
- *Refer to the separate controller manual for information on transporting, unpacking, and carrying the controller packing box included in the packing box.*

4.2 Storage

Transfer to the storage site carefully so as to ensure that the robot is not damaged. Avoid storing the robot and controller without using them for long periods of time after unpacking them.
Also, select a storage location that meets the environmental requirements in 6.1 Installation Environment.

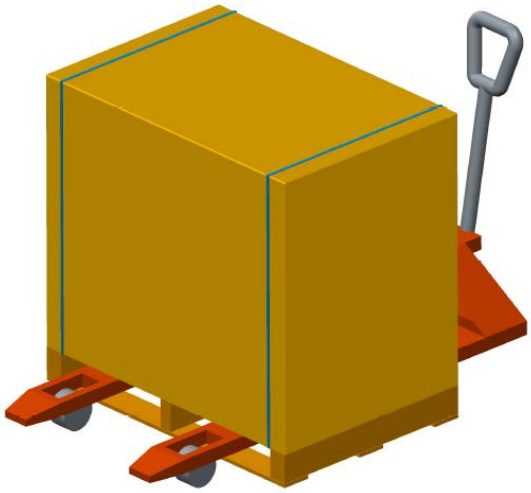


Fig. 4.2 Example of transfer method when storing

⚠ WARNING

- Never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - Use a hand lifter or similar equipment.
  - Do not allow people to approach the transportation route.

NOTICE

- Never carry the packing boxes by hand.
  The worker may be unable to support the packing box and could fall over, resulting in damage to the robot.
  - Use a hand lifter or similar equipment.
  - Do not allow people to approach the transportation route.

Avoid storing the robot and controller for long periods of time after unpacking them.
If this is unavoidable, be sure to strictly observe the following storage precautions.

NOTICE

- Never store the robot at a location exposed to direct sunlight, high temperature, or high humidity. Otherwise, the cable ducts, timing belt, and resin parts could deteriorate.
- DO NOT store the robot at a place where it is exposed to direct rain or water splashes, or at a place containing any toxic gas or liquid. Otherwise, the parts could deteriorate prematurely.
- When storing the robot, seal it in a plastic bag, and put a desiccant inside the bag.
- This prevents the entry of small animals and prevents rust from forming and dust from adhering to the ball screw spline unit.
  - Before storing the robot, apply grease to the ball screw spline shaft.
  - For the application method, see "13.2.2 Greasing the Ball Screw Spline Unit and Applying Corrosion Inhibitor."
  - Also apply grease to the ball screw spline shaft before use.
- Before operating a robot that was stored, be sure to perform operation at low speed (about 20% of override) for about four hours.
  During storage, the grease viscosity increases. If the robot is suddenly operated at a high speed, the poor lubrication could result in failure.
- Before operating a robot that was stored, be sure to replace the position detector batteries. (See "14.3 Replacing the Position Detector Backup Batteries")
  An alarm occurs before periodic inspection.
  - If power is not supplied to the robot for an extended period of time due to storage or other reasons, the data retention period by the position detector backup batteries will be shorter.
- When storing the robot, use the supplied fixtures to secure the arm and base, and then firmly secure the mounting sections. If placed directly on the floor, the robot will be unstable and will fall over.

5. Transfer

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

⚠ WARNING

- Never use an awkward posture when taking out the product from the packing box.
  Overexertion could injure the worker's back.
  - Provide space where work can be performed easily.
- Be sure to always store floor-routed cables in ducts and similar protection.
  Otherwise, someone could trip and fall on them.
  - Workers should wear a helmet to protect their head.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

5.1 Transfer

Transfer the robot while being careful that it is not subjected to any impacts or vibrations. If the robot will be subjected to vibrations over a long period of time, be sure to fully tighten all the fixing jigs and base set bolts.

In principle, the robot should be conveyed in the state shown in Fig. 5.1.
The arm is folded and is secured by the supplied fixing jig. (The robot is packed in this state, and even after it is unpacked, it should be transferred in this state.) When securing in place, be careful not to apply a large force to the ball screw spline shaft.

5.2 External Dimensions

The robot dimensions are shown in Fig. 5.1.

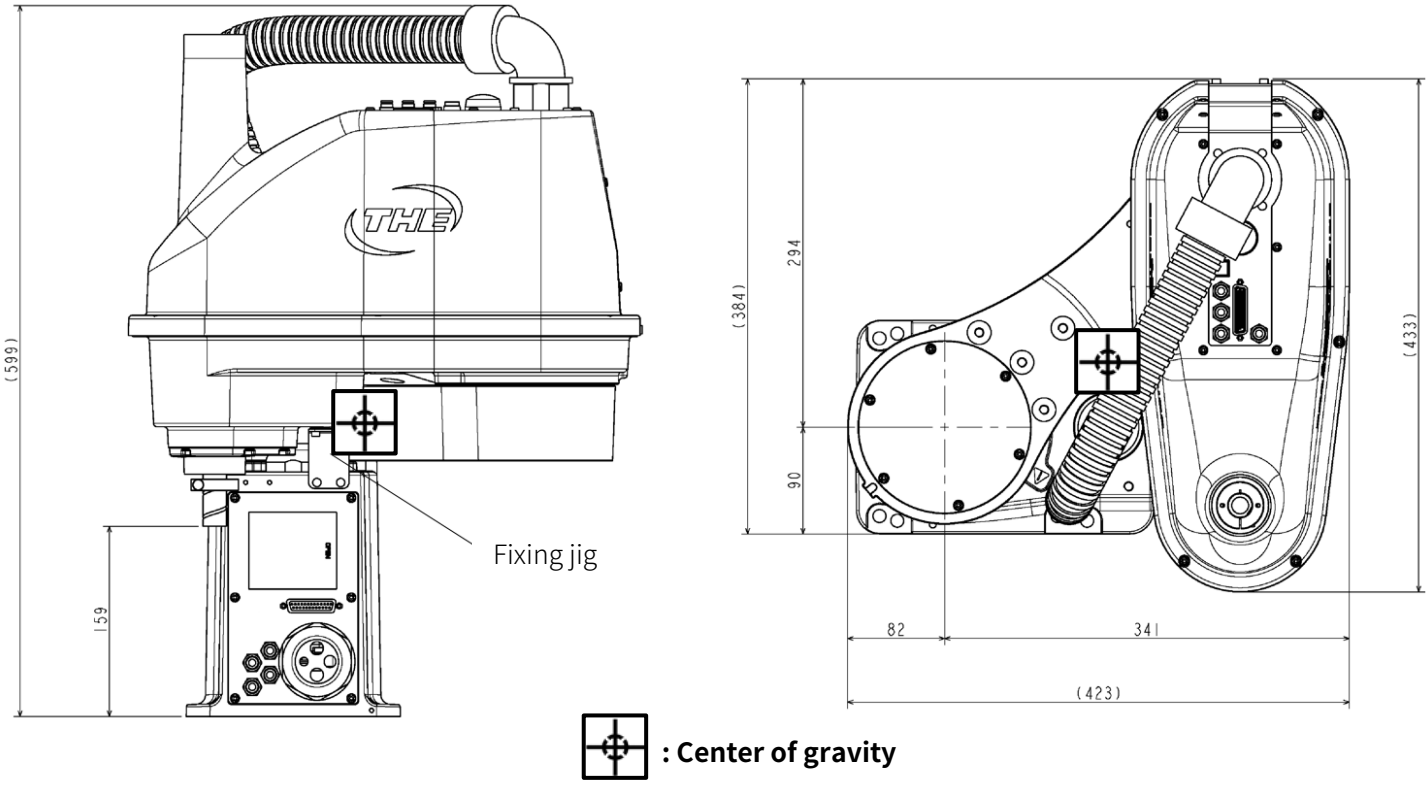


Fig. 5.1 Dimensions at transfer (THE600-E)

⚠ CAUTION

- When transferring the robot, be sure to always secure in place using the supplied fixing jigs. Otherwise, a worker could get caught in an arm that moved.
  - Check that no bolts of the jig are loose.
  - The weight of the arm section is approximately 17 kg.
  - Wear protective gloves to prevent slippage and to protect the hands.

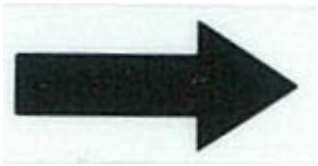
5.3 Transferring the Robot by Crane

⚠ WARNING

- When hoisting the robot, be sure to always follow the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the **supplied** eyebolts or eyebolts with a working load of 0.79 kN or more when suspended at 45°.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
  - Be sure to always hoist and lower the robot slowly.
  - Clear a transfer route, and make sure that no one goes under the suspended load.

The robot can be hoisted and transferred.

The following arrow marks are affixed to the base and 1st arm of the robot. Mount the two supplied M8 eyebolts into the tapped holes indicated by the arrow marks as shown below in Fig. 5.2, secure the wire to the eyebolts, and slowly hoist the unit by two-point suspension.

Label	Label description
	- Mount the supplied eye bolts into the tapped holes indicated by the arrow mark.

- When hoisting and transferring the robot by crane, be sure to mount the supplied eyebolts into the tapped holes indicated by the arrow marks. Otherwise, the robot will be unstable and could crash into a worker.

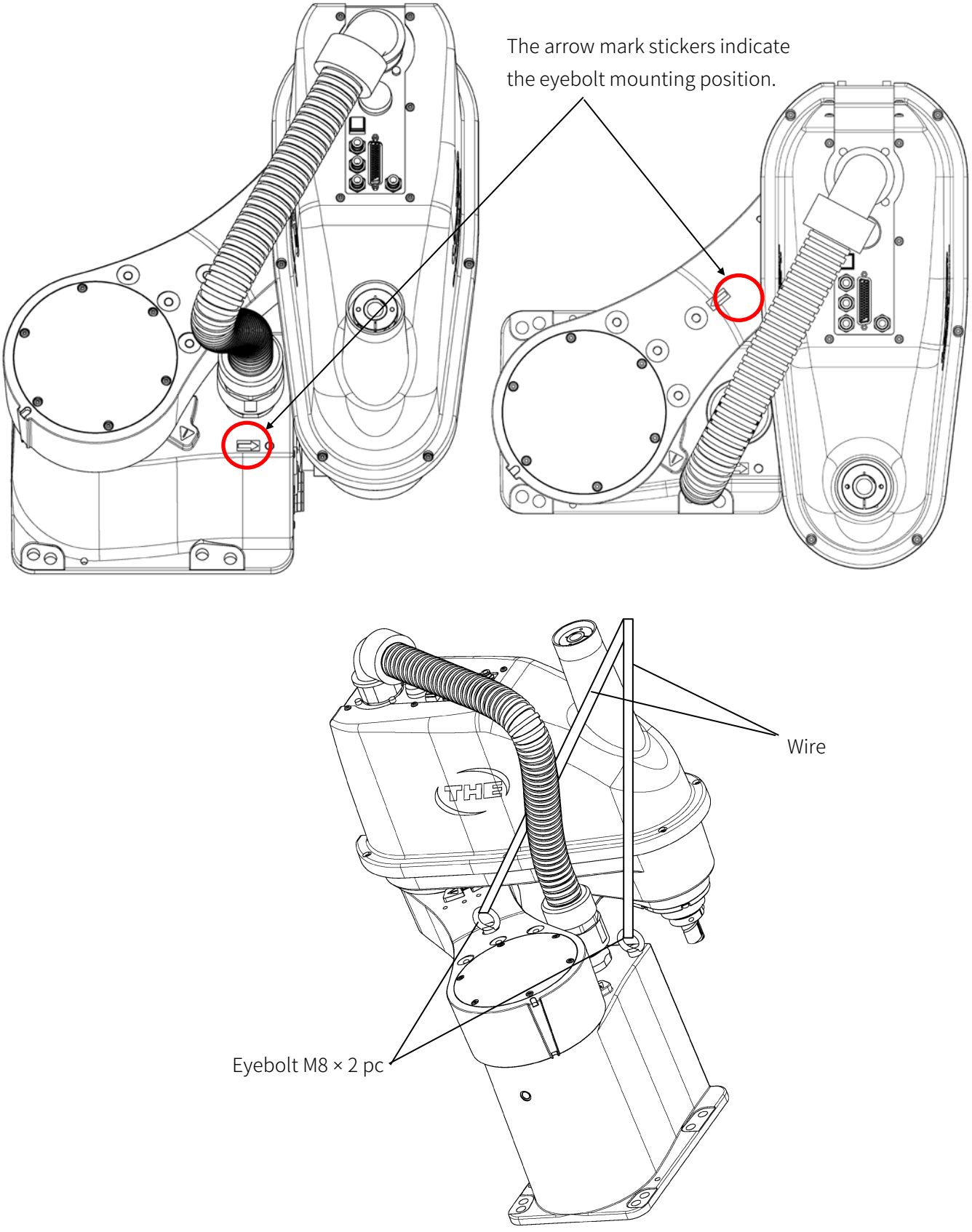


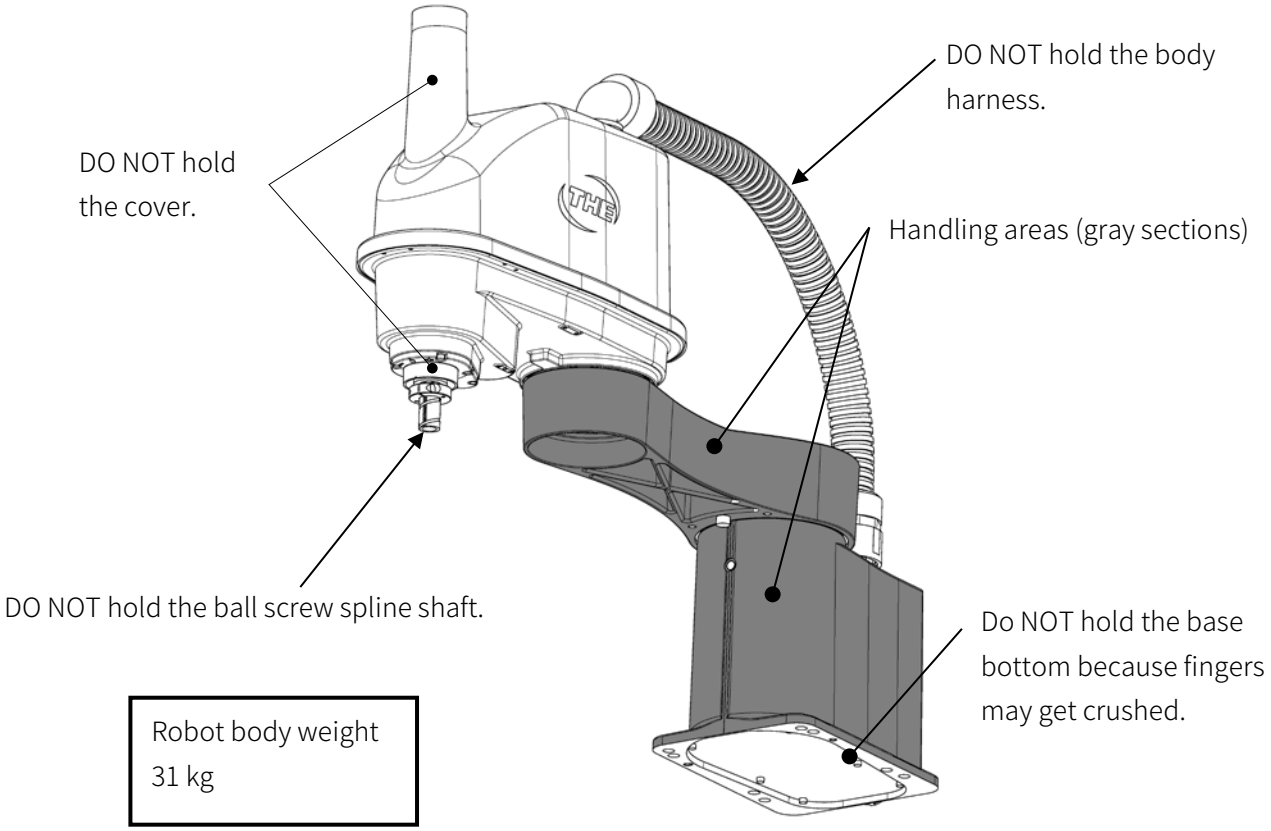
Fig. 5.2 Lifting up the robot

NOTICE

- When hoisting and transferring the robot, mount the supplied eyebolts into the tapped holes indicated by the arrow marks, and use the eyebolts to hoist the robot. Otherwise, the robot could lose its balance and fall over.
  - When hoisting with eyebolts provided by the customer, select a product with a working load of 0.79 kN or more when suspended at 45°.
- Never touch the ball screw spline shaft with your bare hands.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to hoist and lower the robot slowly.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- After installation is completed, be sure to remove the fixing jig and eyebolts.
  If the robot is moved while they are mounted, the robot could be damaged or fail.
  - The removed eyebolts will be used when the robot is transferred again, and so keep them in a safe place for future use.

5.4 Transferring the robot by Workers

If lifting up the robot by workers, see Fig. 5.3 to hold the robot by hands.



MEMO

This figure shows the handing areas, and it does not show the transfer orientation. For details on the transfer orientation, see Fig. 5.1.

Fig. 5.3Handling areas of the robot (THE600-E)

⚠ WARNING

- The robot should never be transferred by only one worker.
  Otherwise, the robot cannot be fully supported and could drop.
  - The robot weight is 31 kg.
  - Lift the robot using more than one person.
  - Use a cart or similar device to the transfer location.
  - Clear a transfer route, and make sure that a dropped robot would not hit anyone.
  - Wear protective gloves and safety shoes.

NOTICE

- The robot should never be transferred by only one worker.
  Otherwise, the worker may be unable to support the robot and it could fall over.
  - Do not use a product that has been dropped.
- Never touch the ball screw spline shaft with your bare hands.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- When lifting up the robot by workers, be sure to hold the gray portions shown in Fig. 5.3 by hands.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.
- After installation is completed, be sure to remove the fixing jig.
  If the robot is operated while the jig is mounted, the robot may be damaged or fail.

6. Installation of the Robot

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO

- *Before starting work, check the information in 10.1 Rescuing a Person Trapped by a Robot, and conduct a rescue drill.*
- *The 1st, 2nd, and 4th axes of THE600 can be moved by turning off the drive power (servo off) and pushing them by hand. By releasing the brake while the 3rd axis is powered on, it can be pushed and moved by hand without turning on the drive power supply (turning on the servo). For details, see 9 Teaching.*
- *We have prepared various methods for guiding the robot, which is done by turning on a drive power supply (turning on the servo). For details, refer to the following the separate manuals.*
  - TP5000 Operation Manual
  - TCmini (Internal PLC) Manual
  - Communication Manual
  - TSAssist User's Manual

6.1 Installation Environment

Table 6.1 shows the environmental conditions for the location in which the robot is to be installed.

Table 6.1 Environmental conditions for robot

Item	Specifications
Temperature	During operation: 0 to 40°C During transport and storage: -10°C to 50°C
Humidity	During operation, transport, and storage: 20% to 80% (no condensation) DO NOT store or install the equipment in locations where it will be exposed to water or other fluids.
Altitude	1000 m or less
Vibrations	During operation and storage: 0.98 m/s^2 or less
Dust	The environment must not be exposed to inductive dust. Consult with SHIBAURA MACHINE beforehand if the equipment will be used in a dusty environment.
Gas	The environment must not be exposed to corrosive or combustible gas.
IP (Ingress Protection) rating	IEC60529-1:1989, IP2X
Overvoltage category	IEC60664-1:2020, Class III
Protection against electric shock	IEC61140:2016, Class I
Pollution degree	IEC60664-1, Pollution degree 3
Sunlight	The environment must not be exposed to direct sunlight.
Power noise	No sources of excessive noise must be in the surrounding area.
Magnetic field	No devices that generate strong magnetic fields must be in the surrounding area
Other ambient environmental requirements	The environment must not be exposed to iron powder, oil, salt, or organic solvents.

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

NOTICE

- Never turn on the controller in a location where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Otherwise, the robot could fail.
- When operating the robot after an extended stop at a low temperature (10°C or less), be sure to perform operation at a low speed (about 20% of override) for four (4) hours.
  The grease viscosity increases in low-temperature environments. If the robot is suddenly operated at a high speed, the poor lubrication could result in failure.
  - Apply grease to the ball screw spline shaft before use.

6.2 Installation

Before the robot is installed, the installation layout must be examined by taking into account the working envelope, the coordinate system, and the maintenance clearance.

⚠ WARNING

- Be sure to always store floor-routed cables in ducts and similar protection.
  Otherwise, someone could trip and fall on them.
  - Workers should wear a helmet to protect their head.
- Never modify parts except those that are expressly permitted.
  Otherwise, the robot could fail, and cause an expected accident.

⚠ CAUTION

- Be sure to connect the cables to the controller after the robot is installed.
  Otherwise, a worker could receive an electric shock due to touching a cable that was broken due to a carried object that was dropped.
  - If a cable is broken, do not turn on the power.
- When removing the fixing jig, never grasp the edges strongly.
  If excessive force is applied, a hand or finger could be cut on the fixing jig.
  - Wear protective gloves.
  - The bolts that are fastened to the robot may become hardened due to vibration during transportation or transfer.

NOTICE

- Be sure to route the cables after installation is completed.
  Otherwise, a carried object could drop on a cable that was already routed, resulting in breakage.
  - Store the wiring in ducts or other protection.
- The cable between the robot and controller is not a movable cable. If the robot is mounted on a traveling axis or similar structure, the cable will be damaged prematurely.
- Never modify parts except those that are expressly permitted.
  Otherwise, the performance and lifespan could be reduced and the robot could malfunction.

6.2.1 Installing the Robot

⚠ WARNING

- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal operation (Table 6.2).

NOTICE

- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot could vibrate during operation. The robot will malfunction.
  - Prepare the frame by referring to the load applied during horizontal movement (Table 6.2).
- Be sure to install the robot on a level surface.
  Otherwise, the performance and lifespan of the robot could be adversely affected. The robot will malfunction.
- Be sure to secure the frame where the robot is installed to a floor or wall.
  Otherwise, the frame will move due to the force applied during robot operation.
- The robot should never be installed by only one worker.
  Otherwise, the worker may be unable to support the robot and it could fall over, resulting in breakdown.

The robot is secured in place using the mounting holes on the base (four locations).
Use the supplied hex socket head cap screws M10 × 25 to secure in place.
Table 6.2 shows the load on the frame during horizontal movement, and Fig. 6.1 shows the robot installation dimensions. Pinholes are provided on the base unit. When aligning the position of robot base coordinate system or when replacing the robot, the robot can be positioned using the pinholes.

Table 6.2 Load on frame during horizontal movement

Model	Load on frame during horizontal movement (Nm)	Robot weight (kg)
THE600-E	655	31

*This value is a reference value. When designing the frame, take the safety factor into consideration.

Also, use a frame having the flatness and surface roughness shown in listed in Table 6.3 for the frame where the robot is installed.

Table 6.3 Flatness of robot installation frame

Flatness of robot installation surface	Surface roughness of robot installation surface
0.02	Ra 6.3

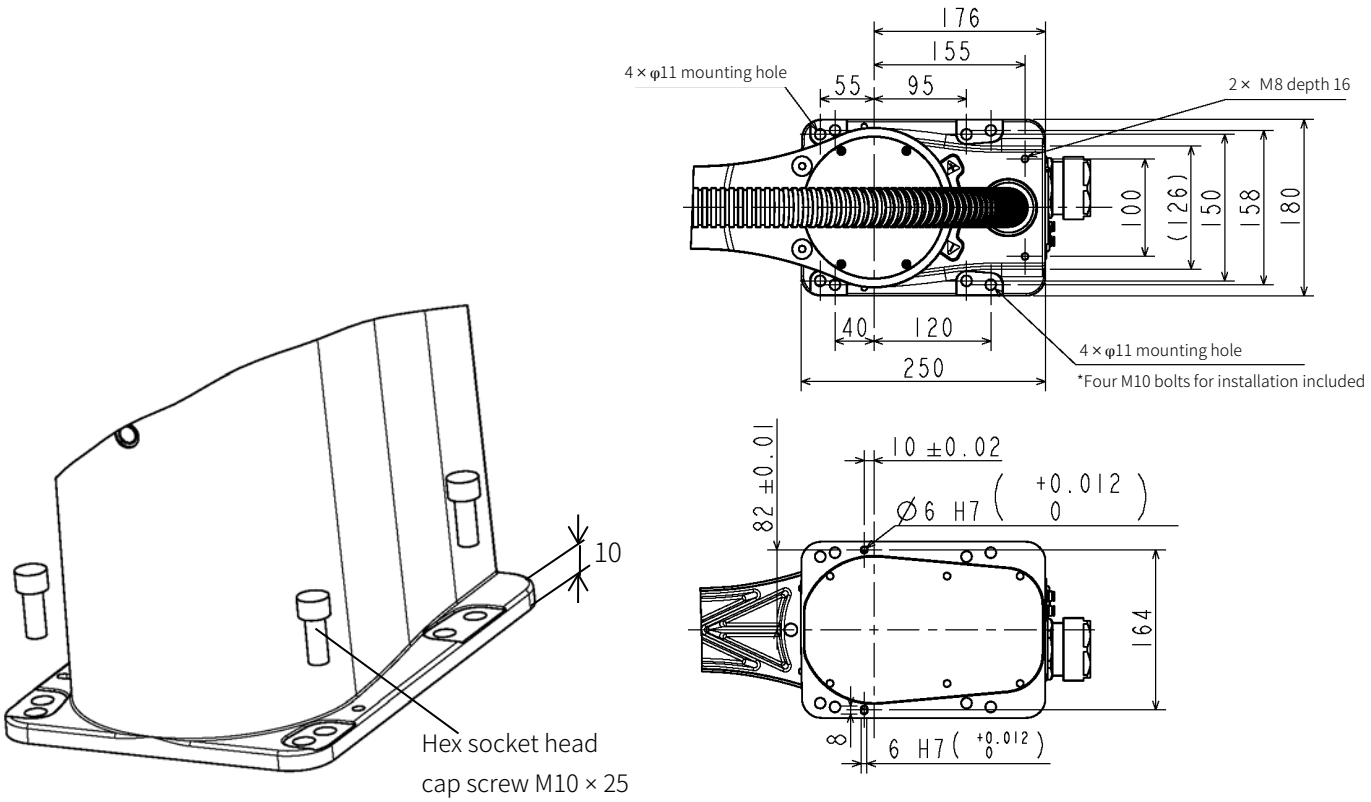


Fig. 6.1 Installation Method (THE600-E)

The robot base has a total of eight mounting holes for installation. Use four of the eight mounting holes to install the robot on the frame. An example of combining the mounting holes is shown in Fig. 6.2.

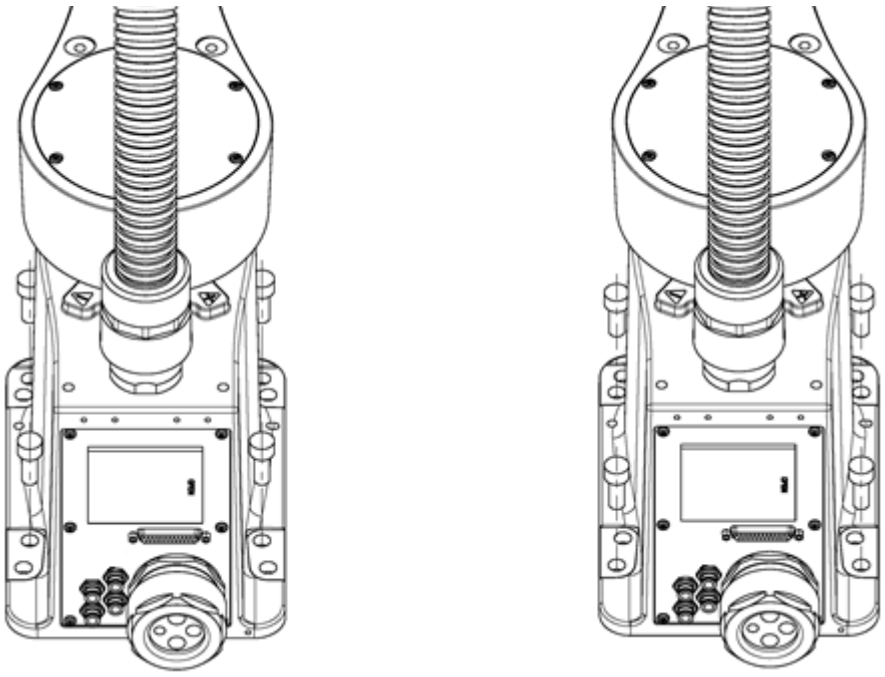


Fig. 6.2 Example of Combining Mounting Holes

6.2.2 Grounding the Robot

As shown in Fig. 6.3, the robot has a tap hole for grounding, and a PE label is affixed to mark the location.

When grounding the robot, use this tap hole to secure a ground wire terminal with a bolt.

⚠ DANGER

- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.

NOTICE

- Be sure to always connect a ground wire.
  Otherwise, the robot could fail due to noise.

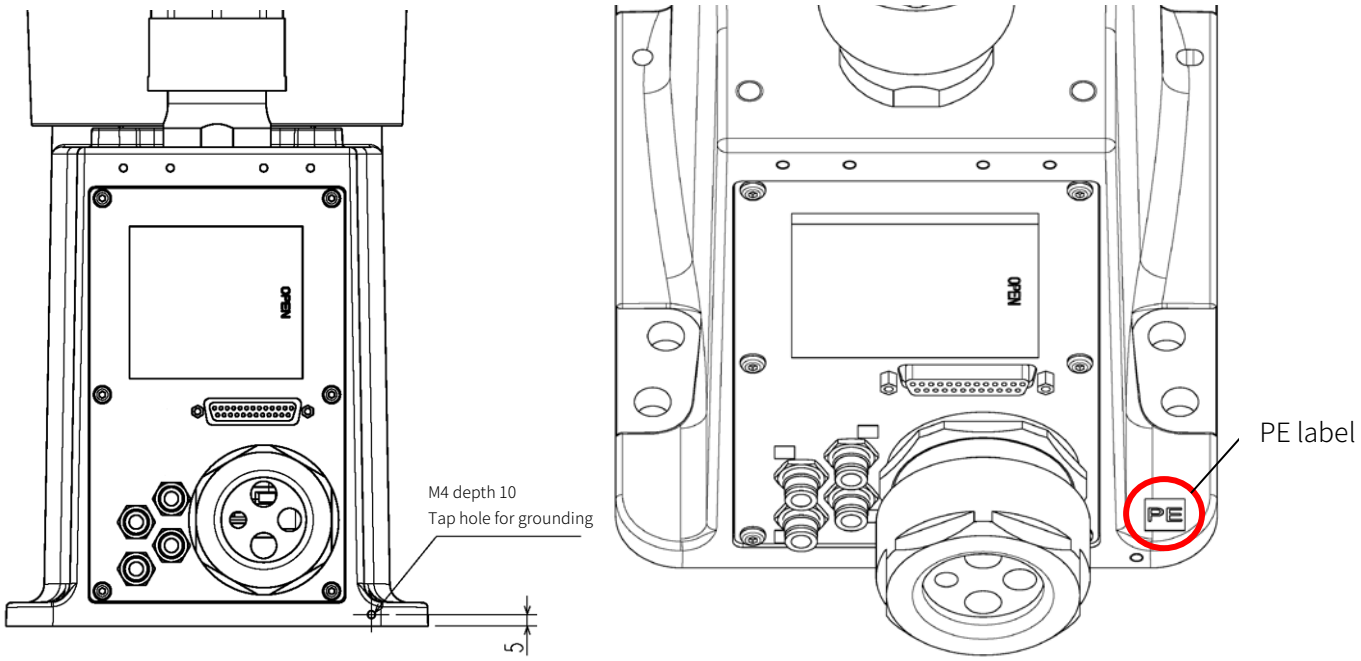


Fig. 6.3 Tap hole for grounding

MEMO

For the connecting ground wire, use a 2.0 mm² thick cable with a combination of green and yellow wires, and connect with a cable length not exceeding 1 meter.

6.3 Working Envelope

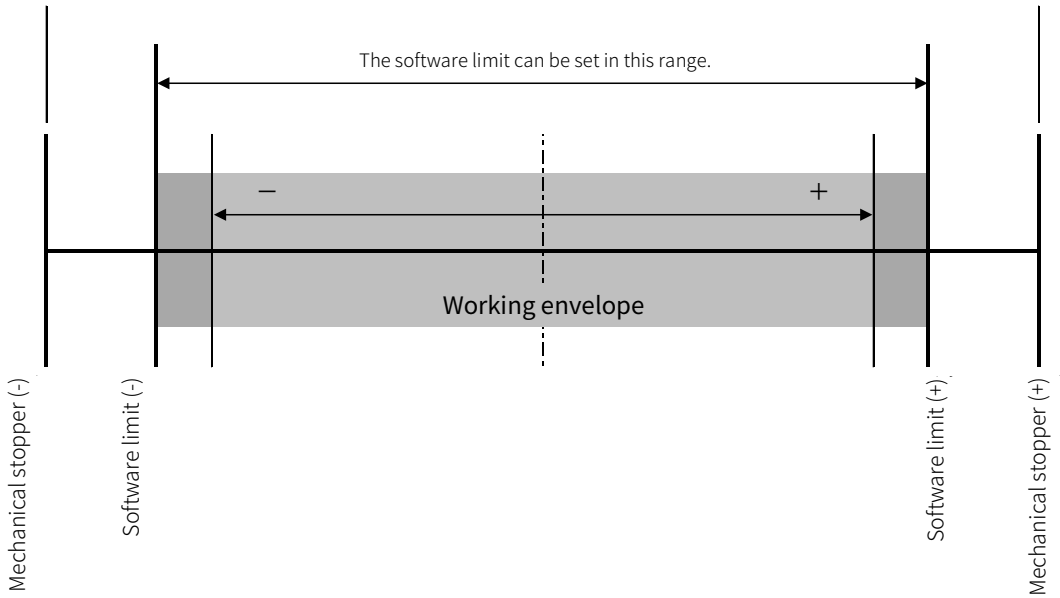
Fig. 6.4 shows the operating range.

Each joint axis can operate within the working envelope. To prevent the robot from operating outside the working envelope due to mistaken operation, it has two types of movement limit functions.

The 1st joint axis, 2nd joint axis, and 3rd joint axis each have mechanical stoppers on the outer side of the working envelope.

In addition, there are software limits that can be set by the customer.

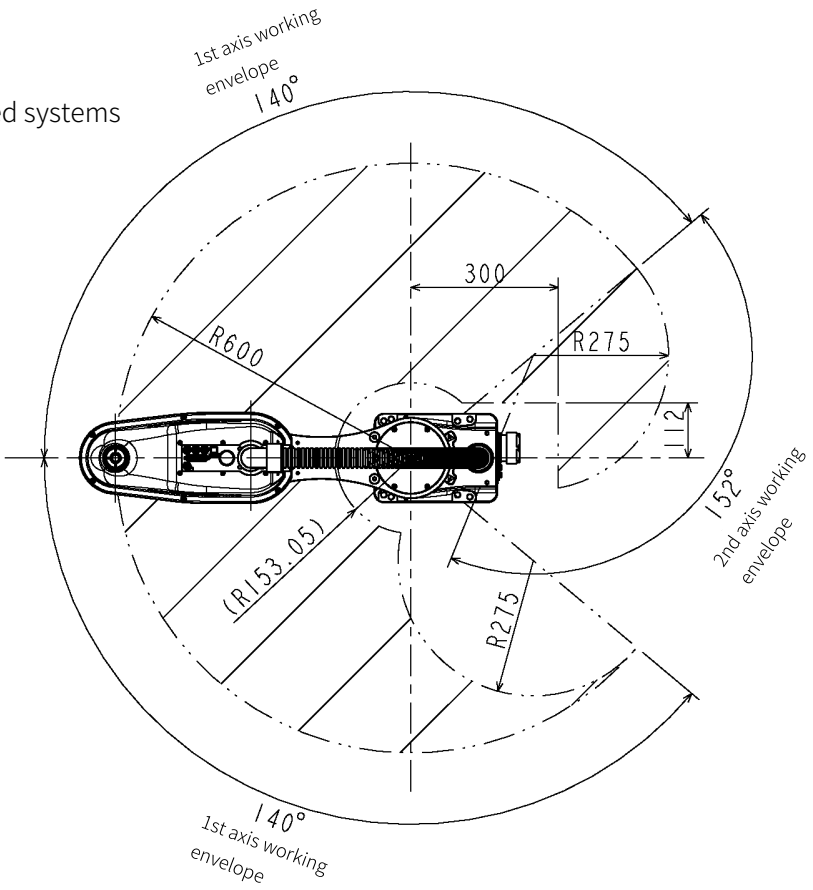
For details, refer to the separate instruction manual “STEA1226 User Parameters Manual.”



MEMO

The software limit cannot be set to a range that is wider than the value set at shipment. The software limit can be set to a range that is narrower than the value set at shipment, and so if changing the mechanical stopper position and narrowing the working envelope, be sure to also change the software limit.

For left-handed systems



For right-handed systems

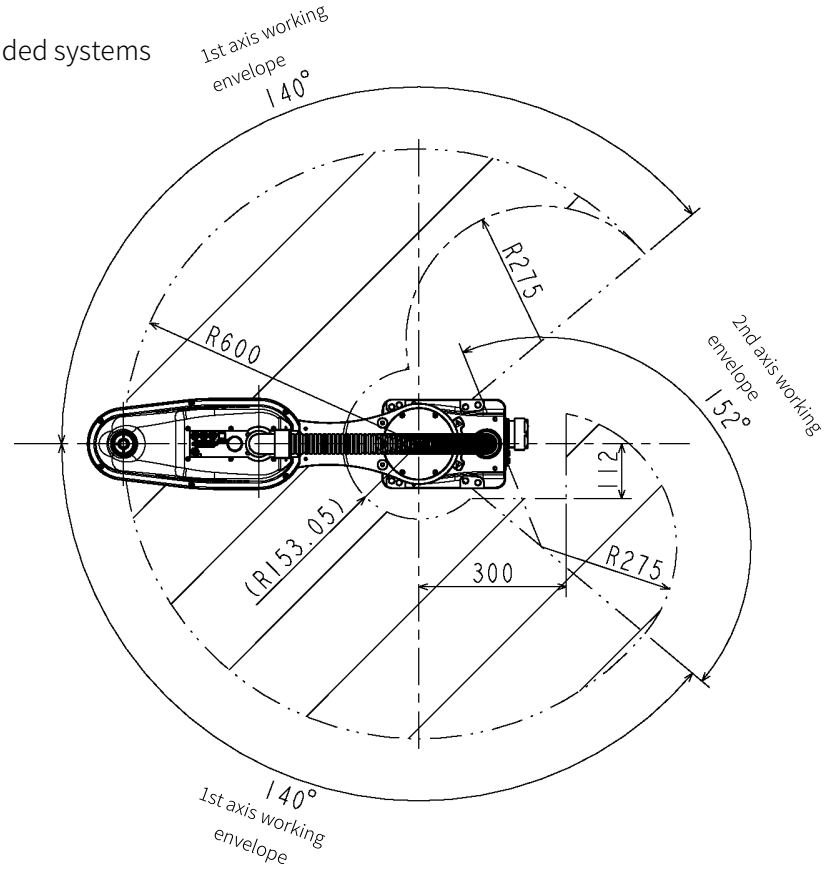


Fig. 6.4 Working envelope

6.3.1 Changing the 1st and 2nd Axis Working Envelope

The robot is equipped with mechanical stoppers that mechanically limit the working envelope of each joint axis. Changing the mechanical working envelope of the robot by changing these mechanical stoppers is referred to as “changing the working envelope.”
This section explains how to change the working envelope for the joint axes of the robot.

NOTICE

- Never reuse a mechanical stopper that has collided with the arm and was used to change the working envelope. A damaged mechanical stopper will not function properly.
- When designing, manufacturing, and mounting the mechanical stoppers, be sure to observe the following points.
  - Be sure to set the changed working envelope to be within the range shown in the specification table and external view drawing.
    The robot cannot be operated in a working envelope that is outside the specifications.
  - Set the origin position information again by following the procedure in “14.8 ③ Other Origin Position Settings.”
    If a mechanical stopper is changed, the function described in “14.8.1.4 Restoring Data by SYSTEM1 or SYSTEM2” is disabled. At installation, enter the original position data by following the procedure in “14.8.1.2 Setting HOME1 and HOME2.”
  - A function is also available to restore the origin position using a setting jig provided by the customer. If using the procedure in “14.8.1.3 Restoring Data by HOME1 or HOME2” enter the origin position data using the procedure in 14.8.1.2 Setting HOME1 and HOME2” during the installation process.
  - A robot failure caused by a mechanical stopper is outside the scope of the warranty.
- Be sure to set the software limits inside the mechanical stoppers.
  If the software limits are not set inside the mechanical stoppers, the robot will collide with the mechanical stoppers during operation.
- If the robot collides with a mechanical stopper and operation is stopped, immediately turn off the controller power, and contact the SHIBAURA MACHINE Service Department (see Service Network).

The 1st axis working envelope can be changed by changing the position of mechanical stoppers as shown in Fig. 6.5.

Table 6.4 Axis-1 working envelope before and after change

		Before change	After change
1st axis working envelope	+ direction	140°	110°
	- direction	140°	110°

Mount hex socket head cap screws (M8 × 12) into the outside tap holes.

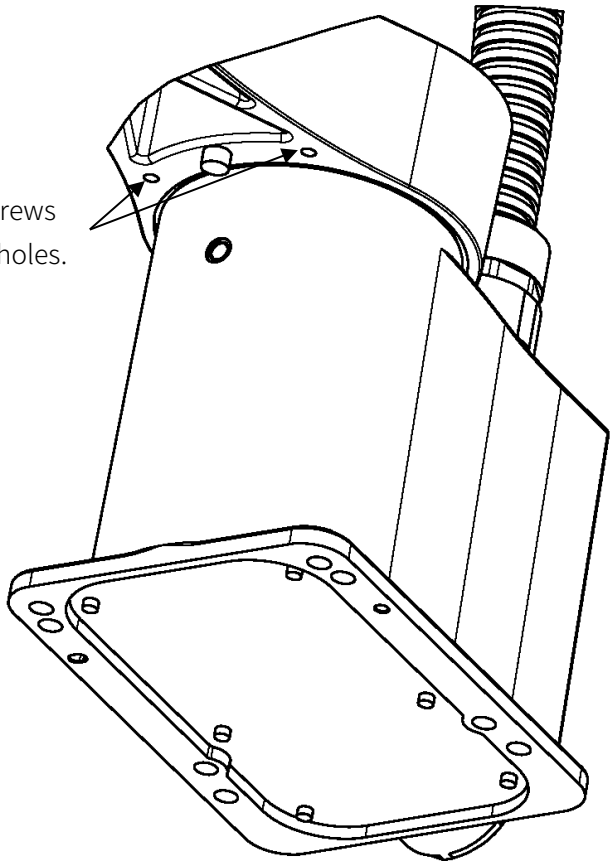


Fig. 6.5 Example of changing the 1st axis working envelope

When the 2nd axis working envelope is changed, it can be set to any working envelope by changing the shapes of the mechanical stoppers. Fig. 6.6 shows the mounting position of the 2nd axis mechanical stopper, and Fig. 6.7 and Fig. 6.8 show examples of changing the 2nd axis mechanical stopper. The customer should design and manufacture the mechanical stoppers as needed.

Table 6.5 Axis-2 working envelope before and after change

		Before change	After change
2nd axis working envelope	+ direction	152°	120°
	- direction	152°	152°

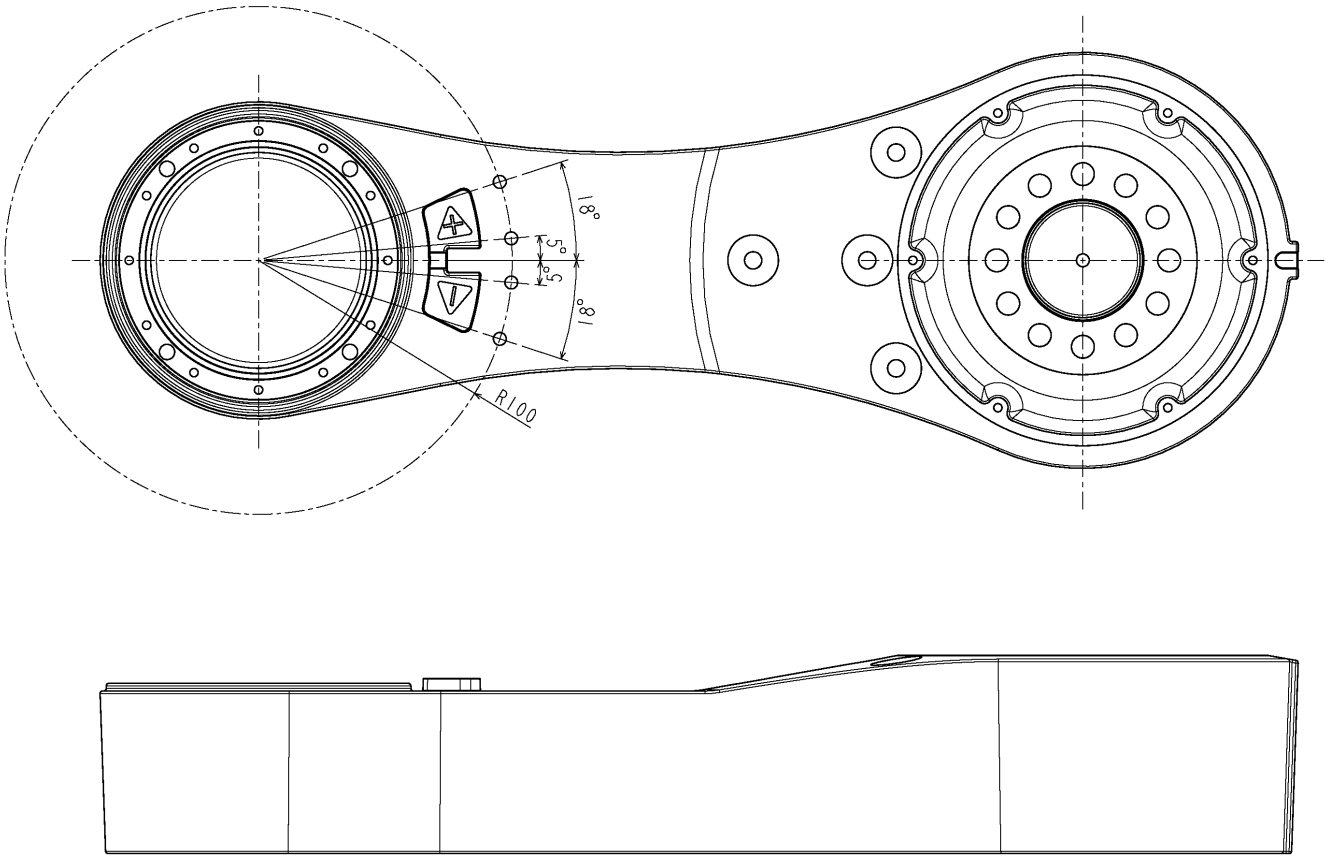


Fig. 6.6 Mounting position of axis-2 mechanical stopper

Example of changing the 2nd axis mechanical stopper
Working envelope settings: 120°, -152°

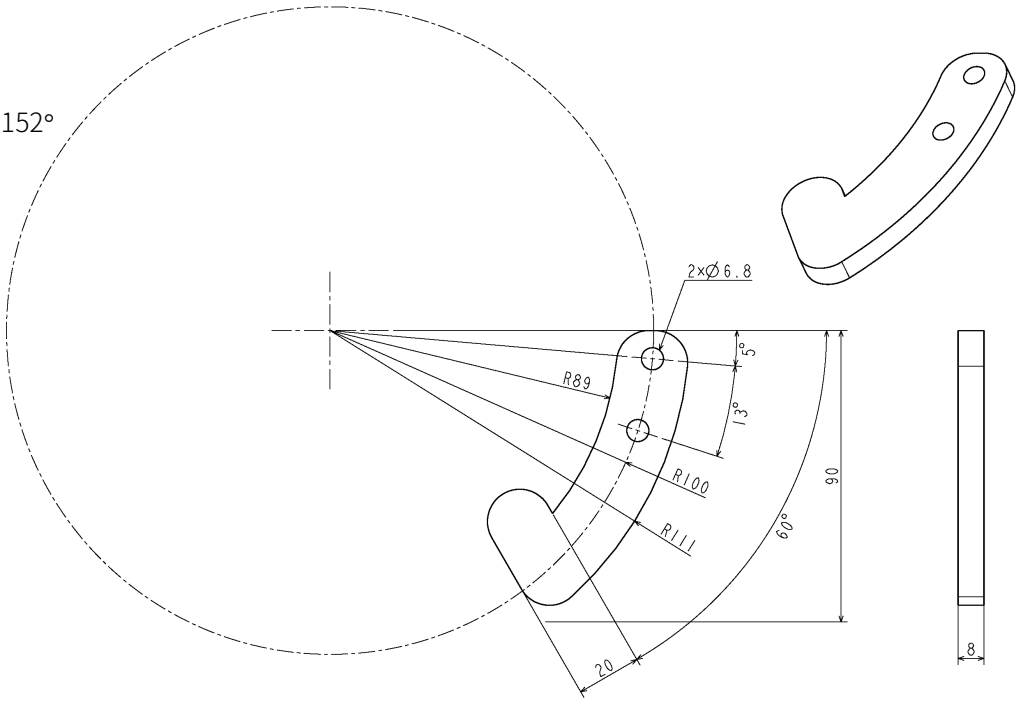


Fig. 6.7 Example of changing the 2nd axis mechanical stopper

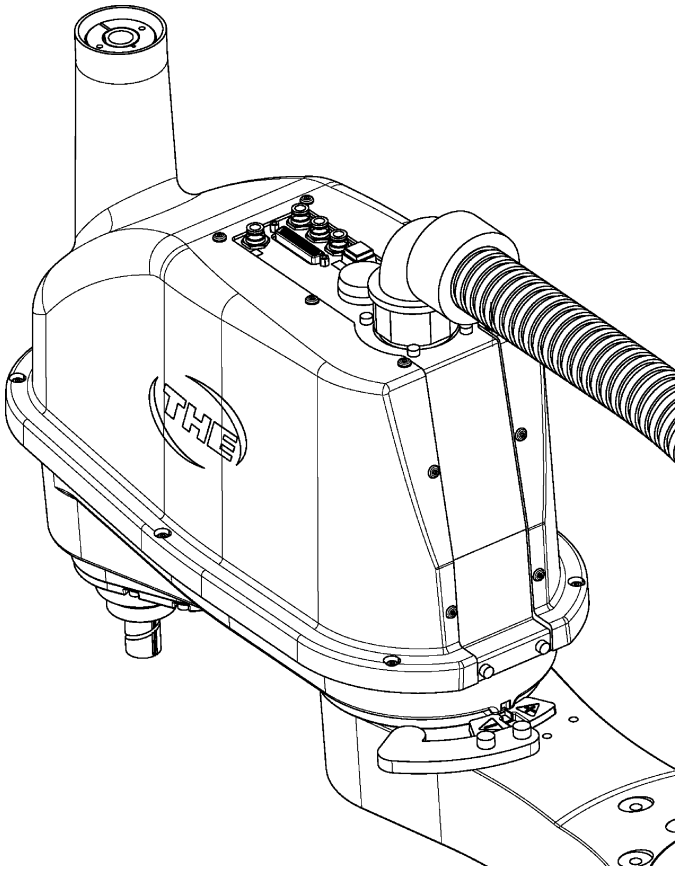


Fig. 6.8 Example of changing the 2nd axis working envelope

If a working envelope is changed, the user parameters must also be changed.
For the procedure on changing the software limits, see "6.3.3 Changing the Software Limit."

6.3.2 Changing the 3rd Axis and 4th Axis Working Envelope

The 3rd axis is equipped with a mechanical stopper, but when applying a limit to the working envelope, be sure to also change the software limit. The working envelope for the 4th axis does not use a mechanical stopper and is limited by the software limit only. For details on changing the 3rd axis and 4th axis working envelope, see "6.3.3 Changing the Software Limit."
To change the working envelope of the 3rd axis mechanical stopper, contact the SHIBAURA MACHINE Service Department.

⚠ WARNING

- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

6.3.3 Changing the Software Limit

NOTICE

- Be sure to set the software limit inside the mechanical stopper.
  Otherwise, the robot could collide with the mechanical stopper during operation.
- If the robot collides with a mechanical stopper and operation is stopped, immediately turn off the controller power, and contact the SHIBAURA MACHINE Service Department (see Service Network).

If the mechanical stopper is changed, be sure to change the software limit as well.
The software limit can be changed using one of the following two methods.

[1] Using the teach pendant to change the software limit from the software limit setting screen
For details, refer to section “8.2.1.1 Setting the software limit” in the separate instruction manual “STEA1162 TP5000 Operation Manual.”

[2] Changing the setting value of the user parameter file (file name: USER.PAR)
The customer can set some software limits. For details, refer to section “[U14] SOFTWARE LIMIT” in the separate instruction manual “STEA1226 User Parameters Manual.”

When the software limit is changed using one of the methods above, the software limit value set in the default user parameter file (file name: USER.PAR) is changed and overwritten.
Therefore, **before changing the software limit, be sure to back up the “user parameter file”** so that the software limit value set in the factory-set user parameter file can be identified.

If the software limit is changed using the method in [2], be sure to turn off the power and then turn it on again.
The changes to the user parameters are not applied unless the robot is restarted.

7. Setting the Hand

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.

Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO

Before starting work, check the information in 10.1 Rescuing a Person Trapped by a Robot, and conduct a rescue drill.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

7.1 Mounting the Robot Hand

7.1.1 Selecting the Robot Hand

A robot hand provided by the customer can be mounted on this robot. For the shape of the hand mounting surface, see "7.1.2 Hand Mounting Section Specifications."
Also, select or fabricate a hand that meets the following requirements.

- Make sure that there are no protrusions or other hazardous parts, except for those necessary for work.
- The structure should be able to grasp the target object stably even if the robot stops suddenly due to a power failure, breakdown, or emergency stop during operation.
- Hand that meets the allowable load requirements in Table 7.2 while grasping the target object

7.1.2 Hand Mounting Section Specifications

The hand is mounted to the lower end of the shaft or onto the tool flange. Fig. 7.1 shows the dimensions of the shaft lower end and the tool flange. If the hand is mounted onto the tool flange, the tool is centered using the φ12H7 mating section. The hand direction is adjusted by the 4 × 4 keys and secured in place by the four (4) M4 bolts. The tool flange is optional.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal "Output signal initialization" or "RESET DOUT" command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the robot hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.

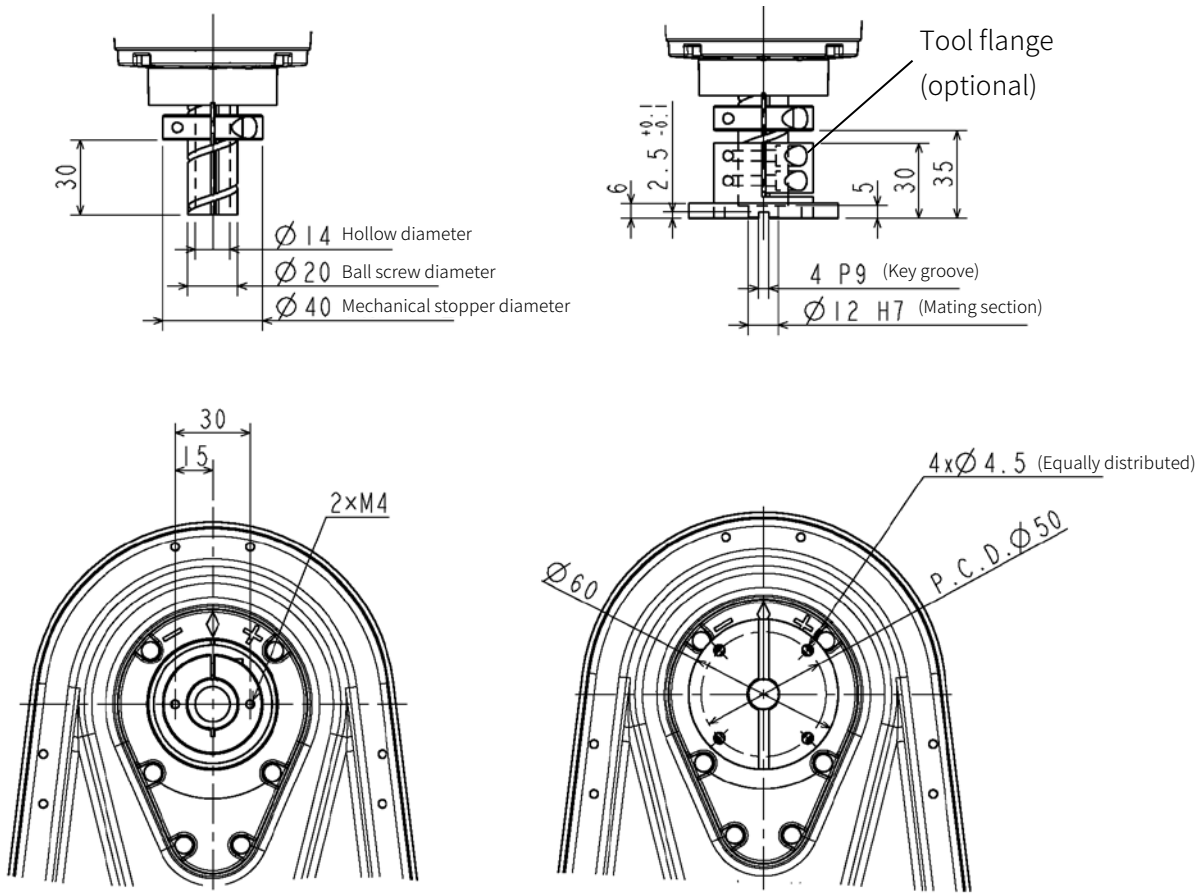


Fig. 7.1 Hand mounting dimensions

7.2 Mounting Camera and Air Valve

Tap holes like those shown in Fig. 7.2 are provided in the 2nd arm bottom surface and the base. These tap holes are used when a camera, air valve, or other device is mounted on the arm. Fig. 7.3 shows a mounting example.

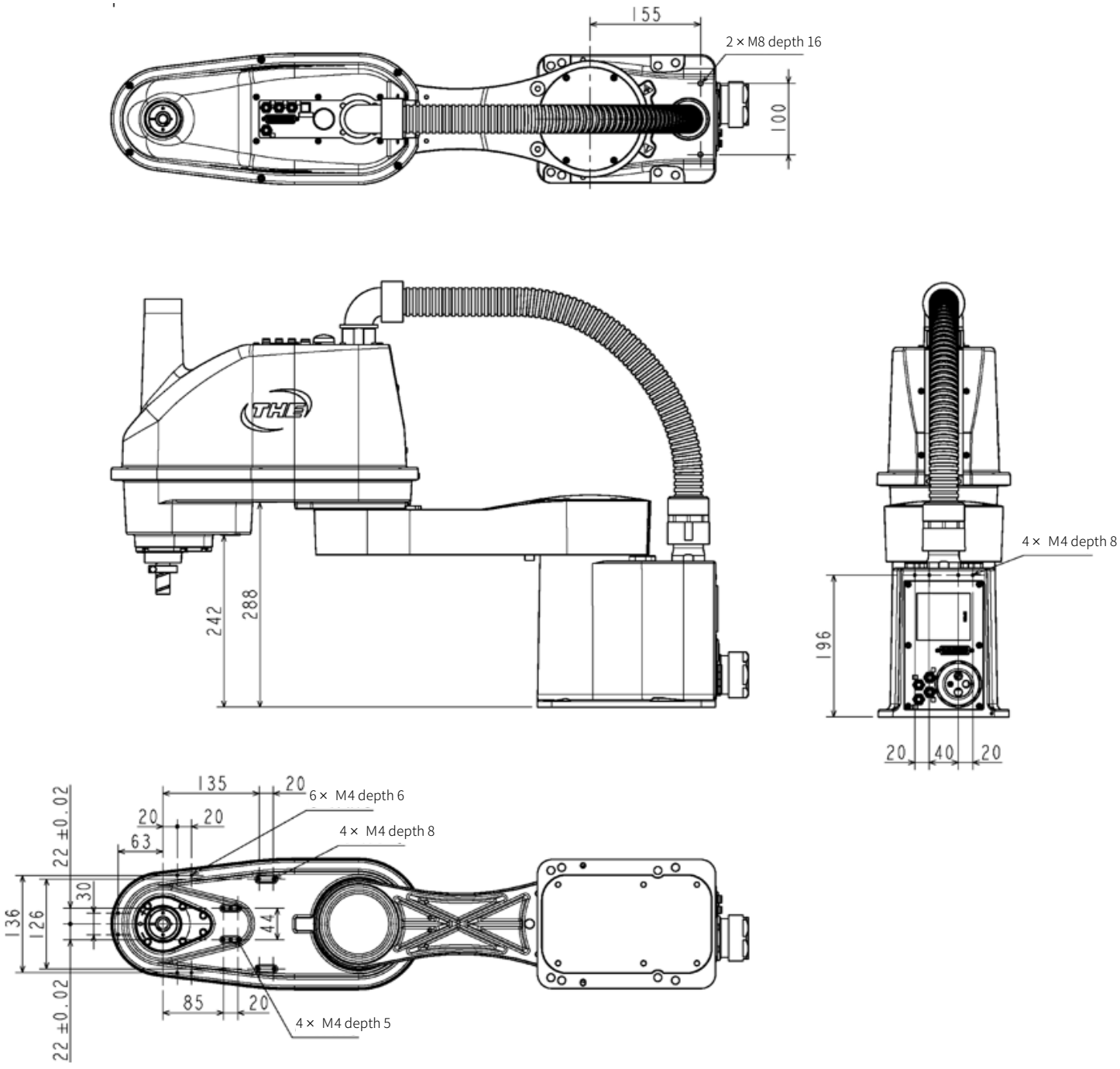
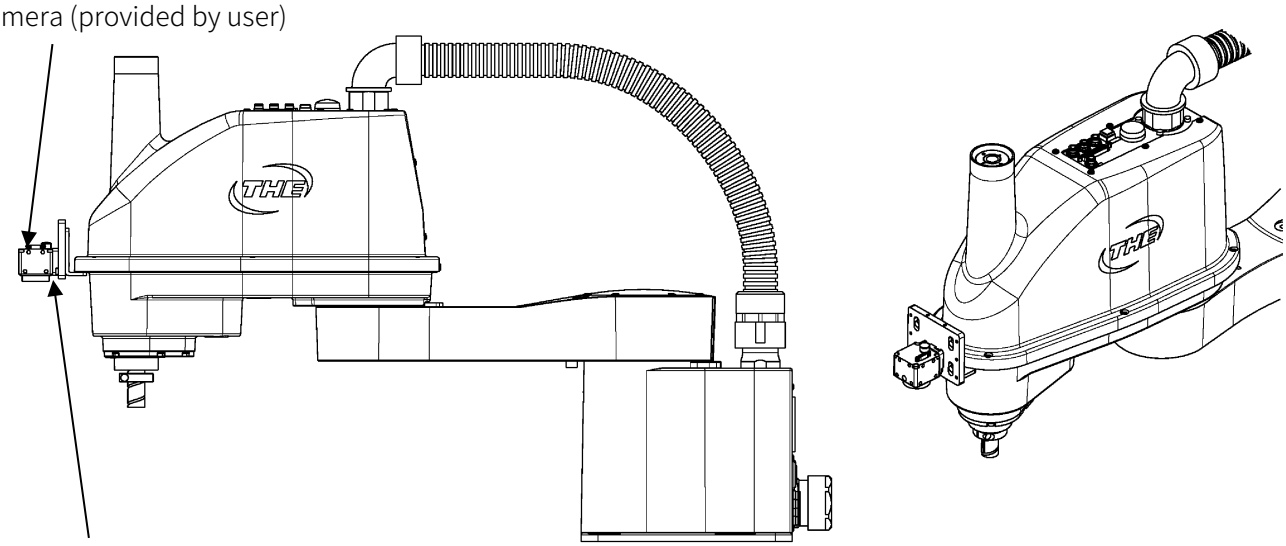


Fig. 7.2 Service tap dimensions

Camera (provided by user)



Camera mounting bracket (provided by user)

The camera mounting plate shown in this figure is available as an option.

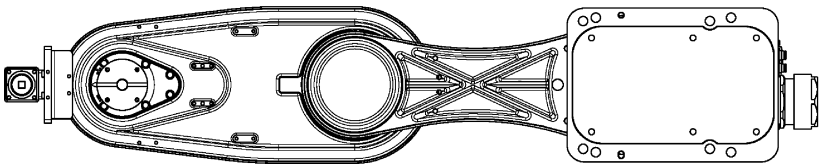


Fig. 7.3 Camera mounting example

NOTICE

- If the service tap is used to mount parts, be sure to always check whether the parts will interfere with the robot or peripheral equipment during operation. If such interference occurs, this could result in a breakdown of the robot or peripheral equipment.

Table 7.1 shows manufacturers of compatible camera mounting plates (option).

Table 7.1 Camera mounting plate (option)

Manufacturer	Product name	Example of camera manufacturer model	SHIBAURA Dwg. No.
Keyence	For camera mounting Bracket	CV-S035CH/CV-S035MH	L23754G01
		CV-S200CH/CV-S200MH	
		CA-HX200C/CA-HX200M	
		CA-HX500C/CA-HX500M	
OMRON		FH-SC/FH-SM	L23754G02
		FH-SC02/FH-SM02	
		FH-SC04/FH-SM04	
		FH-SM05R/FH-SC05R	
Panasonic		ANPVC2040	L23754G03
		ANPVC1040	
		ANPVC5030	
		ANPVC1470	
Basler		GigE Vision air sensor Ace series	L23754G04

7.3 Hand Air Piping

Four hand piping lines (Φ6 × 4) are available. The lines are connected on the base rear and the 2nd arm top surface.

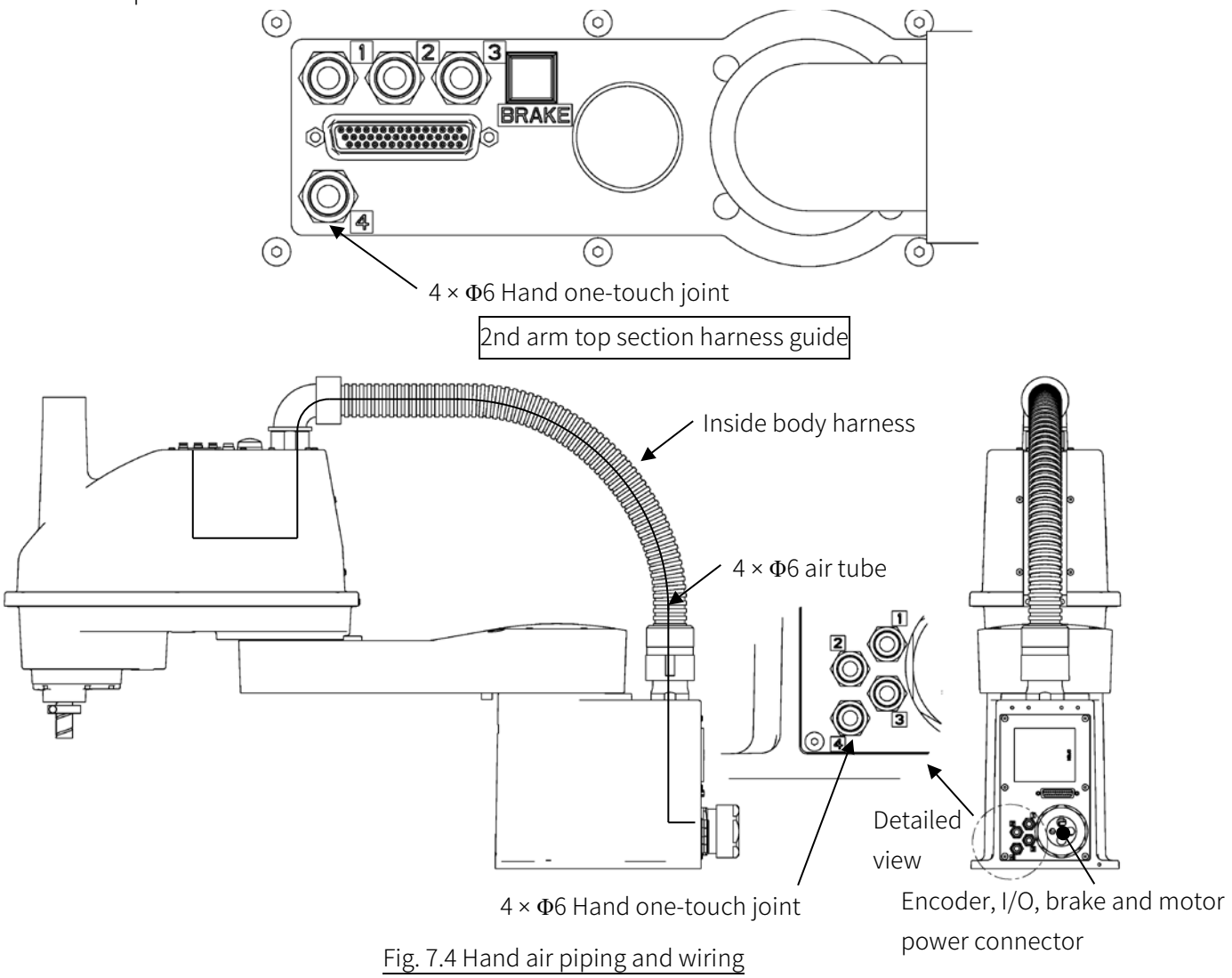


Fig. 7.4 Hand air piping and wiring

MEMO

Numbers are assigned to the joints mounted on the robot.

When mounting the hand air piping, it is recommended that piping numbers be affixed to the mating air tubes to prevent piping mounting mistakes when piping is removed during maintenance work.

NOTICE

- Do not route piping together with the body harness.
  Otherwise, excessive force could be applied on the body harness, resulting in breakage of the internal cable.
- The solenoid valve air tubes are not supplied with the robot.
  They should be provided by the customer.

7.4 Allowable Load Requirements and Program Settings

This section describes the allowable load requirements of the robot and how to set up the program based on the load.

7.4.1 Allowable Load Requirements

The robot load conditions are defined by the hand mass, moment of inertia and offset value of hand gravity center from the hand shaft center as shown in Fig. 7.5.
The allowable load conditions are shown in Table 7.2.

Table 7.2 Allowable load requirements

Requirement	Allowable value
Weight	Max. 12 kg (rated 2 kg)
Load moment of inertia	Up to 0.25 kg·m^2
Load center of gravity offset value	Max. 100 mm

NOTICE

- Never use the robot under conditions outside the specification values.
  Otherwise, performance and lifespan could be reduced.
- Be sure to use the PAYLOAD command.
  Otherwise, performance and lifespan could be reduced.
  - Even if the load center of gravity offset value is 100 mm or less, on hands with a long projection length from the 4th axis rotation center, micro-vibrations at the axis center can be amplified to cause large vibrations on the hand and workpiece tip.
- Even if the weight is within the maximum payload weight, when a heavy object is held and the 3rd axis is lowered, vibrations could occur for certain robot orientations. In these cases, use the SPEED command, ACCEL/DECEL command, LOCUS command, and similar commands to adjust the operating speed and acceleration/deceleration so that vibration does not occur.
  For the SPEED command, ACCEL/DECEL command, and LOCUS command, refer to “3 Explanation of Command Words” in the separate instruction manual “Robot Language Manual.”

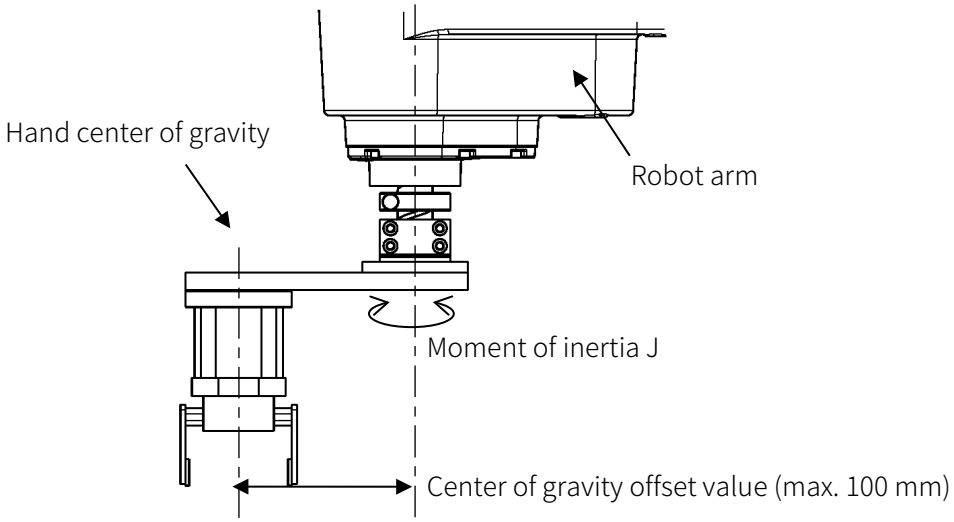


Fig. 7.5 Robot tool

7.4.2 Load Requirements and Program Settings

This robot has a function for automatically changing the maximum speed, acceleration/deceleration, and servo gain by using the PAYLOAD command in the program based on the load conditions.

Be sure to always use the PAYLOAD command.

The specific method for using this function is explained below.

PAYLOAD command format

The PAYLOAD command format is written as shown below for a tool weight of M kg and a center of gravity offset of L mm.

```
PAYLOAD= {M, L}
    M: Load weight (kg)
    L: Center of gravity offset (mm)
```

The PAYLOAD command has the following functions.

The maximum speed and acceleration/deceleration of each robot axis are automatically changed based on the load conditions that were set.

The servo gain of each robot joint axis is automatically changed based on the load conditions that were set.

Program examples

Examples of basic programs using the PAYLOAD command are shown below.

For further information, refer to “3 Explanation of Command Words” in the separate instruction manual “STEA1227 Robot Language Manual.”

Program example 1:

The robot is moved under the load conditions of 5 kg load weight and 100 mm center of gravity offset.

```
PROGRAM SAMPLE
    SPEED= 100
    PAYLOAD=  {5,100}
    MOVE P1
    MOVE P2
    STOP
END
```

Program example 2:

Suppose that the hand has a weight of 3 kg and a center of gravity offset of 30 mm, and when a workpiece is held, the hand has a weight of 5 kg and a center of gravity offset of 50 mm.

Pick-and-place operation is executed under the above conditions.

```
PROGRAM SAMPLE
   PAYLOAD=  {3,30}
   ACCUR=FINE
   ENABLE NOWAIT
   RESET DOUT
   MOVE P0
   DOUT(1)
   WAIT DIN(1)
LOOP:
   MOVE P1+POINT(0,0,100)
   IF DIN(-1)THEN GOTO FIN
   MOVE P1
   WAIT MOTION>=100
   DOUT(213)
   DELAY 1
   PAYLOAD=  {5,50}
   MOVE P1+POINT(0,0,100)
   MOVE P2+POINT(0,0,100)
   MOVE P2
   WAIT MOTION>=100
   DOUT(-213)
   DELAY 1
   PAYLOAD=  {3,30}
   MOVE P2+POINT(0,0,100)
   GOTO LOOP
FIN:
   MOVE P0
   DOUT(1)
   STOP
END
```

Program example 3:

Vibration occurs when moving from P1 to P2 under the conditions of program example 1, and the acceleration is adjusted using the ACCEL/DECEL command to reduce the vibration.

```
PROGRAM SAMPLE
   SPEED= 100
   PAYLOAD=  {5,100}
   MOVE P1
   MOVE P2 WITH ACCEL = 80 WITH DECEL = 80
   STOP
END
```

Program example 4:
Vibration occurs during linear motion from P1 to P2, and the LOCUS command is used to adjust so that vibration is reduced.

```
PROGRAM SAMPLE
   SPEED= 100
   PAYLOAD=  {5,100}
   MOVES P1
   MOVES P2 WITH LOCUS = 0
   STOP
END
```

Setting the PAYLOAD command

In the default state, that is, when the PAYLOAD command is not used, the maximum speed and acceleration/deceleration are set to 100% and the servo gain is set to the value at a minimum load. See "7.4.3 Setting the Maximum Speed and Acceleration/Deceleration for the Load Conditions."

NOTICE

- Be sure to always use the PAYLOAD command.
  Otherwise, vibration or overrun could reduce the lifespan of the robot or damage the robot.
  - The moment of load inertia should be used within the tolerance listed in Table 7.2.
  - Even if there is no load center of gravity offset, when the moment of inertia is large, the robot could vibrate. In these cases, calculate the virtual center of gravity offset L (mm) from the moment of inertia (kg·m^2) and weight M (kg).
    $L=\sqrt{(J \times 10^6/M)}$
    Enter this calculation result.
    PAYLOAD = {M,L}
  - Micro-vibrations could occur depending on the robot orientation.
  - If the vibrations or overrun does not subside, use together with the SPEED command or DECEL command.
- Even if the weight is within the maximum payload weight, vibrations could occur depending on the robot orientation, such as when a heavy object is held and the 3rd axis is lowered. In these cases, use the SPEED command, ACCEL/DECEL command, LOCUS command, or similar commands to adjust the operating speed and acceleration/deceleration so that vibration does not occur.
  For details on the SPEED command, ACCEL/DECEL command, and LOCUS command, refer to "3 Explanation of Command Words" in the separate instruction manual "STEA1227 Robot Language Manual."
- If the robot vibrates when using guidance, be sure to set the PAYLOAD in manual mode by referring to "3.1.7 Setting load for manual guide" in the separate instruction manual "STEA1162 TP5000 Operation Manual."
  Otherwise, vibration could reduce the performance and lifespan of the robot.
  - If the load weight or center of gravity offset is large, the robot could vibrate.

Weight of load mounted on the arm

If a camera, air valve, or other device is mounted on the arm, the weight of the load is converted to the equivalent weight of the shaft and is added to the load weight for setting the PAYLOAD command.

Equivalent weight formulas

If load is mounted on the tip from the axis-3 center: $W_M = M(A_{M2})^2/(A_2)^2$

If load is mounted between axis-1 center and axis-3 center: $W_M = M(A_{M1})^2/(A_1+A_2)^2$

W_M : Equivalent weight

M: Camera weight

A_1 : Arm-1 length

A_2 : Arm-2 length

A_{M1} : Distance from axis-1 rotation center to gravity center of camera

A_{M2} : Distance from axis-2 rotation center to gravity center of camera

Examples are shown in Fig. 7.6 and Fig. 7.7.

Example: When a camera of 1 kg is mounted on the 2nd arm tip and a hand of 2 kg is mounted on the shaft tip

Equivalent mass W_M of the camera mounted on the arm 2 tip is

$W_M = M (A_{M2})^2/(A_2)^2$

$=1 \times 345^2/275^2 = 1.57$

Thus, the total load weight

$W + W_M = 2 + 1.57 = 3.57$

is rounded up to become **3.6 kg**.

Set "**3.6**" in load mass of the PAYLOAD command.

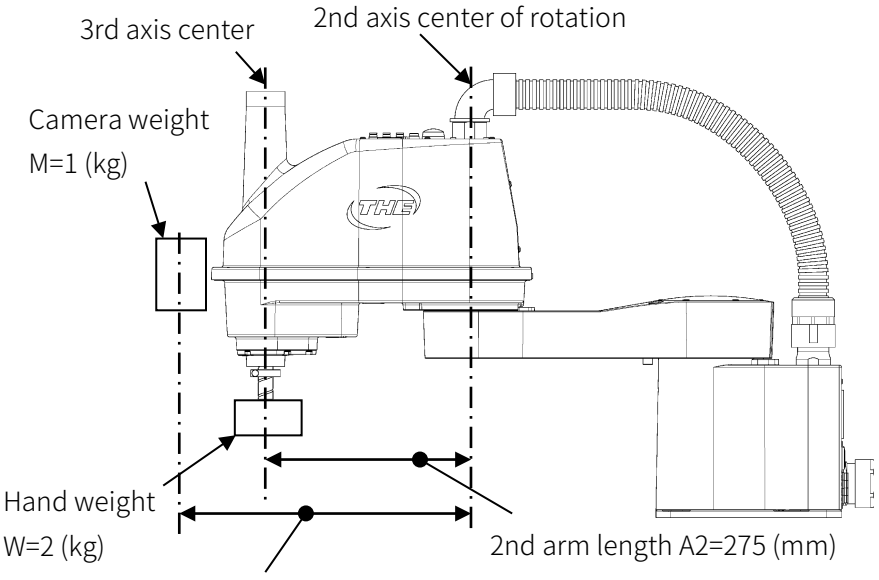


Fig. 7.6 Mounting on the tip from 3rd axis center

Example: When solenoid valve of 1 kg is mounted in the middle of 2nd arm and hand of 2 kg is mounted on the shaft tip

The equivalent weight W_M of a solenoid valve mounted between the 1st axis center of rotation and the 3rd axis center is found using

$W_M = M (A_{M1})^2/(A_1 + A_2)^2$
$= 1 \times 400^2/(325 + 275)^2 = 0.44$
Thus, the total load weight
$W + W_M = 2 + 0.44 = 2.44$
is rounded up to become **2.5 kg**.

Set "**2.5**" in load mass of the PAYLOAD command.

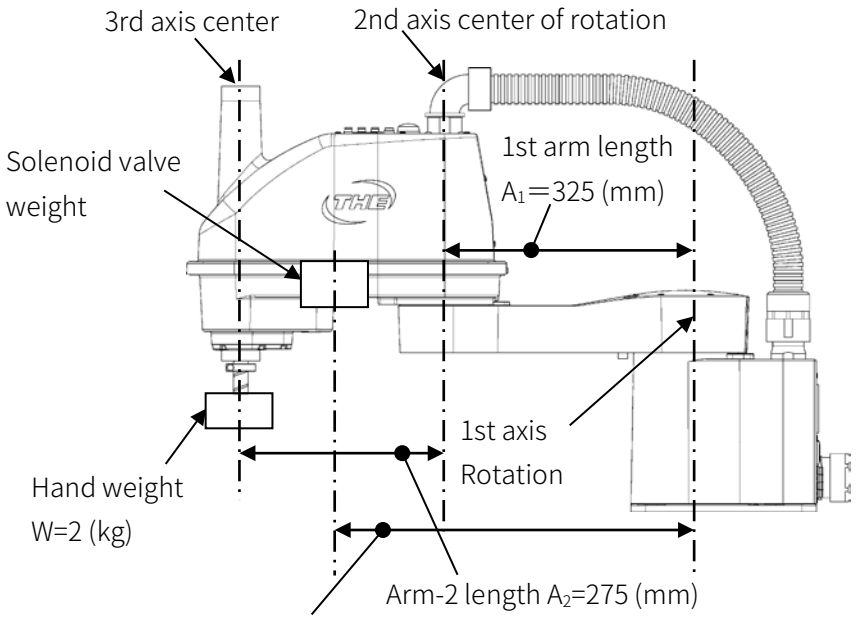
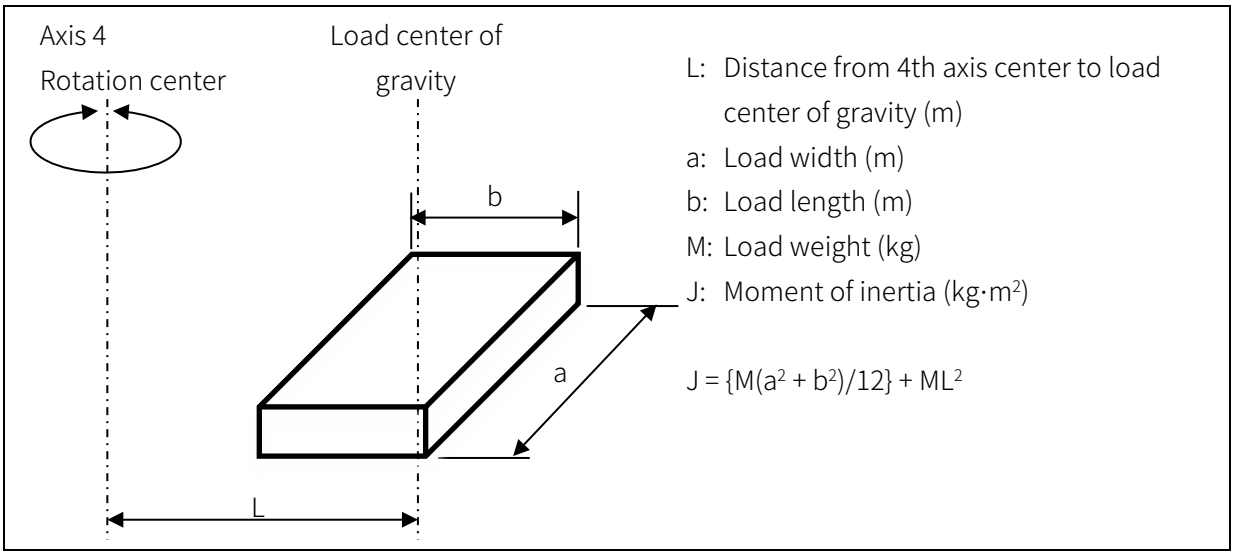


Fig. 7.7 Mounting between 1st axis center of rotation and 3rd axis center

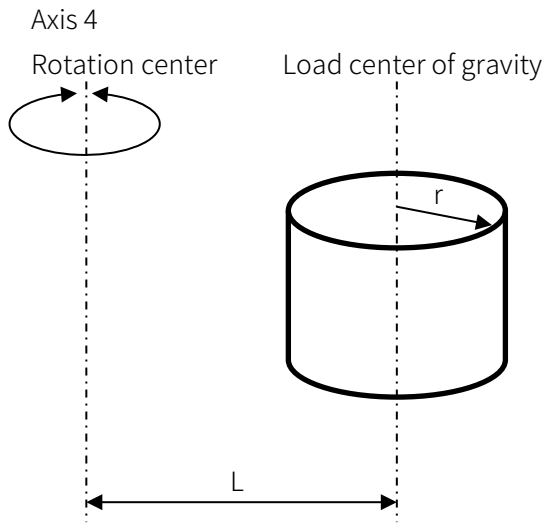
Calculating the moment of inertia

A simplified model of the robot and load is used to show an example of the formula for the load moment of inertia below.

1) When the load shape is rectangular



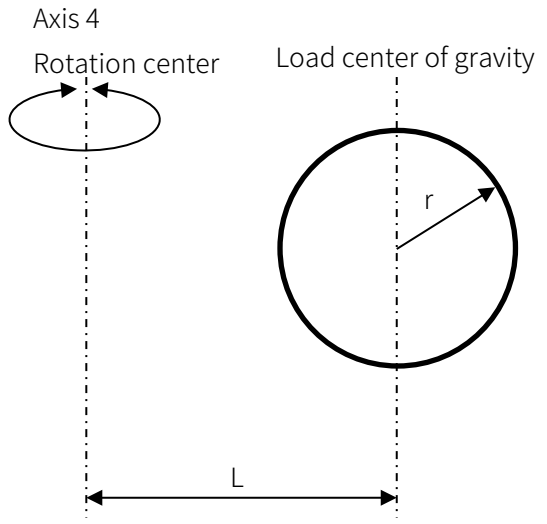
2) When the load shape is columnar



L: Distance from 4th axis center to load center of gravity (m)
r: Load radius (m)
M: Load weight (kg)
J: Moment of inertia (kg·m^2)

$J = Mr^2/2 + ML^2$

3) When the load shape is spherical



L: Distance from 4th axis center to load center of gravity (m)
r: Load radius (m)
M: Load weight (kg)
J: Moment of inertia (kg·m^2)

$J = 2Mr^2/5 + ML^2$

Moment of inertia calculation examples

Fig. 7.8 and Fig. 7.9 show calculation examples of the moment of inertia.
The moment of inertia of the entire load can be obtained by finding the total moment of inertia of the individual parts.

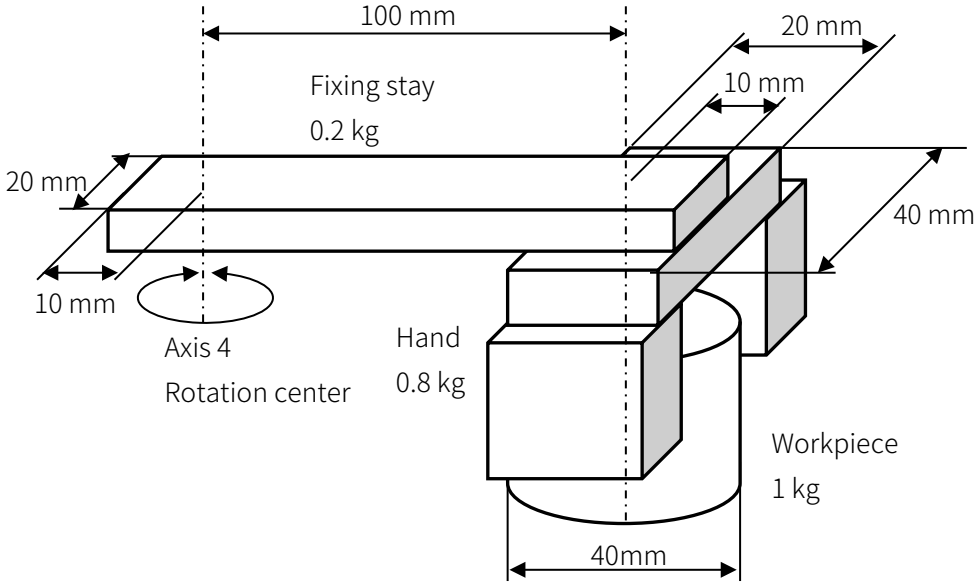
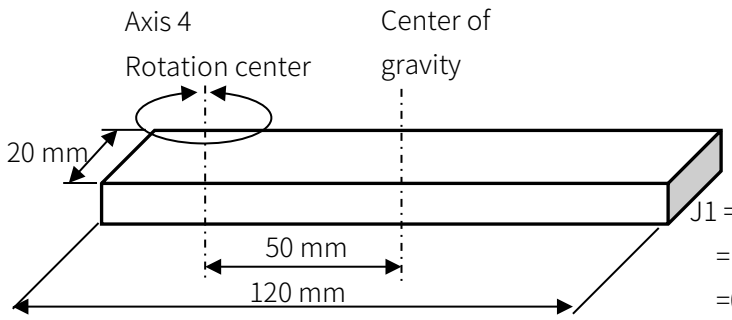


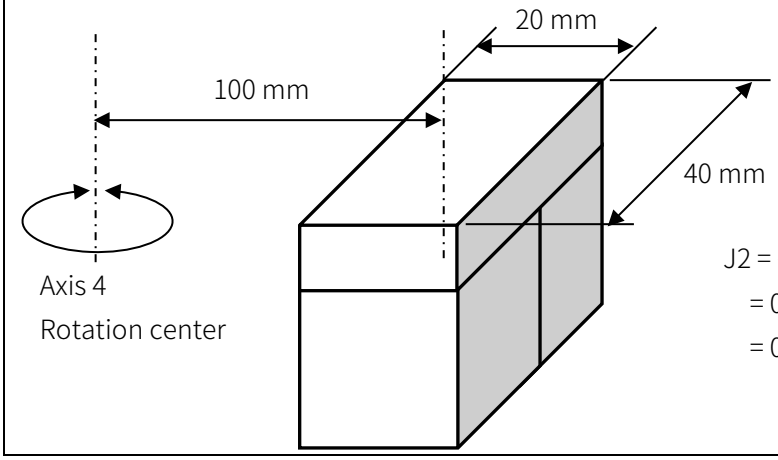
Fig. 7.8 Moment of inertia calculation example 1

1) Moment of inertia of fixing stay



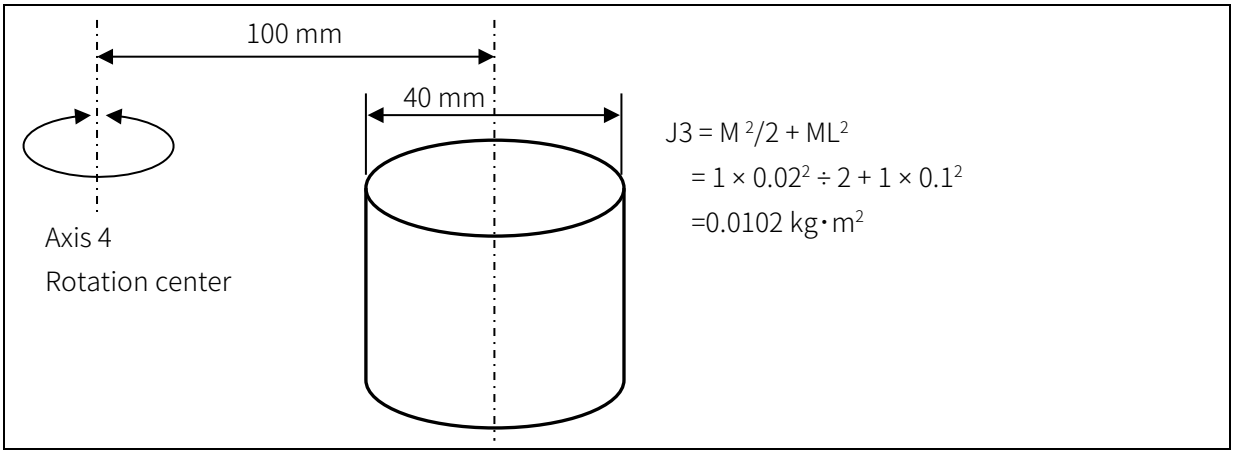
$J1 = \{M(a^2 + b^2)/12\} + ML^2$
$= 0.2 \times (0.02^2 + 0.12^2) \div 12 + 0.2 \times 0.05^2$
$=0.000746\ kg \cdot m^2$

2) Moment of inertia of hand



$J2 = \{M(a^2 + b^2)/12\} + ML^2$
$= 0.8 \times (0.04^2 + 0.02^2) \div 12 + 0.8 \times 0.1^2$
$= 0.00813\ kg \cdot m^2$

3) Moment of inertia of workpiece



The entire moment of inertia is the sum of J1+J2+J3 to result in 0.019076 kg·m².

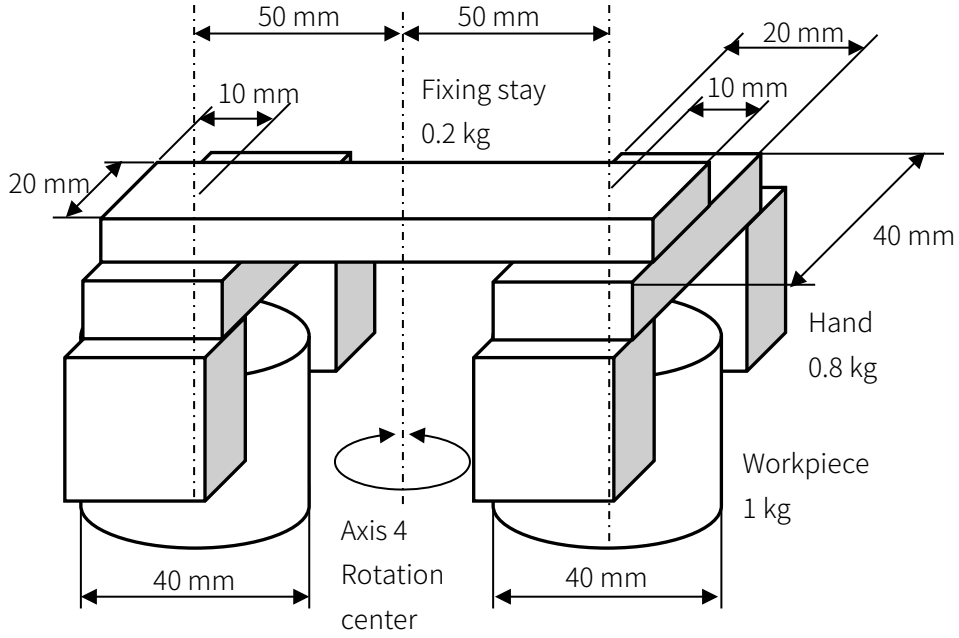
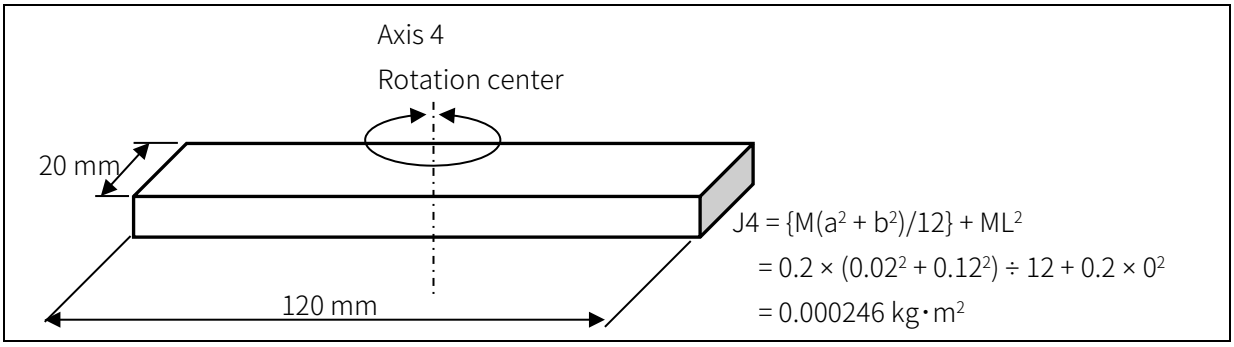
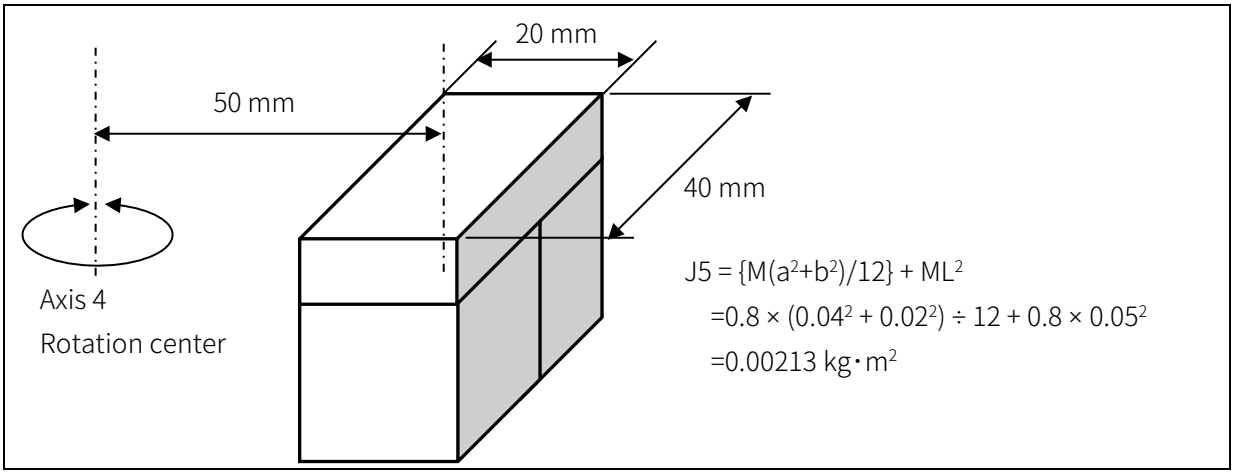


Fig. 7.9 Moment of inertia calculation example 2

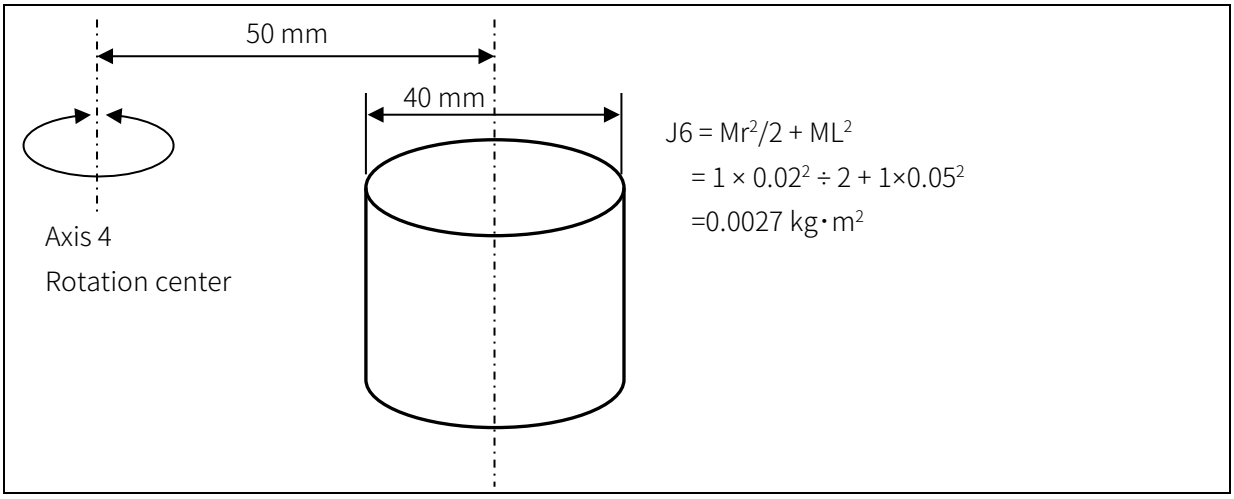
4) Moment of inertia of fixing stay



5) Moment of inertia of hand



6) Moment of inertia of workpiece



The entire moment of inertia is the sum of J4 + J5 × 2+J6 × 2 to result in 0.009906 kg·m².

To obtain virtual gravity center offset L from the moment of inertia, see "Setting the PAYLOAD command"

In moment of inertia calculation example 2, the center of gravity is at the 4th axis center of rotation. When the virtual center of gravity is obtained from the moment of inertia, the result is as follows.

$L=\sqrt{(J \times 10^6/M)}$

$=\sqrt{\ (0.009906 \times 10^6/3.8)}$

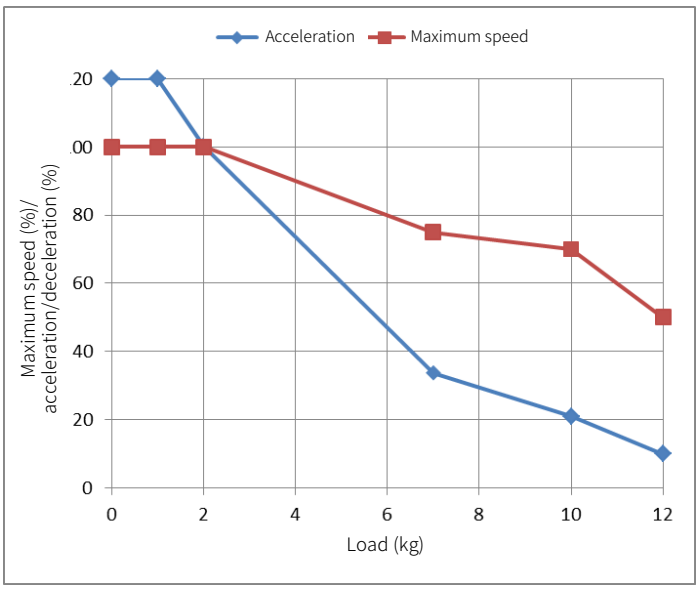
=51.057 (unit: mm)

The virtual gravity center offset is rounded up to become "51.1 mm."

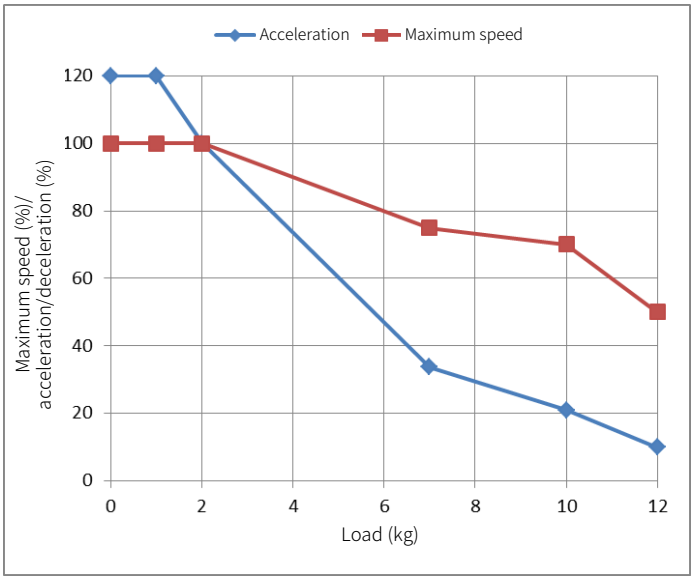
The PAYLOAD command is set as PAYLOAD={3.8，51.1}.

7.4.3 Setting the Maximum Speed and Acceleration/Deceleration for the Load Conditions

When the PAYLOAD command is used, the maximum speed and acceleration/deceleration of the robot are automatically changed based on the load conditions. The maximum speed and acceleration/deceleration change based on the load weight as shown in Fig. 7.10 to Fig. 7.11.

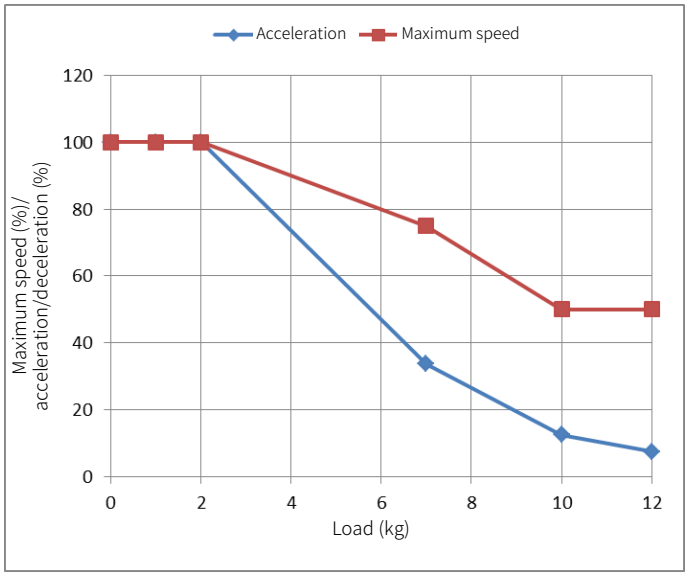


Setting the maximum speed and acceleration/deceleration for the load weight (1st axis)

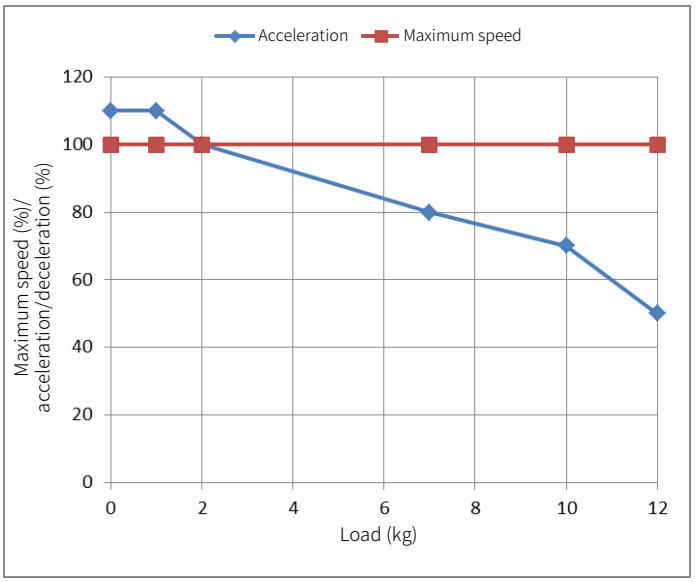


Setting the maximum speed and acceleration/deceleration for the load weight (2nd axis)

Fig. 7.10 Setting the maximum speed and acceleration/deceleration for the load weight (1st and 2nd axes)



Setting the maximum speed and acceleration/deceleration for the load weight (3rd axis)



Setting the maximum speed and acceleration/deceleration for the load weight (4th axis)

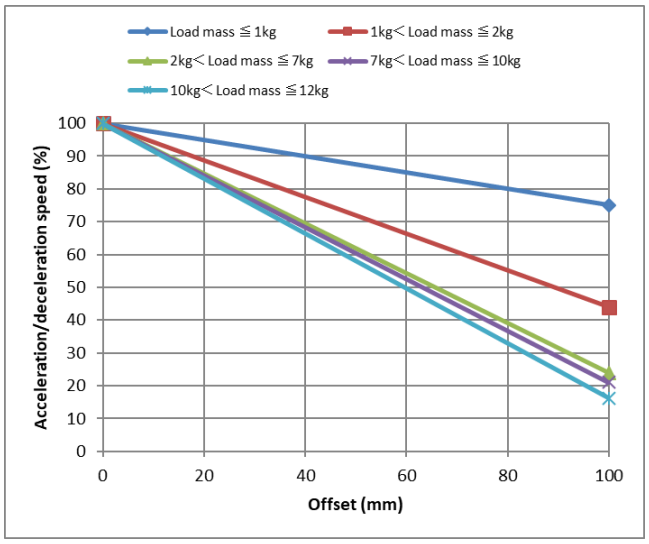
Fig. 7.11 Setting the maximum speed and acceleration/deceleration for the load weight (3rd and 4th axes)

To set the acceleration to 120% when the load is 1 kg or less, “ACCEL = 120” must be written in the program.
An example of a program for setting the acceleration to 120% is shown below.

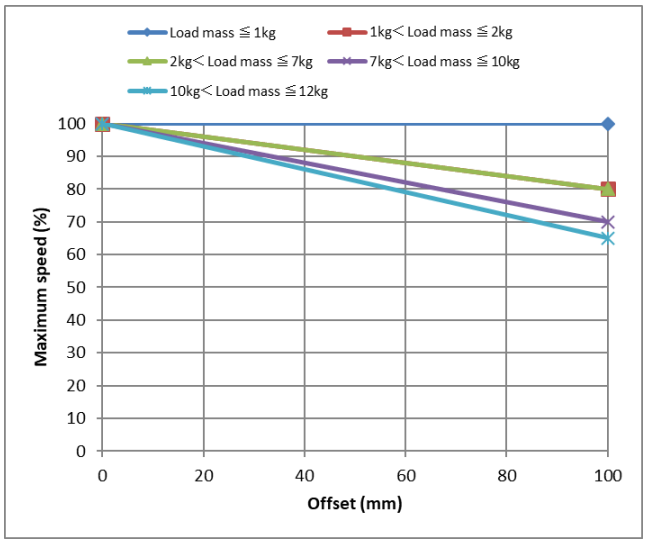
Program example for setting acceleration to 120%

```
PROGRAM SAMPLE
  SPEED= 100
  PAYLOAD=  {1,0}
  ACCEL=120
  MOVE P1
  MOVE P2
  STOP
END
```

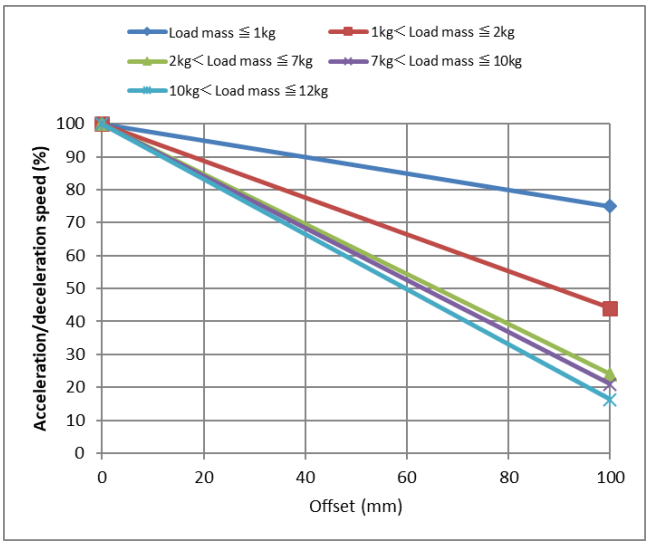
In addition, when there is a load center of gravity offset, the acceleration/deceleration and maximum speed are changed as shown in Fig. 7.12 and Fig. 7.13 .



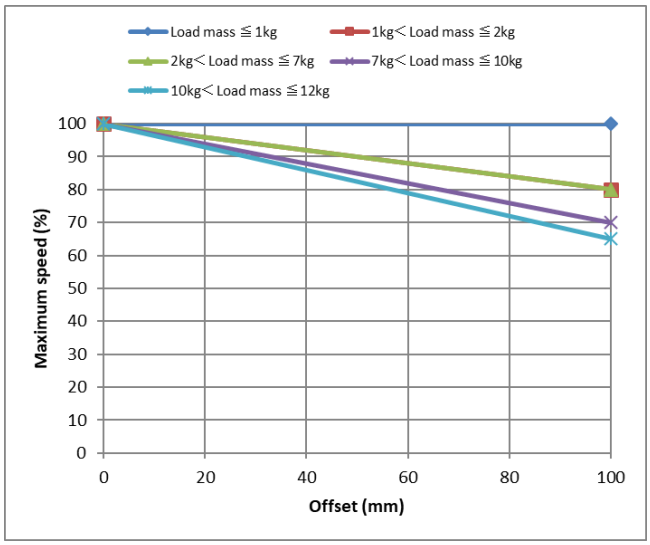
Setting of acceleration/deceleration in relation of offset (Axis 1)



Setting of maximum speed in relation of offset (Axis 1)

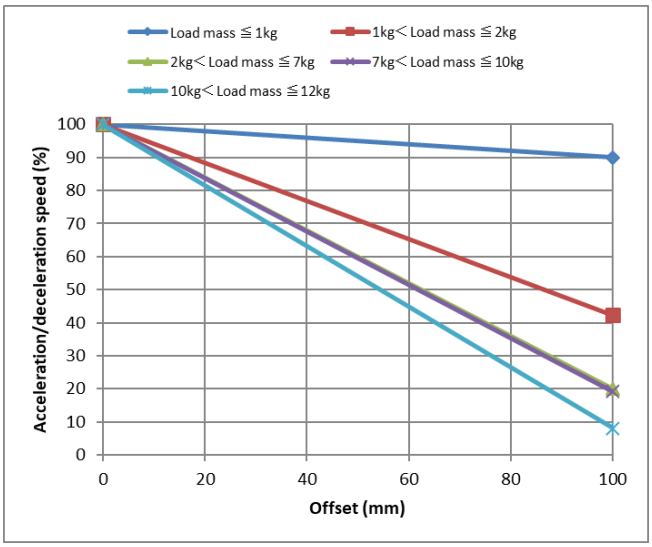


Setting of acceleration/deceleration in relation of offset (Axis 2)

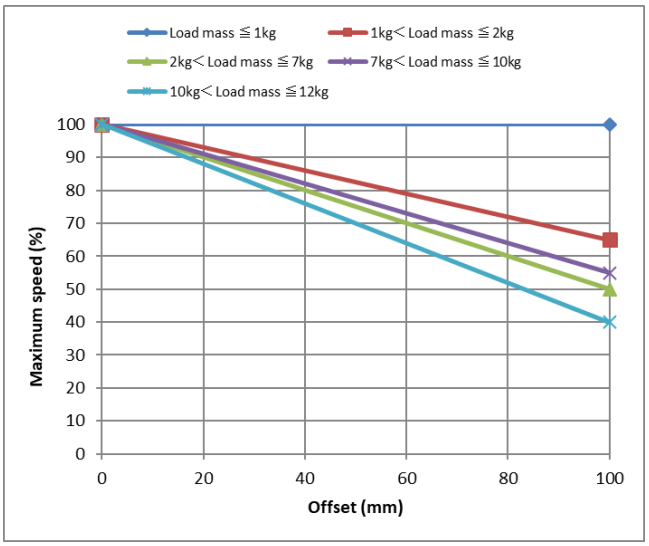


Setting of maximum speed in relation of offset (Axis 2)

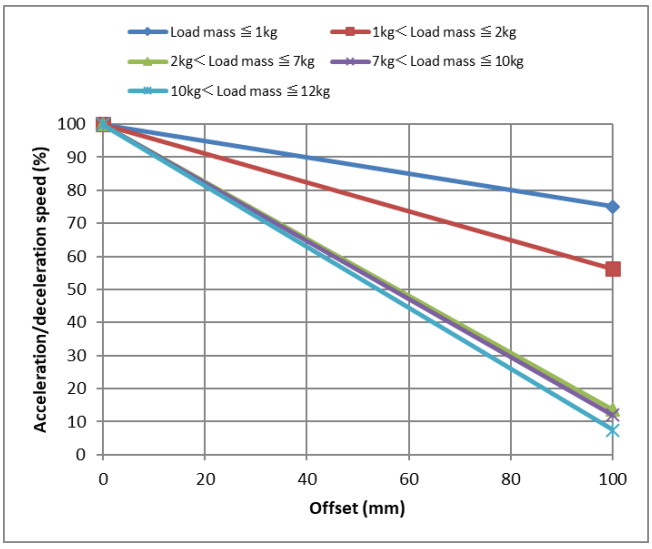
Fig. 7.12 Setting of acceleration/deceleration in relation to gravity center offset (Axes 1 and 2)



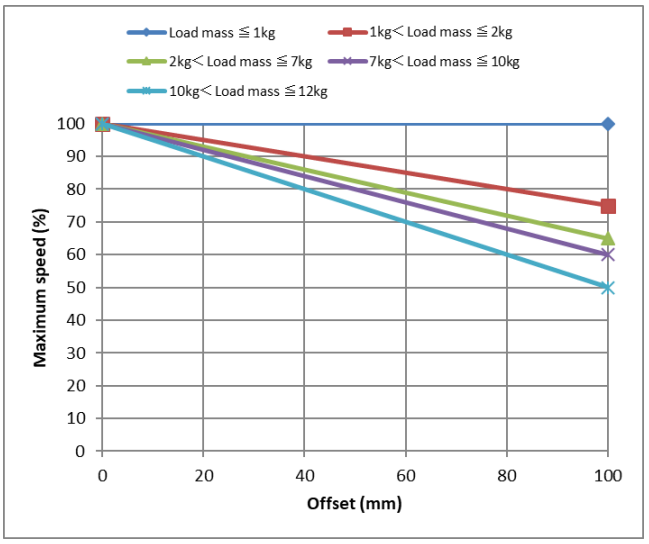
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)



Setting of acceleration/deceleration in relation to offset (Axis 4)



Setting of maximum speed in relation to offset (Axis 4)

Fig. 7.13 Setting of acceleration/deceleration in relation to gravity center offset (Axes 3 and 4)

8. User Wiring

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO

- *Before starting work, check 10.1 Rescuing a Person Trapped by a Robot, and conduct a rescue drill.*
- *The 1st, 2nd, and 4th axes of THE600-E can be moved by turning off the drive power (servo off) and pushing them by hand. By releasing the brake while the 3rd axis is powered on, it can be pushed and moved by hand without turning on the drive power supply (turning on the servo). For details, see "9. Teaching."*
- *We are preparing various methods for guiding the robot, which is done by turning on a drive power supply (turning on the servo). For details, refer to the following the separate manuals.*
  - TP5000 Operation Manual
  - TCmini (Internal PLC) Manual
  - Communication Manual
  - TSAssist User's Manual

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

8.1 Hand Wiring

The hand wiring is controlled from the controller or from self-standing PLC or other device (provided by the customer).

⚠ DANGER

- When fabricating the hand wiring cable, be sure to use the compatible wire.
  Otherwise, the cable could overheat or short circuit between the terminals, resulting in a fire.
- Never connect a power supply with a voltage exceeding the specified range to the user wiring connector.
  This could cause the cable to overheat, resulting in a fire.
  - Before connecting, use the tester or similar device to check the voltage of the power supply.

NOTICE

- Never mistake the terminal arrangement for the hand wiring.
  Otherwise, the connected parts could fail.
  - Before usage, use the tester or similar device to check the terminal arrangement.
- When fabricating the hand wiring cable, be sure to use the compatible wire.
  Otherwise, overheat or short circuit could occur, resulting in burning of the cable.
- Be sure to use a highly flexible and twisting-resistant robot cable for the hand wiring cable.
  If a general-purpose cable is used, the wire may break during robot operation.
  - Secure the cable to the arm using a saddle or other mechanism to prevent the cable from being swung around.
- Never connect the hand wiring cable to a power supply whose voltage exceeds the specified range.
  Otherwise, the cable could overheat, resulting in a fire.
  - Before connecting, use the tester or similar device to check the voltage of the power supply.
- Be sure to perform hand wiring and piping while checking the following status during robot operation.
  Otherwise, the robot and hand will become faulty early.
  - Check whether a load is being applied to the hand connector.
  - Check whether the hand wiring and piping are contacting the robot or peripheral equipment and are worn.

8.1.1 Hand Control Wiring

Eight input signals, eight output signals, and 24 VDC/PGND are provided for hand wiring and are connected to the controller. Fig. 8.7 and Fig. 8.8 are the wiring diagrams. The wiring is connected to the connector on the 2nd arm top surface.
The user should provide a connector to connect the cable.

Connector (standard) model:	Hood:	09670250494 (manufacturer: HARTING)
	Connector:	09563005601 (manufacturer: HARTING)
	Terminal:	09560008177 (manufacturer: HARTING)

Cross-sectional area of compatible wire: AWG 24 to 26 (plain braided wire and twisted wire)

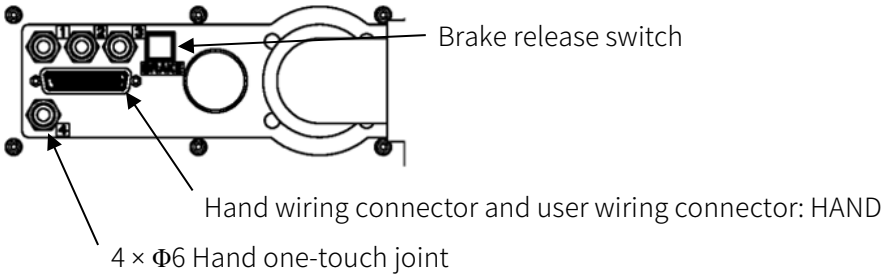
*Connecting the crimp terminal and wire requires a special crimping tool.
Tool model:Crimp tool: 09990000596 (Manufacturer: HARTING)

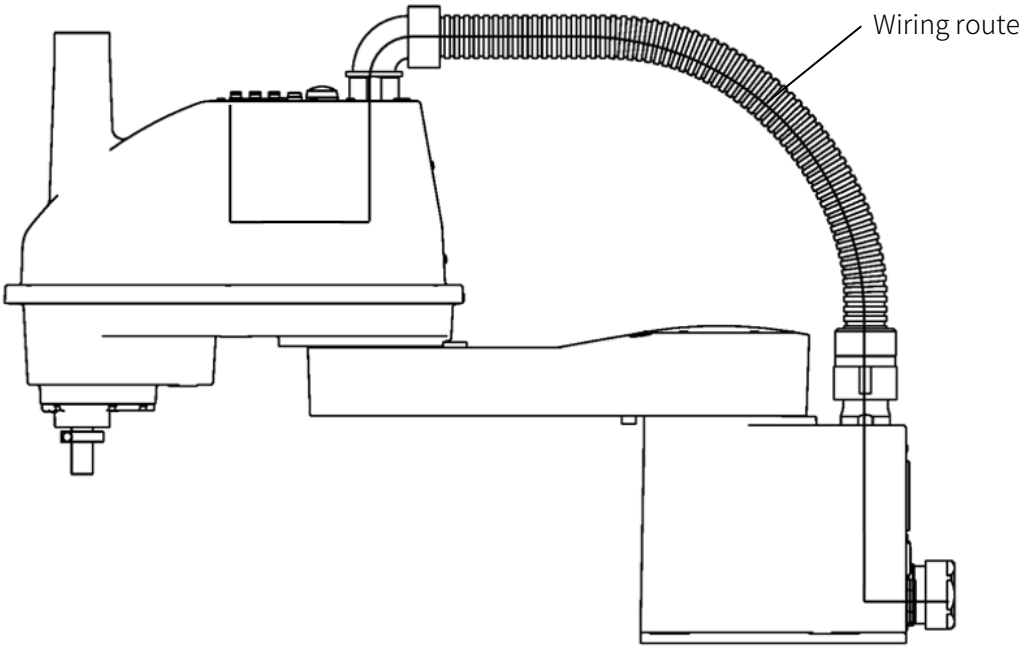
MEMO

A hand I/O cable is also available as an option. For details on the optional hand I/O cable, see "8.2 Hand I/O Cable (Option)."

A signal line from the controller passes inside the robot and is connected to the hand wiring connector “HAND.”

Connect the fabricated cable connector to “HAND.”





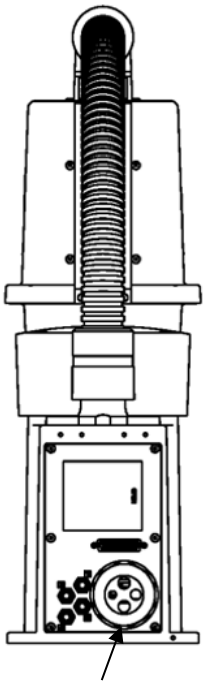


Fig. 8.1 Hand wiring

8.1.2 User Wiring

Sixteen (16) signals are provided for user wiring that uses a self-standing PLC or other device to control a motorized gripper.
The wiring is connected to the connector on the 2nd arm top surface and on the base rear surface.
The user should provide a connector to connect the cable.

Connector (standard) type:	Hood:	XM2S-2511 (manufacturer: OMRON)
	Connector:	XM3A-2521 (manufacturer: OMRON)

Cross sectional area of compatible wire: AWG 24 to 26 (plain braided wire and twisted wire)
*Connect the connector and wire by soldering.

For the 2nd arm-side connector, see the connector in "8.1.1 Hand Control Wiring."

The signal line from the base rear connector "USER" passes inside the robot and is connected to the hand wiring connector "HAND."
Connect the fabricated cable connector to "USER" and "HAND."

> *MEMO*
> *A hand I/O cable that is connected to the connector "HAND" is also available as an option. For details on the optional hand I/O cable, see "8.2 Hand I/O Cable (Option)."*

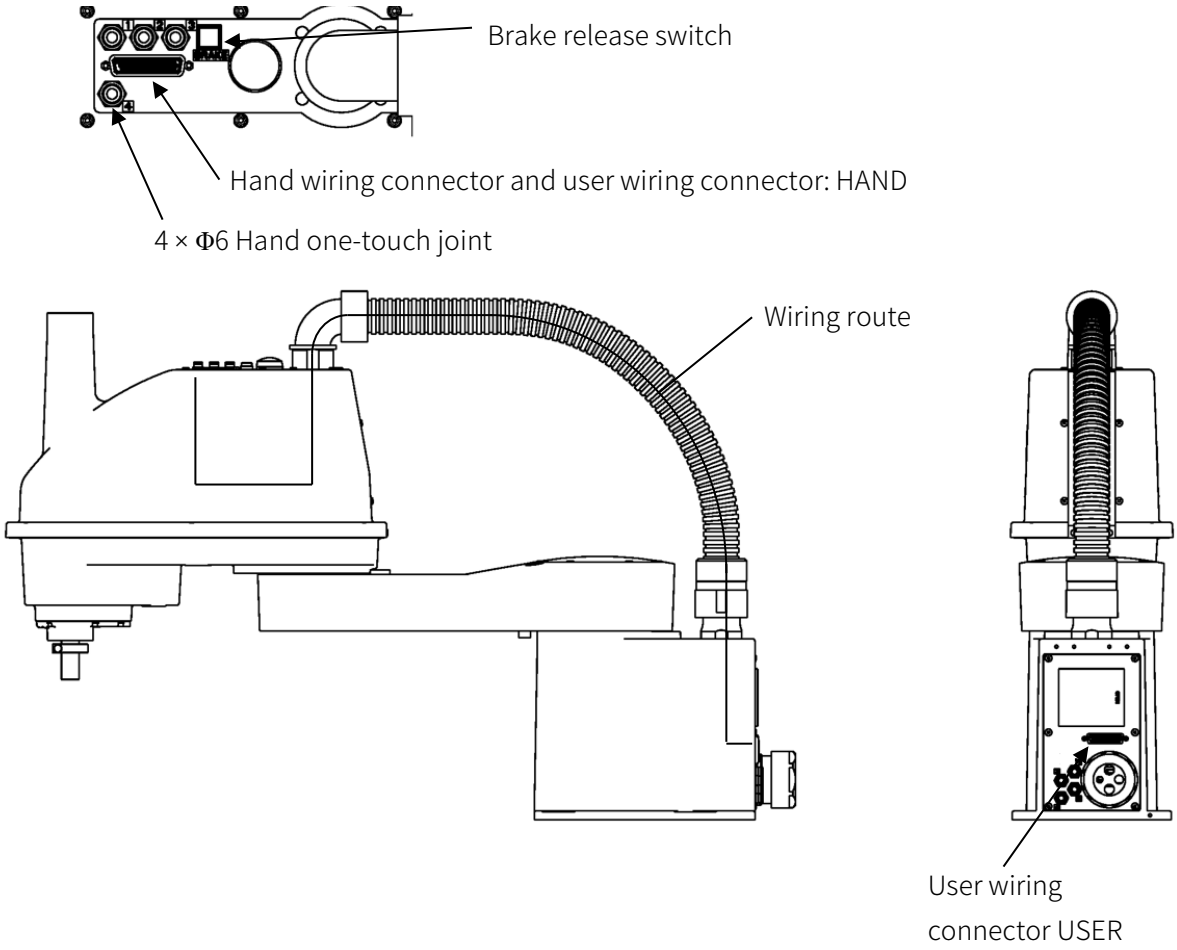


Fig. 8.2 Wiring to PLC

User wiring is routed as shown in Fig. 8.3.

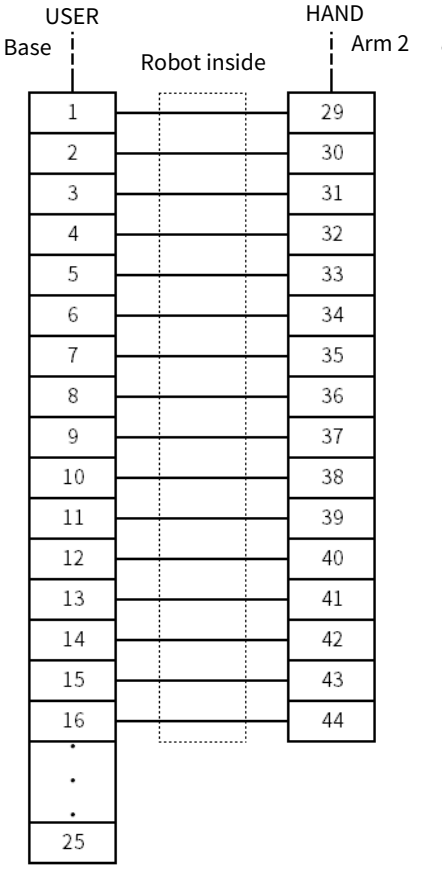
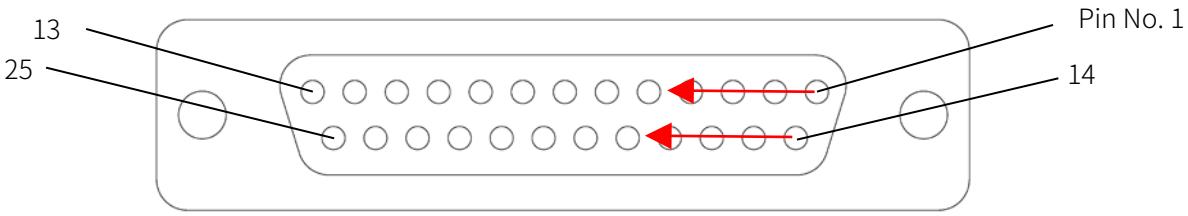


Fig. 8.3 User wiring diagram



Pin arrangement as viewed from the connection port

Fig. 8.4 User connector pin numbers

> *MEMO*
>
> *For the pin arrangement on the HAND connector side, see Fig. 8.5.*

8.1.3 Specifications for Hand Connector HAND

Table 8.1 Input/output signal connector HAND (TS5000-EMS, Type-N)

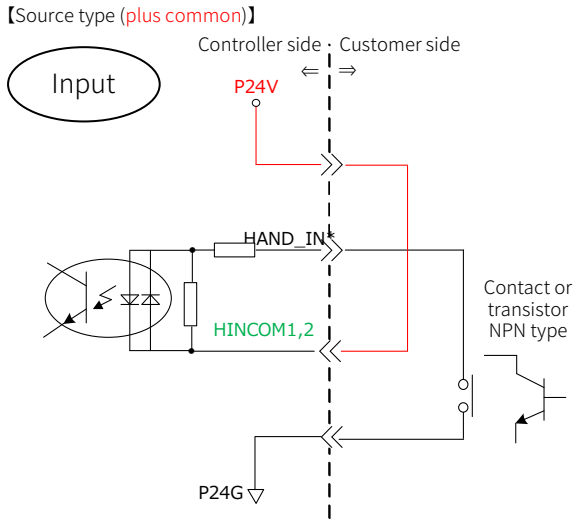
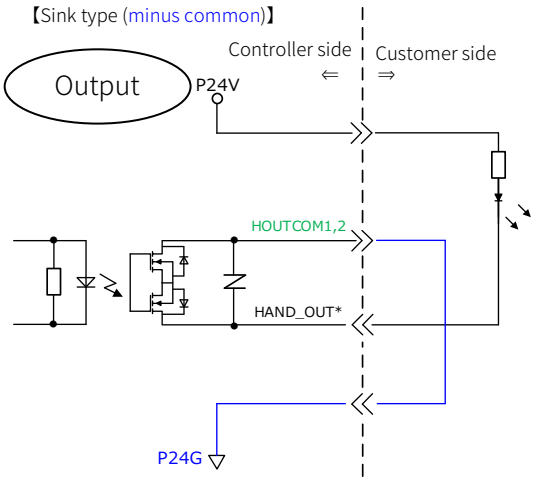
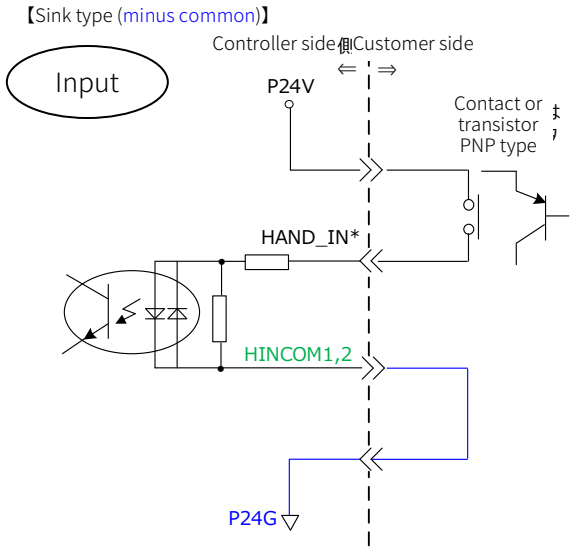
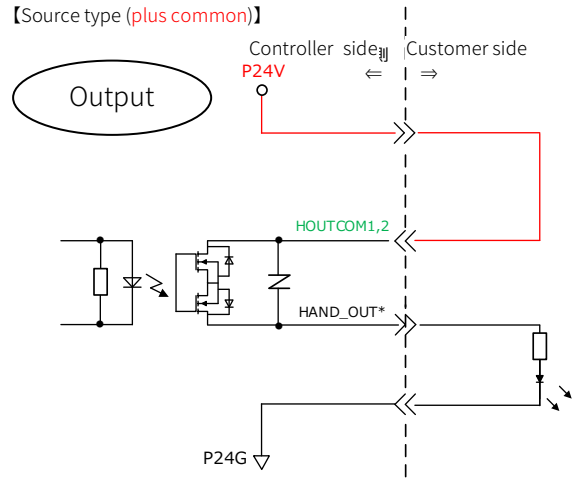
Pin	Signal name		Signal No.	Input/output signal circuit and connection example
1	HAND-IN0	Input signal 1	201	
2	HAND-IN1	Input signal 2	202	
3	HAND-IN2	Input signal 3	203	
4	HAND-IN3	Input signal 4	204	
5	HAND-IN4	Input signal 5	205	
6	HAND-IN5	Input signal 6	206	
7	HAND-IN6	Input signal 7	207	
8	HAND-IN7	Input signal 8	208	
9	P24V			
10				
11	HINCOM1			
12	HINCOM2			
13	P24VGND			
14				
15	HAND-OUT0	Output signal 1	201	
16	HAND-OUT1	Output signal 2	202	
17	HAND-OUT2	Output signal 3	203	
18	HAND-OUT3	Output signal 4	204	
19	HAND-OUT4	Output signal 5	205	
20	HAND-OUT5	Output signal 6	206	
21	HAND-OUT6	Output signal 7	207	
22	HAND-OUT7	Output signal 8	208	
23	P24V			
24				
25	HOUTCOM1			
26	HOUTCOM2			
27	P24VGND			
28				
29 to 44	User wiring			

Table 8.2 Input/output signal connector HAND (TS5000-EMS, Type-P)

Pin	Signal name		Signal No.	Input/output signal circuit and connection example
1	HAND-IN0	Input signal 1	201	
2	HAND-IN1	Input signal 2	202	
3	HAND-IN2	Input signal 3	203	
4	HAND-IN3	Input signal 4	204	
5	HAND-IN4	Input signal 5	205	
6	HAND-IN5	Input signal 6	206	
7	HAND-IN6	Input signal 7	207	
8	HAND-IN7	Input signal 8	208	
9	P24V			
10				
11	HINCOM1			
12	HINCOM2			
13	P24VGND			
14				
15	HAND-OUT0	Output signal 1	201	
16	HAND-OUT1	Output signal 2	202	
17	HAND-OUT2	Output signal 3	203	
18	HAND-OUT3	Output signal 4	204	
19	HAND-OUT4	Output signal 5	205	
20	HAND-OUT5	Output signal 6	206	
21	HAND-OUT6	Output signal 7	207	
22	HAND-OUT7	Output signal 8	208	
23	P24V			
24				
25	HOUTCOM1			
26	HOUTCOM2			
27	P24VGND			
28				
29 to 44	User wiring			

Input signal

No-voltage contacts or transistor open collector inputs are used for input signals.

No-voltage contact specifications:

Contact rating:	24 VDC, 10 mA or more (circuit current: approx. 7 mA)
Minimum contact current:	24 VDC, 1 mA
Contact impedance:	100Ω or less

Transistor specifications:

Withstand voltage between collector and emitter:	30 V or more
Current between collector and emitter:	10 mA or more (circuit current: approx. 7 mA)
Leak current between collector and emitter:	100 μA or less

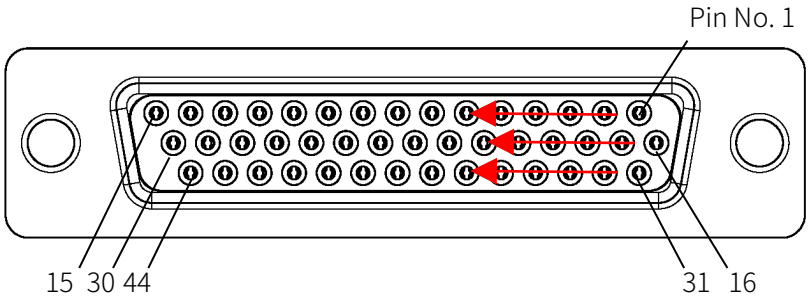
Output signal

The P24V of the controller can be used to drive a relay, solenoid valve, or other device.

To use an external power supply, connect GND of the external power supply with the GND (P24G) of the controller (GND common connection).

Transistor specifications:	Rated voltage:	24 VDC
	Rated current:	100 mA

- Limit the P24V supplied from the robot controller to a maximum of 3 A total.
- To use an external power supply, limit the total current value to a maximum of 3 A total.
- To connect a relay, solenoid valve, or other device, use a surge killer or diode to absorb the surge voltage.



Pin arrangement as viewed from the connection port

Fig. 8.5 Hand connector pin arrangement

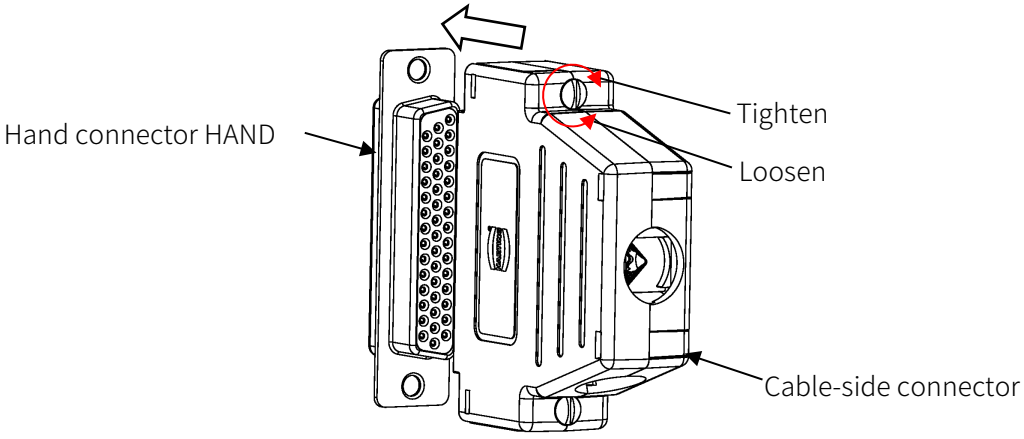


Fig. 8.6 Connecting/disconnecting hand input/output signal connector

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and could collide with a worker.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or “RESET DOUT” command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- Be sure to lock the hand connector, and be sure to secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.
- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

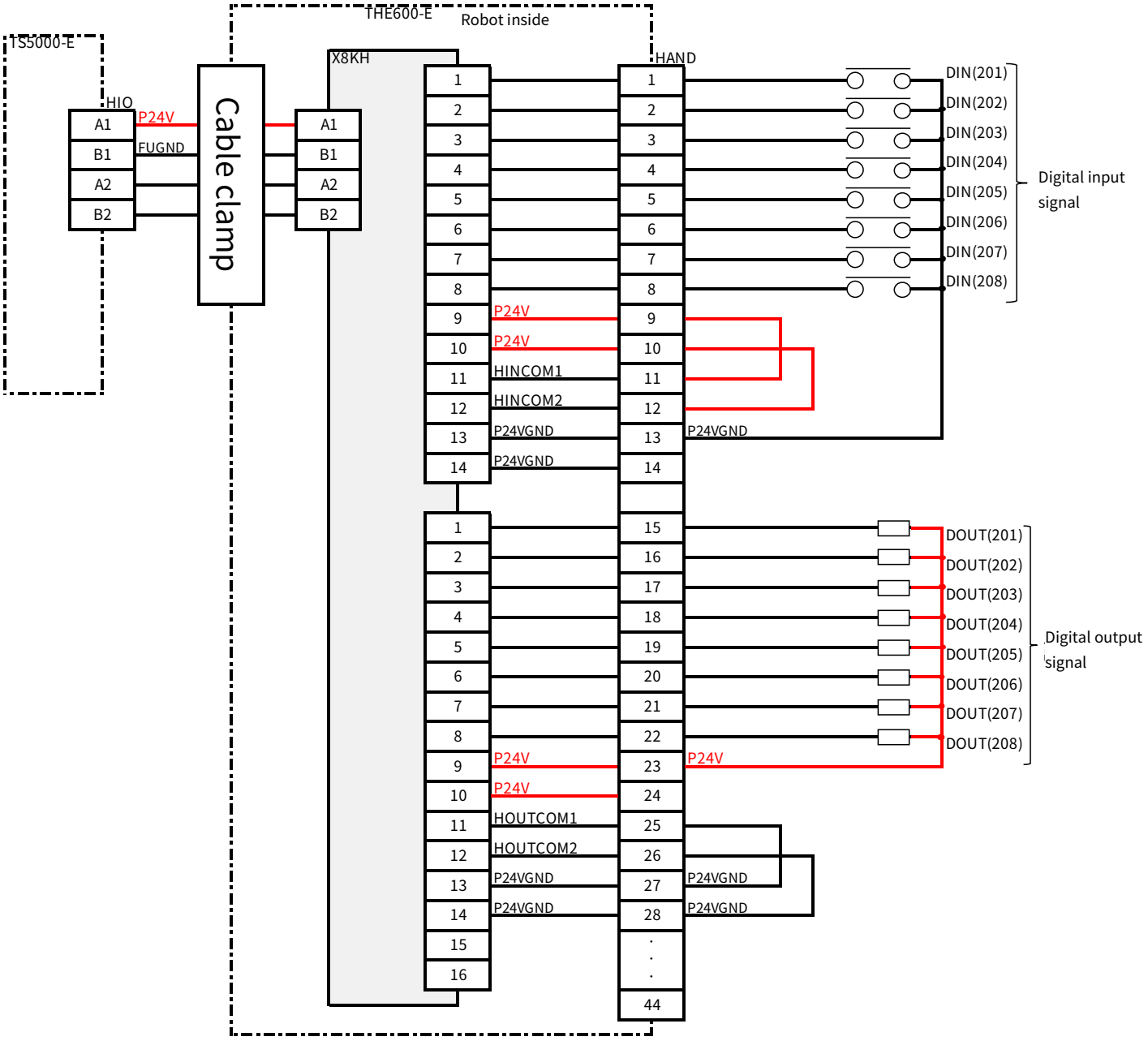


Fig. 8.7 Hand wiring diagram (Type N)

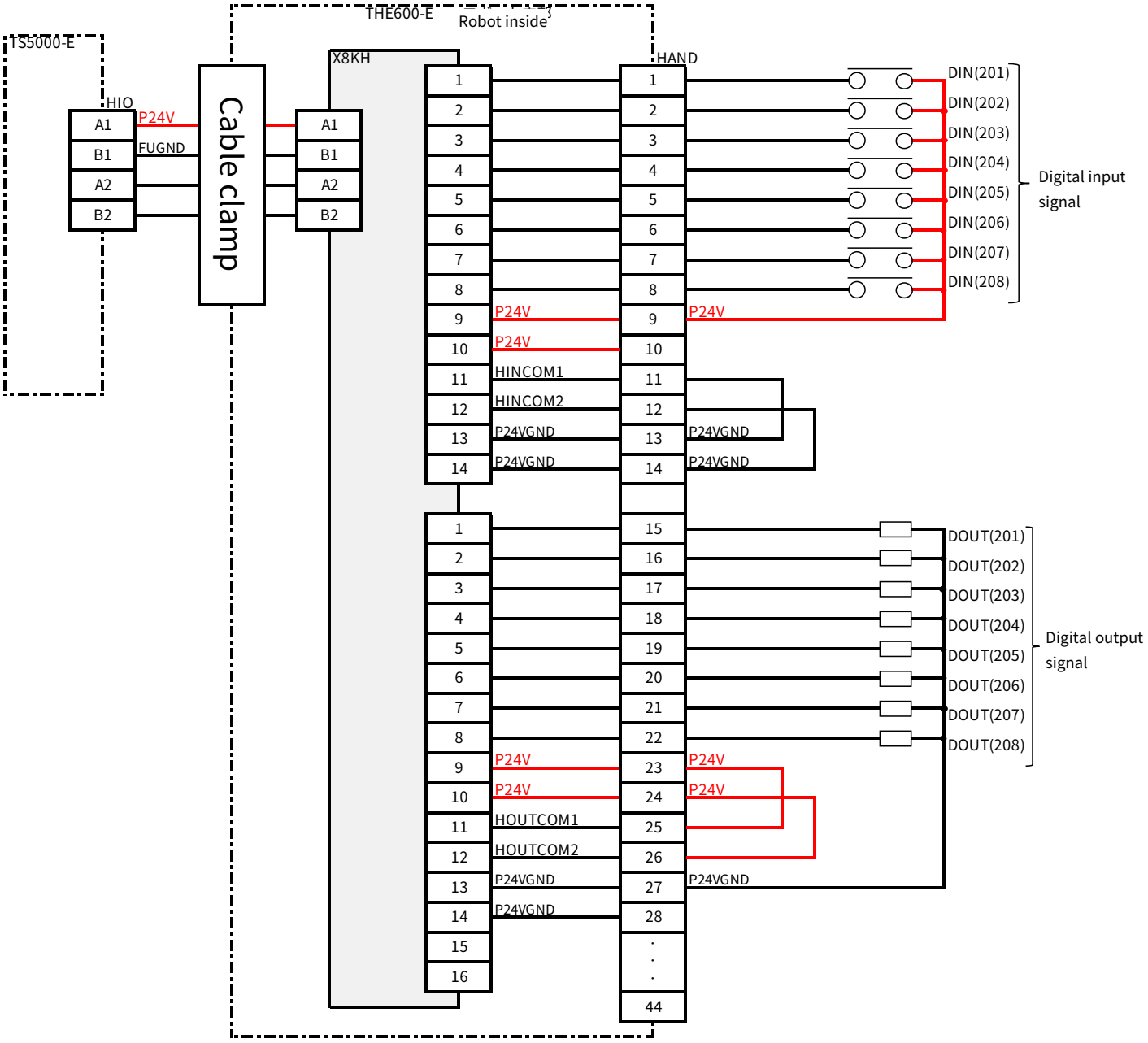


Fig. 8.8 Hand wiring diagram (common to Type-N)

8.1.4 Connection Example

Fig. 8.9 shows an example of hand wiring.

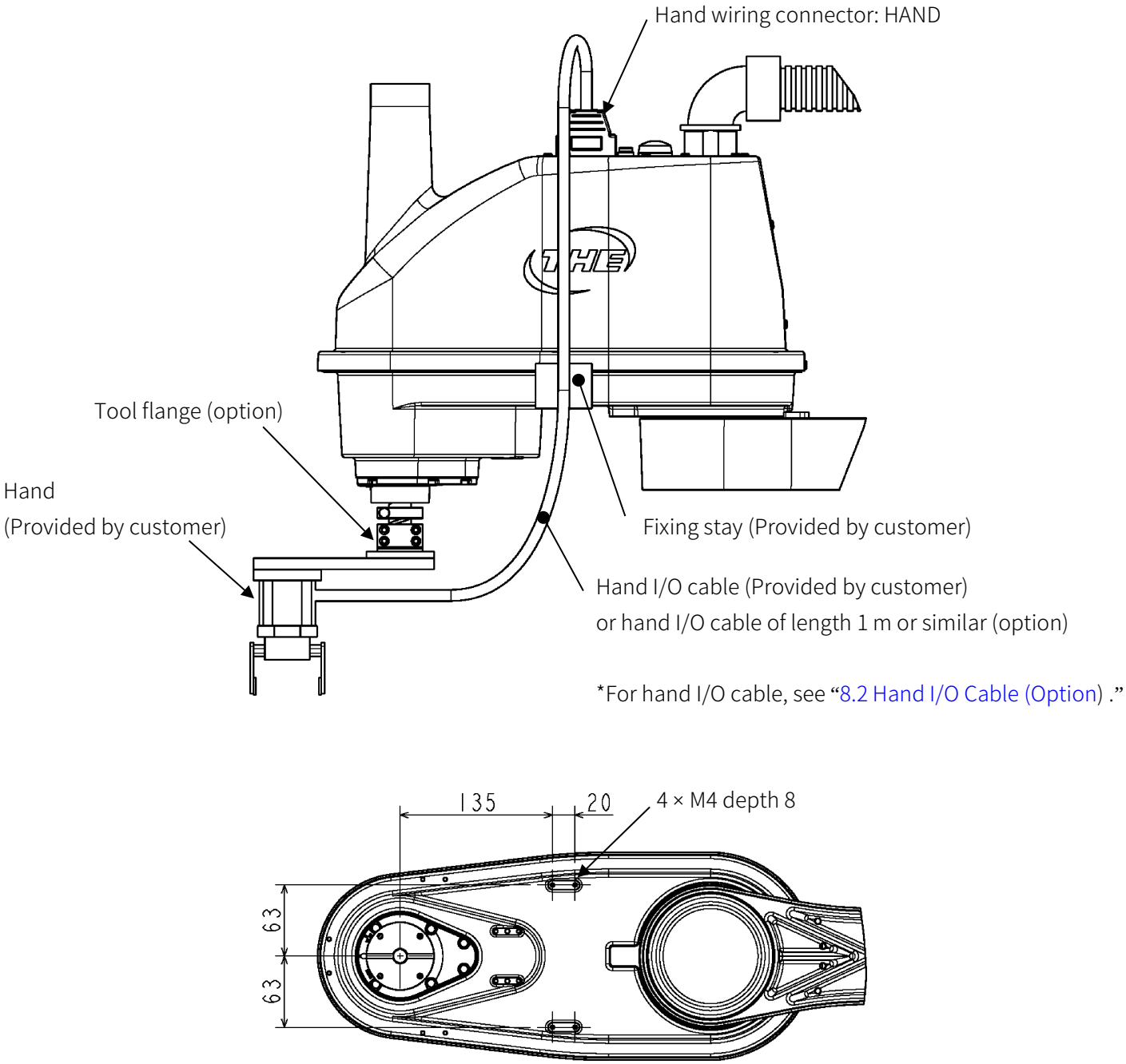


Fig. 8.9 Example of hand I/O cable wiring to arm-2 top

For the dimensions of the other service taps, see Fig. 7.2.

Fig. 8.10 shows a hand piping connection example.

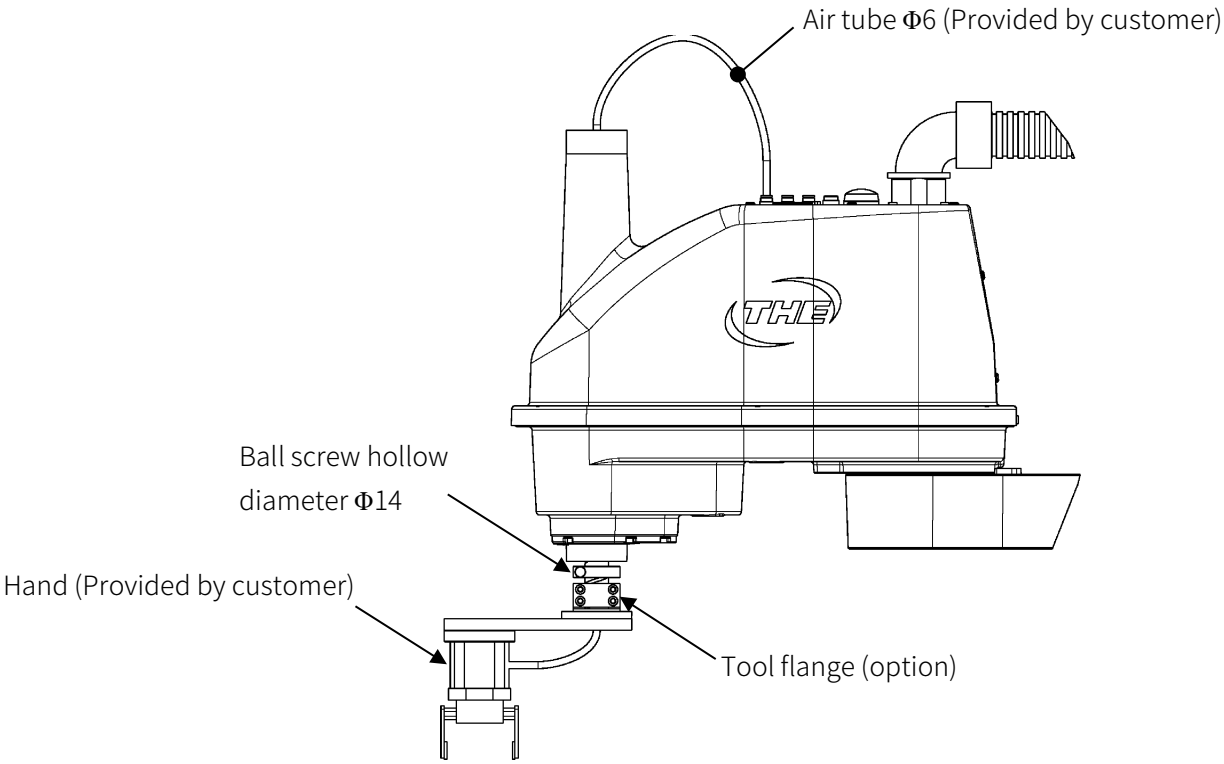


Fig. 8.10 Example of air tube piping to arm-2 top

NOTICE

- Be sure to perform hand wiring and piping while checking the following status during robot operation. Otherwise, the robot and hand will become faulty early.
  - Check whether a load is being applied to the hand connector.
  - Check whether the hand wiring and piping are contacting the robot or peripheral equipment and are worn.
  - Check whether the fixing stay interferes with the 1st arm or other parts

8.2 Hand I/O Cable (Option)

For reference, Fig. 8.11 and Fig. 8.12 shows the external view and wiring diagram for the hand I/O cable (option). Table 8.3 shows the parts used, and Table 8.4 shows the SHIBAURA drawing number for the hand I/O cable (option).

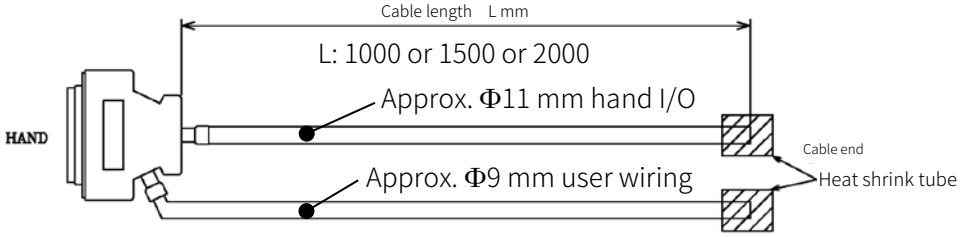


Fig. 8.11 External view drawing of hand I/O cable (option)

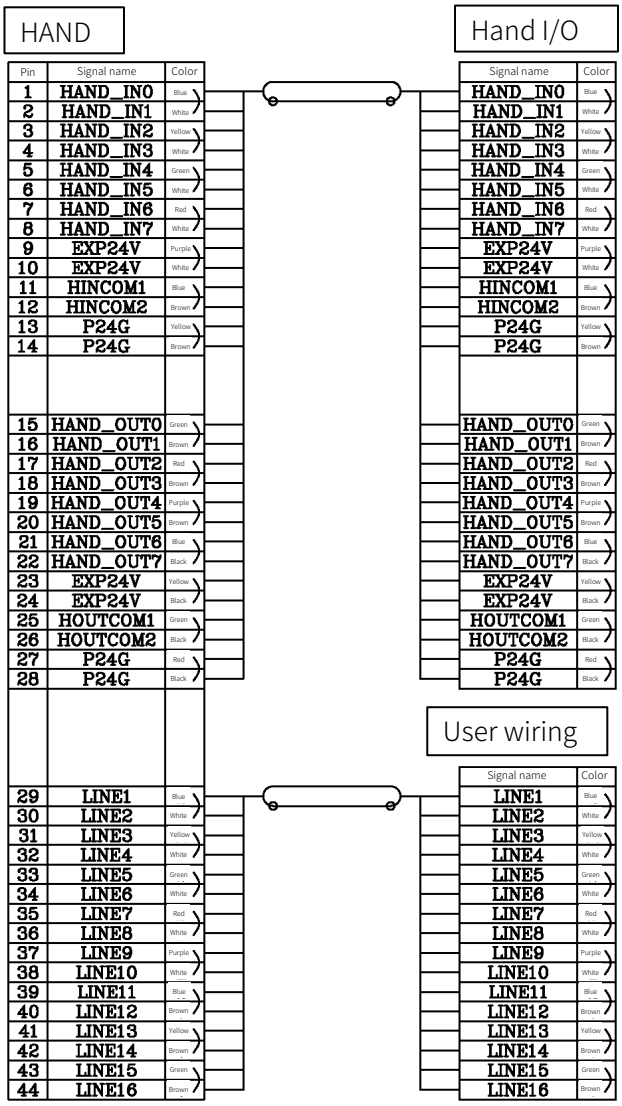


Fig. 8.12 Hand I/O cable (option) wiring diagram

Table 8.3 Hand I/O cable parts used

No.	Product name	Model	Maker	Qt'y used
1	Cable	RMFEV-SB(CL3) 0.2SQ-15P	DYDEN Corp.	1
2		RMFEV-SB(CL3) 0.2SQ-8P		1
3	D-Sub connector	09563005601	HARTING	1
4	D-Sub hood	09670250494		1
5	D-Sub contact	09560008177		44

Table 8.4 Hand I/O cable (option)

Product name	Type	Cable length	SHIBAURA Dwg. No.
Hand I/O cable		1 m	396N7701
		1.5 m	396N7702
		2 m	396N7703

9. Teaching

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.
THE600-E does not have a safety-compliant low-speed control function. (It does not have a function to monitor that it continues to operate at a low speed in MANUAL mode.) Do not perform work in the hazardous area when teaching.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO

- *We have prepared various methods for teaching and guiding the robot, which is done by turning on a drive power supply (turning on the servo). For details, refer to the following the separate manuals.*
  - TP5000 Operation Manual
  - TCmini (Internal PLC) Manual
  - Communication Manual
  - TSAssist User's Manual
- *Before starting work, check the information in 10.1 Rescuing a Person Trapped by a Robot, and conduct training.*

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a malfunction due to noise could cause an expected accident.
- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.
- Be sure to set up a safety fence or door, and perform teaching outside the robot maximum movable range. If you perform teaching within the robot's maximum movable range, the robot could collide with you.
- When turning on the servo, be sure to always be outside the robot maximum movable range.
  Otherwise, a collision could occur due to an unexpected movement by the robot.

⚠ WARNING

- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal movement (Table 6.2).
- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- Be sure to lock the hand connector, and be sure to secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.
- Never enter the robot maximum movable range during test operation.
  A collision could occur with the robot.

9.1 Guiding the Robot by Pushing by Hand

The 1st, 2nd, and 4th axes of THE600-E can be moved by turning off the drive power (servo turned off) and pushing them by hand. If the controller is turned on, by releasing the brake, the 3rd axis can be pushed and moved by hand without turning on the drive power supply (turning on the servo).

⚠ WARNING

- Before turning on the servo for the robot, always make sure that there is no one inside the robot's safety protection zone.
  Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, carry a teach pendant set to MANUAL mode, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

NOTICE

- Never touch the ball screw spline shaft with your bare hands.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could reduce the performance and lifespan of the robot. The robot will malfunction.

9.1.1 Guidance for 1st Axis and 2nd Axis

MEMO

The 1st and 2nd axes can be pushed and guided by hand without releasing the brake.

The 4th axis can also be guided without releasing the brake, but due to the structure of the ball screw spline, the 3rd axis will move up and down at the same time, and so we recommend releasing the brake before guiding.

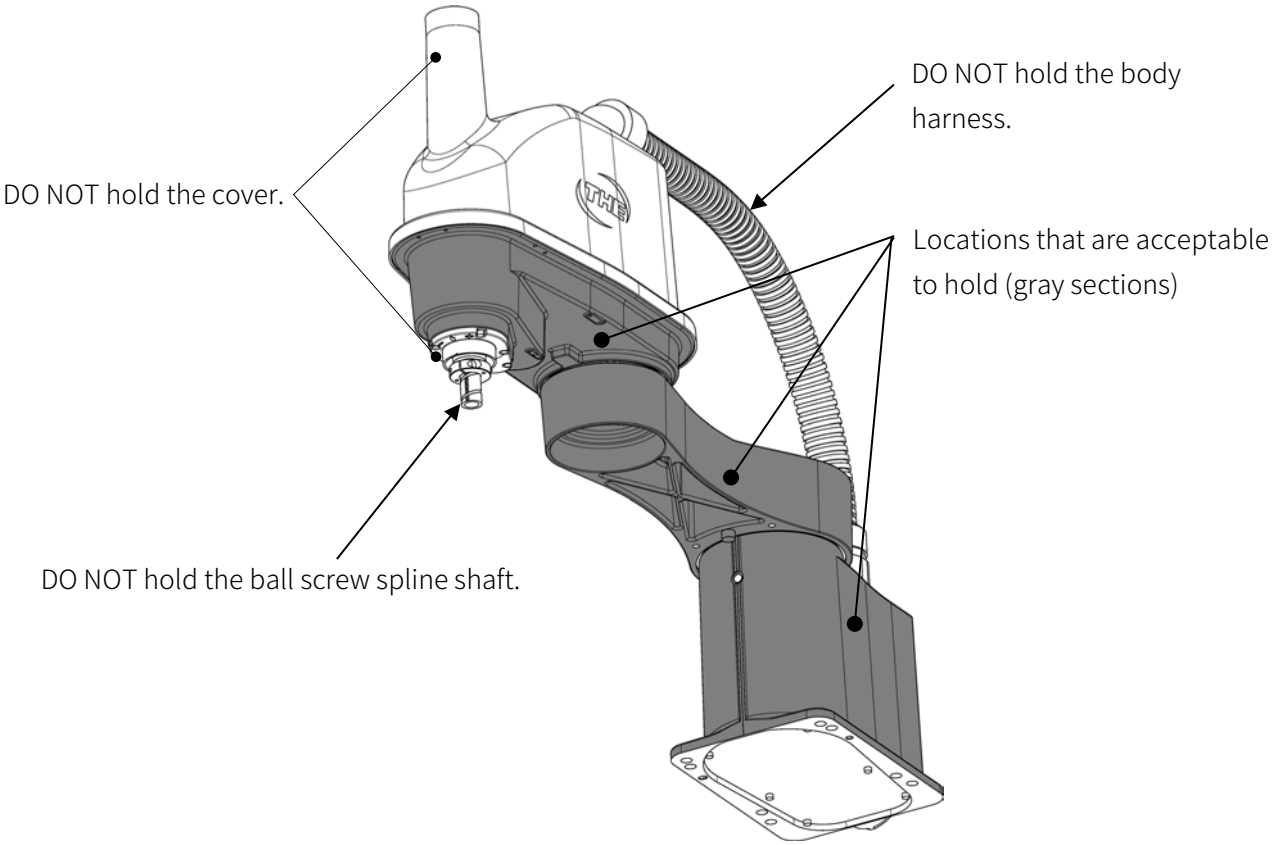


Fig. 9.1 Handling areas of the robot (THE600-E)

NOTICE

- When pressing the robot by hand to move it, be sure to hold the gray sections shown in Fig. 9.1.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.

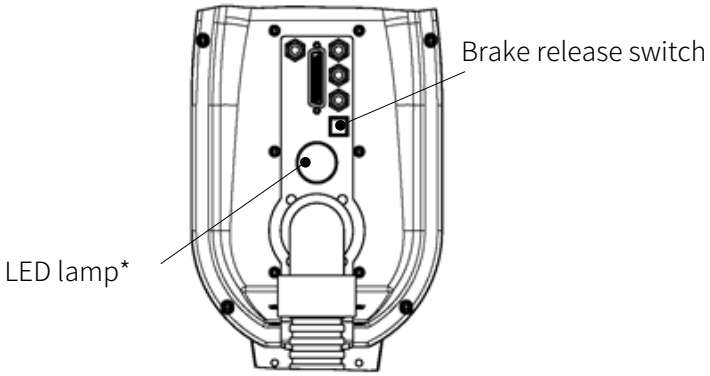
9.1.2 Guidance for 3rd Axis and 4th Axis

⚠ WARNING

- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.



* Make sure that the LED is turned off (servo is turned off).
If the LED is turned on (servo is turned on), the axis will not move even if it is pushed by hand.

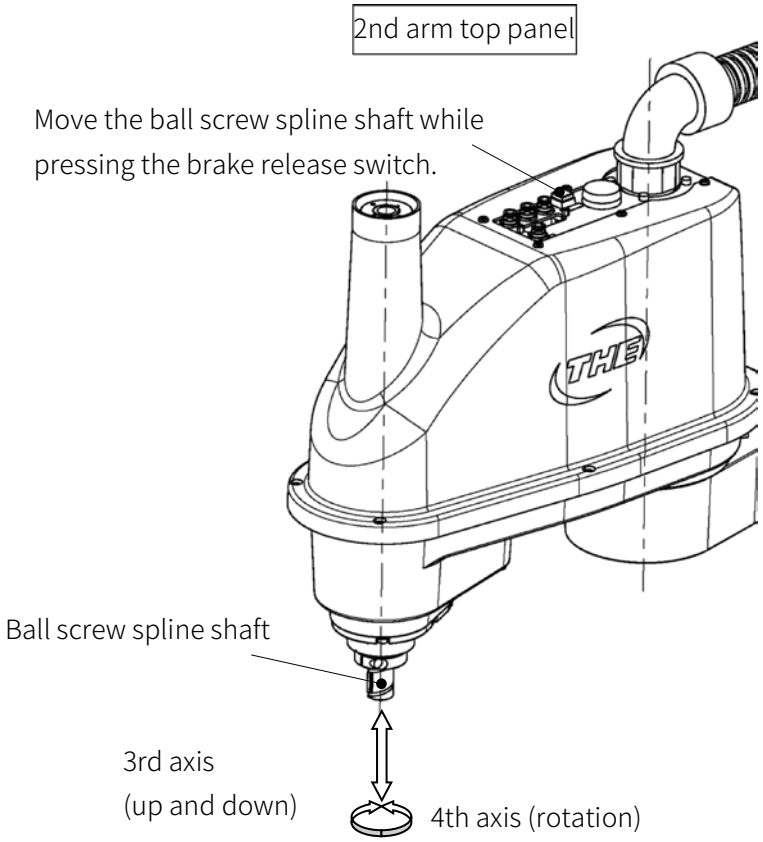


Fig. 9.2 Guiding the 3rd and 4th axes by pushing by hand

MEMO

When guiding the 3rd and 4th axes by pushing them by hand, hold the mechanical stopper mounted to the ball screw spline shaft to guide them.

Direct touching the ball screw spline shaft could result in rusting.

10. Operation

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO

- *We have prepared various methods for operation. For details, refer to the following the separate manuals.*
  - TP5000 Operation Manual
  - TCmini (Internal PLC) Manual
  - Communication Manual
  - TSAssist User's Manual
- *After initial start-up or when the system configuration has been changed, be sure to perform test operation of the created program before putting the robot into the production process to confirm that the robot performs the intended operation and that the robot does not collide with surrounding equipment or objects. For details on test operation, refer to "3.2 Test operation" in the "STEA1162 TP5000 Operation Manual."*
  Also, please check if the emergency stop switch and other safety-related functions are working properly. For details on the check points, refer to "STEA1223 TS5000-EMS Controller Manual."

⚠ DANGER

- Before turning on the controller power, always make sure that there is no one inside the robot's safety protection zone. Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.
- Before turning on the controller power, always make sure that there is no one inside the robot's safety protection zone. Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- Be sure to lock the hand connector, and be sure to secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.

NOTICE

- Before operating the robot, check the information in “6.2.1 Installing the Robot,” and make sure that it is installed correctly.
  If a robot that is not installed correctly is moved, the robot could run by itself and tip over.

10.1 Rescuing a Person Trapped by a Robot

Even with various safety measures in place, there is still a risk of becoming entangled or trapped by a robot.

In order to rescue victims quickly, be sure to train workers on how to rescue victims from the robot before starting their work.

⚠ DANGER

- Switch the master mode key to “MANUAL.” Carry the teach pendant and, if using the TP5000, remove the master mode key, and manage it properly.
  Otherwise, you could be struck by a robot that has been started by someone else.
  - If there are no impediments to rescue, apply an emergency stop.
  - If something is interfering with rescue, leave the safety door open.
  - Do not turn off the power to the controller. This will prevent releasing of the 3rd axis brake.
  - You can check the status of the servo using the LED lamp on the robot.

10.1.1 Rescue Procedure

LED Lamp

* Make sure that the LED is turned off (servo is turned off). If the LED is turned on (servo is turned on), the axis will not move even if it is pushed by hand.

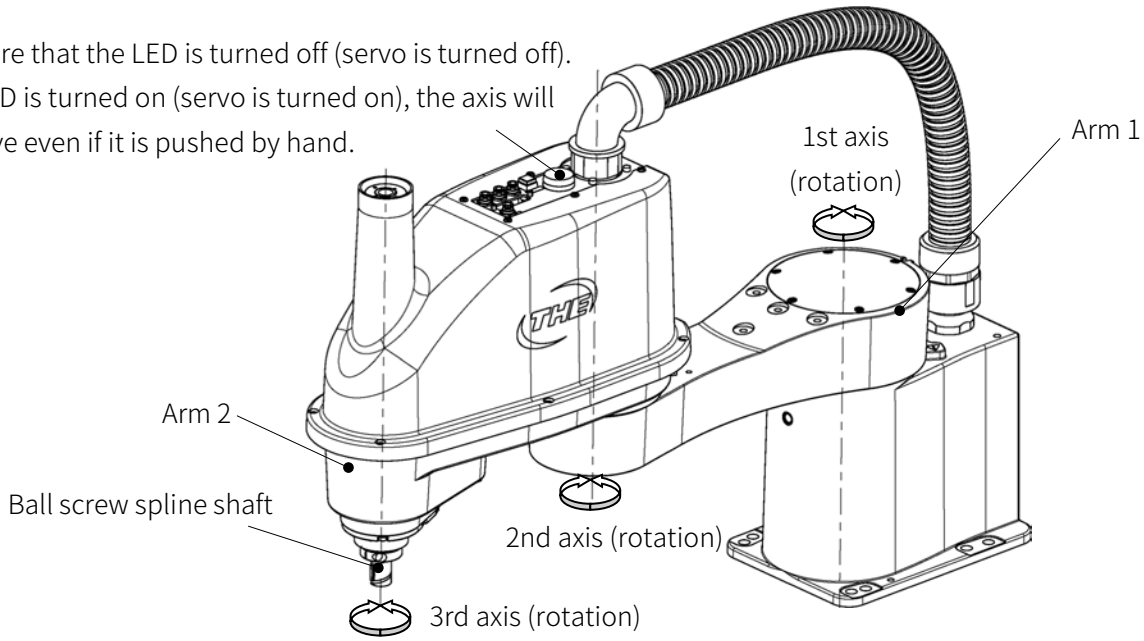


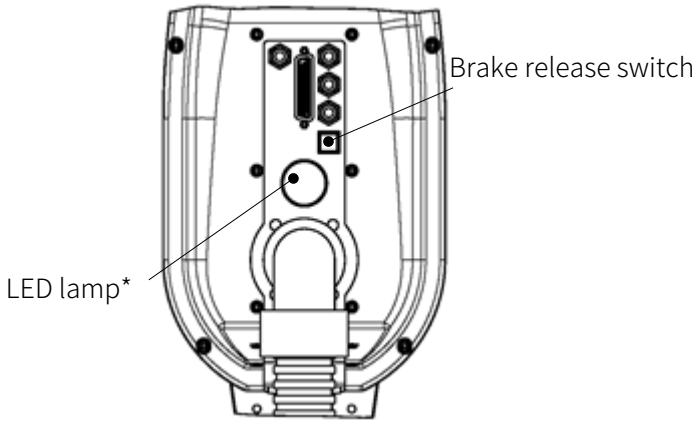
Fig. 10.1 Joint axis moved by pushing by hand when servo is turned off

a) [5] Press the robot arm by hand to move it.
The 1st axis and 2nd axis motors of the THE600-E are not equipped with a brake. By pressing the emergency stop switch or other button to turn off the servo, the 1st and 2nd arms can be pushed by hand to move them.
After turning off the servo, move the 1st and 2nd arms to release the victims from the robot.

b) Press the robot hand by hand to move it.
The 4th axis motor of the THE600-E is not equipped with a brake. By pressing the emergency stop switch or other button to turn off the servo, the 4th axis ball screw spline shaft can be pushed by hand to move it.
After turning off the servo, rotate the robot hand installed on the ball screw spine shaft to release the victims from the robot.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the robot and mechanical stopped mounted on the ball screw spline shaft or under the robot hand.
  Otherwise, your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.



* Make sure that the LED is turned off (servo is turned off).
If the LED is turned on (servo is turned on), the axis will

Fig. 10.2 3rd axis brake release

c) Press the brake release switch, and lift the robot hand by hand to move it.
The 3rd axis motor of the THE600 is equipped with a brake to prevent the ball screw spline shaft and robot hand from falling under their own weight when the servo is turned off. For this reason, the 3rd axis cannot be lifted and moved by hand by simply turning off the servo.
Turn on the controller, make sure that the servo is off, and then release the victim from the robot by lifting the robot hand by hand while pressing the brake release switch.
Be sure to have someone other than the brake release operator standing by to support and move the ball screw spline. When the brake is released, a load of about the weight of the hand is applied.

⚠ WARNING

- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

11. Repacking

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO

- *This describes the key points for repacking. For details on the repacking work, refer to the following chapters.*
  - *Installation*
  - *Transfer*
  - *Unpacking*
  - *Transportation*
- *Before starting work, check 10.1 Rescuing a Person Trapped by a Robot, and conduct training.*

11.1 Safety Information When Removing the Robot

This section provides safety information when removing the robot from the installation site.

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a malfunction due to noise could cause an expected accident.
- Do not remove the ground wire until the power is turned off.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.
- Before turning on the servo for the robot, always make sure that there is no one inside the robot's safety protection zone.
  Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, carry a teach pendant set to MANUAL mode, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the robot hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Do not remove the robot until the controller power is turned off and the cable is removed from the controller. Otherwise, a worker could receive an electric shock due to touching a cable that was broken if the robot is dropped.
- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

NOTICE

- The installed robot should never be removed by only one worker.
  The worker may be unable to support the robot and it could fall over, resulting in breakdown.

11.2 Safety Information When Transferring the Robot

This section provides safety information when carrying the robot to the packing site.

⚠ WARNING

- Gather the cables together before carrying the robot.
  Otherwise, a worker could trip and fall over a cable on the floor.
  - Workers should wear a helmet to protect their head.

⚠ CAUTION

- When transferring the robot, be sure to always secure in place using the supplied fixing jigs.
  Otherwise, a worker could get caught in an arm that moved.
  - The weight of the arm section is approximately 17 kg.
  - Wear protective gloves to prevent slippage and to protect the hands.

11.2.1 Transferring the Robot by Crane

⚠ WARNING

- When hoisting the robot, be sure to always follow the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the attached eyebolts or eyebolts with a working load of 0.79 kN or more when suspended at 45°.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
  - Be sure to always hoist and lower the robot slowly.
  - Provide a transfer route, and make sure that no one goes under the suspended load.

NOTICE

- Never touch the ball screw spline shaft by bare hand.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to hoist and lower the robot slowly.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- When hoisting and transferring the robot, use the eyebolts mounted into the tapped holes indicated by the arrow marks. Otherwise, the robot could lose its balance and fall over.

11.2.2 Transferring the Robot by Workers

⚠ WARNING

- The robot should never be transferred by only one worker.
  Otherwise, the robot cannot be fully supported and could drop.
  - The robot weight is 31 kg.
  - Lift the robot using more than one person.
  - Use a cart or similar device to the transfer location.
  - Clear a transfer route, and make sure that a dropped robot will not hit anyone.
  - Wear protective gloves and safety shoes.
- Never use an awkward posture when inserting the product into the packing box.
  Overexertion could injure the worker's back.
  - Provide space where work can be performed easily.

NOTICE

- The robot should never be transferred by only one worker.
  Otherwise, the worker may be unable to support the robot and it could fall over.
  - Do not use a product that has been dropped
- Never touch the ball screw spline shaft with your bare hands.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- When lifting up the robot by workers, be sure to hold the gray portions shown in Fig. 5.3 by hands.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.

11.3 Safety Information When Repacking the Robot

This chapter provides safety information when repacking the robot.

11.3.1 Repacking

⚠ WARNING

- Be sure to secure the robot to be shipped in a packing box as it was delivered.
  Otherwise, the robot will fall over.
  - The customer must obtain new packing straps (PP bands).
  - If no packing box is available, use the mounting holes in the base to secure in place.

NOTICE

- Be sure to secure the robot to be shipped in a packing box as it was delivered.
  Otherwise, the robot could fall over, resulting in a breakdown.
  - The customer must obtain new packing straps (PP bands).
  - If no packing box is available, use the mounting holes in the base to secure in place.

11.3.2 Moving the Packing Box

⚠ WARNING

- When moving the packing boxes, never carry the packing boxes by hand.
  Otherwise, a packing box could fall on a worker.
  - Use a hand lifter or similar equipment.
  - Do not allow people to approach the transfer route.

NOTICE

- When moving the packing boxes, never carry the packing boxes by hand.
  The packing box may be unable to be fully supported and could drop.

11.4 Safety Information When Transporting the Robot

This section provides safety information when loading the robot from the packing site to the transportation equipment (truck, etc.).

11.4.1 Loading the Packing Boxes

⚠ WARNING

- Never carry the packing boxes by hand when loading them onto a truck or other vehicle.
  Otherwise, a packing box could fall on a worker.
  - Use a forklift or similar equipment.
  - Do not allow people to approach the transportation route.

NOTICE

- Never carry the packing boxes by hand when loading them onto a truck or other vehicle.
  Otherwise, a packing box could fall over, resulting in damage to the robot.
  - Use a forklift or similar equipment.
  - Do not allow people to approach the transportation route.

Maintenance Manual

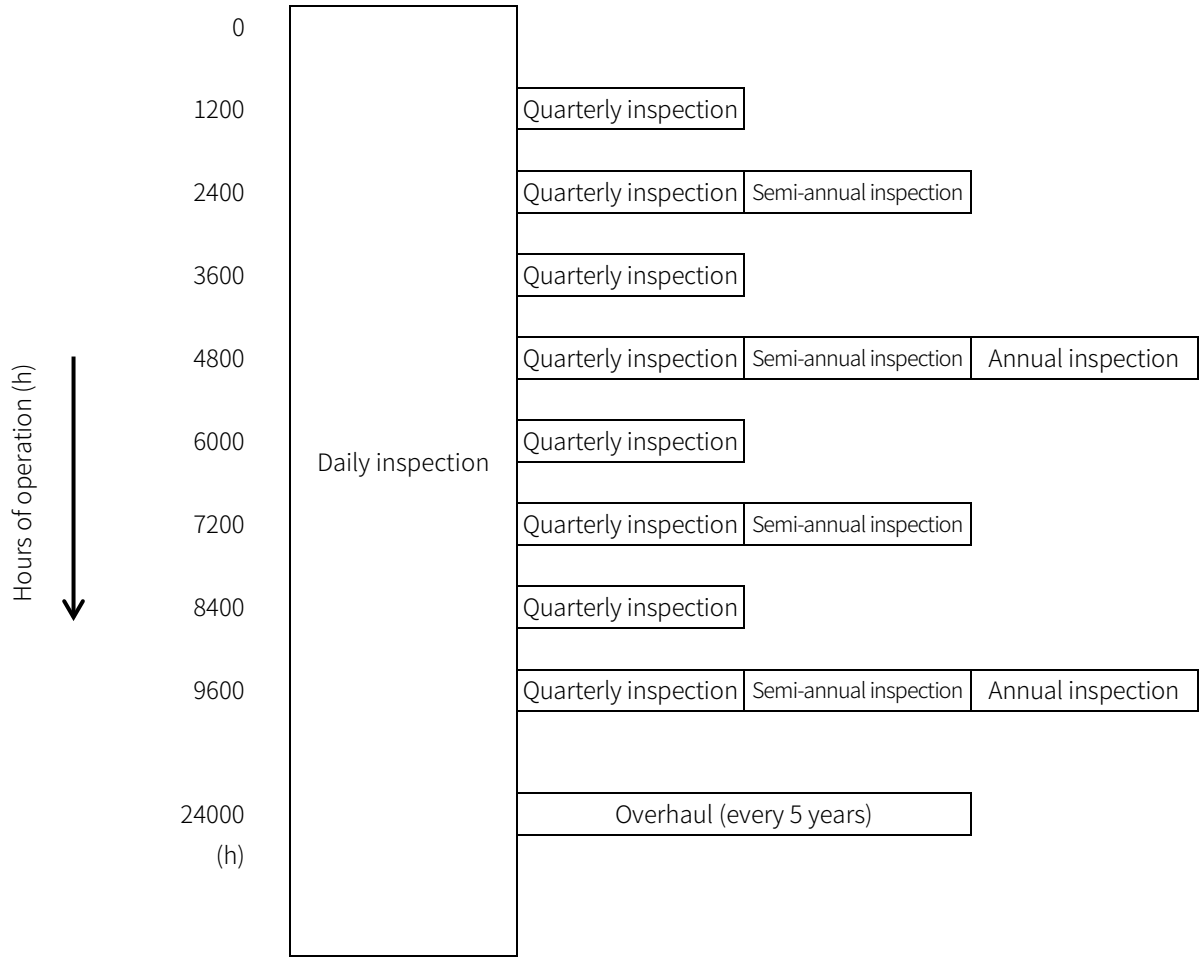
This manual describes how to maintain the various parts of the robot.

12. Maintenance

12.1 Maintenance Schedule

There are two types of maintenance and inspections: Those conducted daily and those performed after a certain period of time.

In the periodic inspection and maintenance, inspection items are added in 1,200-hour units (hours of operation).



MEMO

When the system is operated for 16 hours per day: 1,200 hours ÷ 16 hours ÷ 25 days = 3 months

When the system is operated for 24 hours per day: 1,200 hours ÷ 24 hours ÷ 25 days = 2 months

When the system is operated for 16 hours per day, periodic inspection should be conducted every three months. However, if the hours of operation per day are increased, periodic inspection must be conducted at shorter intervals.

Fig. 12.1 Maintenance schedule

12.2 Maintenance and Inspection Items

There are two types of maintenance and inspections: Those conducted daily and those performed after a certain period of time.

This section describes the maintenance and inspection items. Details on Inspection and replacement are described in their respective sections.

12.2.1 Inspection at Power OFF

Table 12.1 Inspection at Power OFF

Details of Inspection	Inspection location	Daily inspection	Quarterly inspection	Semi-annual inspection	Annual inspection	Refer to
Check for any looseness and rattling of the bolts, and retighten if any looseness or rattling is found.	Robot hand mounting bolts	✓	✓	✓	✓	13.1.1.1
	Robot installation bolt	✓	✓	✓	✓	13.1.1.2
	Covers	✓	✓	✓	✓	13.1.1.3
Check of the wearing out of cables and air tube	Cables and air tubing outside the robot		✓	✓	✓	13.1.2
Check the tightness of cable clamps	2nd arm, base top surface				✓	13.1.3
Inspect for cuts or scratches Clean off any adhering foreign objects	Entire robot	✓	✓	✓	✓	
Check the applied corrosion inhibitor, and apply corrosion inhibitor if needed	Ball screw shaft end face	✓	✓	✓	✓	13.2.2
Check that there is no misalignment between the motion of each joint axis and the origin position by pressing by hand	Each joint axis	✓	✓	✓	✓	14.9

12.2.2 Inspection at Power ON

Table 12.2 Inspection at Power ON

Details of Inspection	Inspection location	Daily inspection	Quarterly inspection	Semi-annual inspection	Annual inspection	Refer to
Check applied grease, and apply grease if needed	Ball screw shaft	✓	✓	✓	✓	13.2.2
Check that the teaching points are not misaligned	Robot tip	✓	✓	✓	✓	
Replacing Position Detector (or Encoder) Batteries	Battery box				✓	14.3.2

12.2.3 Overhaul

When an overhaul must be performed, please contact the SHIBAURA MACHINE Service Department.

Table 12.3 Overhaul

Maintenance description	Maintenance parts	Every 5 years	Refer to
Check the consumable parts and replace if needed	Motors, reduction gears, belts, ball screw, body harness, LED Lamp	✓	

12.3 Preparing for Maintenance Work

Before starting the maintenance work, be sure to provide sufficient space for conducting maintenance. See Fig. 12.2 for the space required for maintenance work.
Do not place any objects in the workspace, and keep cables such as those between the robot and the controller out of the space or protect them with molding or similar material to prevent workers from tripping on them during the work.

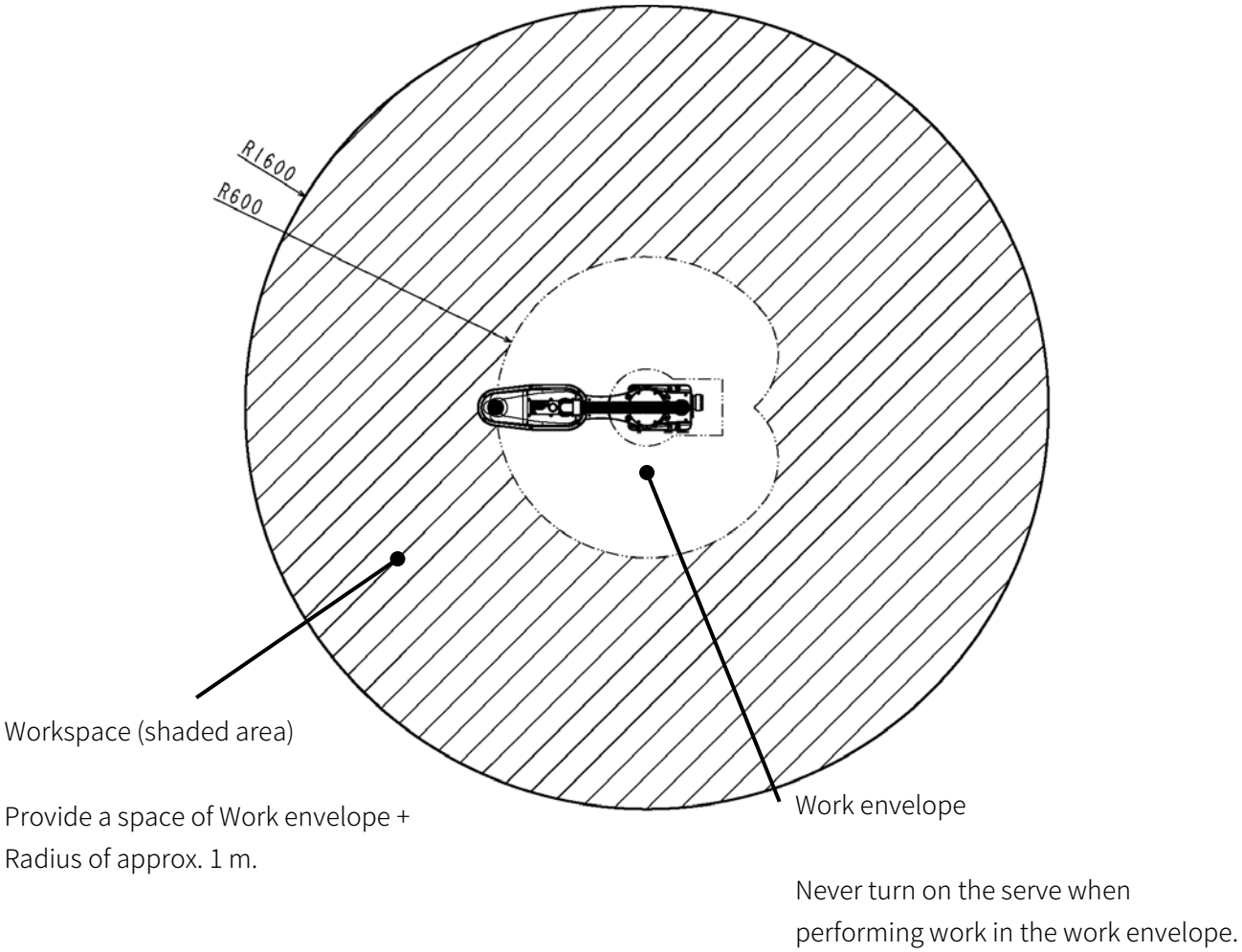


Fig. 12.2 Maintenance workspace

12.4 Safety Information on Maintenance and Inspection

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.
- Before turning on the servo for the robot, always make sure that there is no one inside the robot's safety protection zone.
  Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, carry a teach pendant set to MANUAL mode, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.
- Be sure to always turn off the power before performing maintenance on the robot.
  If work must be performed on the robot within the robot's hazard zone due to unavoidable circumstances, be sure to switch the master mode to “MANUAL.” Carry the teach pendant, and remove the TP5000 master mode key, and manage it properly.
  Otherwise, you could be struck by a robot that has been started by someone else.
  - Post a notice near the controller that work is in progress.
  - Assign a work supervisor who is stationed outside the hazard zone of the robot.
  - The supervisor should ensure the safety of the worker and any other people approaching the robot hazard zone.
  - If the teach pendant is not used, carry an emergency stop device with you when performing work.
  - If the power will be turned on, make sure that the worker and the supervisor receive safety training on tasks related to the inspection of industrial robots and other devices as specified by the laws and regulations of the respective country.
- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the robot hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

NOTICE

- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could reduce the performance and lifespan of the robot. The robot will malfunction.
- Performing maintenance work while the robot is installed on the frame. Otherwise, the robot could lose its balance and fall over and be damaged.

12.5 Maintenance Tools

It is recommended that you obtain the following tools for maintenance.

- Hex key: Nominal size 2.5 mm, 3 mm, 8 mm
- Torque wrench or torque driver (hex bit): Nominal size 2.5 mm, 3 mm, 8 mm
- Wrench: Nominal size 41 mm, 46 mm, 54 mm, 58 mm
- Scale

Recommended items besides tools:

- Alcohol (for cleaning) ● Waste cloth ● Brush (for applying grease to ball screw)
- Corrosion inhibitor, Kluber A20 or WD-40
- TSAssist: Program creation/teaching support software
- 1st axis and 2nd axis origin position setting jig (option)
- 3rd axis and 4th axis origin position setting jig (option)

12.6 Maintenance Contract and Repairs

12.6.1 Maintenance Contract

The SHIBAURA MACHINE Service Department can perform replacement of consumable parts inside the robot. A fee is charged for this service.
We recommend that the customer enter into a service contract with SHIBAURA MACHINE at the time of purchase.

12.6.2 Repairs

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. Provide a description of the fault and the following information indicated on the robot and controller.
(For contact information of SHIBAURA MACHINE Service Department, see Service Network.)

- Robot model name
- Robot serial number
- Controller serial number
- Date of manufacture of robot
- Total hours of operation

For the serial number and the date of manufacture, refer to the following label on the robot base.

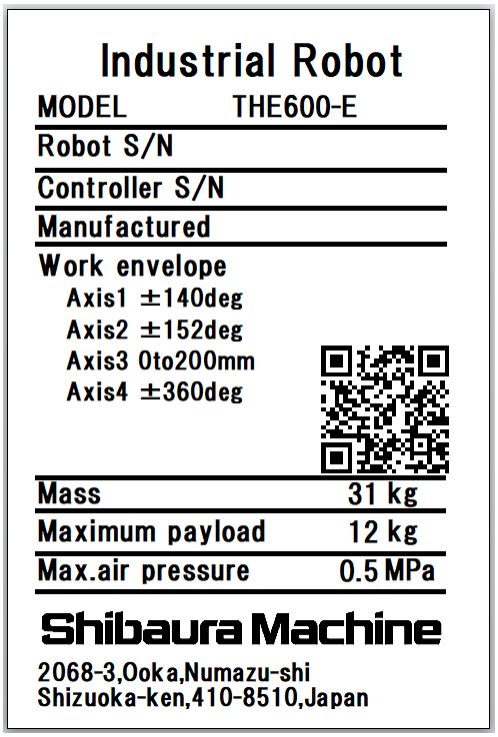


Fig. 12.3 Label

12.6.3 Modifications

This robot and controller MUST NOT be modified or disassembled without a prior written consent from SHIBAURA MACHINE.

⚠DANGER

- Never modify parts except those that are expressly permitted.
  Otherwise, a collision could occur due to an unexpected movement by the robot.

NOTICE

- Never modify parts except those that are expressly permitted.
  Otherwise, the performance and lifespan could be reduced and the robot could malfunction.

13. Robot Maintenance

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.
The SHIBAURA MACHINE Service Department will perform replacement work. A fee is charged for this service. Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO
Before starting work, check the information in *10.1* Rescuing a Person Trapped by a Robot, and conduct training.

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.
- Before turning on the servo for the robot, always make sure that there is no one inside the robot's safety protection zone.
  Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, carry a teach pendant set to MANUAL mode, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.
- Be sure to perform maintenance and inspection periodically.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the robot hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

13.1 Details of Inspection

13.1.1 Bolt Tightness Check

13.1.1.1 Check of Robot Hand Mounting Bolts

When tool flange (option) is not provided

Use the hex key or similar tool to check that the bolts mounting the robot hand used by the customer to the ball screw spline shaft are not loose. Tighten any bolts that are loose.

When tool flange (option) is provided

Use the hex key to check that the four tool flange split bolts (M4) of the tool flange (option) that are split-tightened to the ball screw spline shaft are not loose. Tighten any bolts that are loose.
Use the hex key to check that the four robot hand mounting bolts (M4) that mount the robot hand to the tool flange are not loose. Tighten any bolts that are loose. (Thread locking adhesive is not needed.)

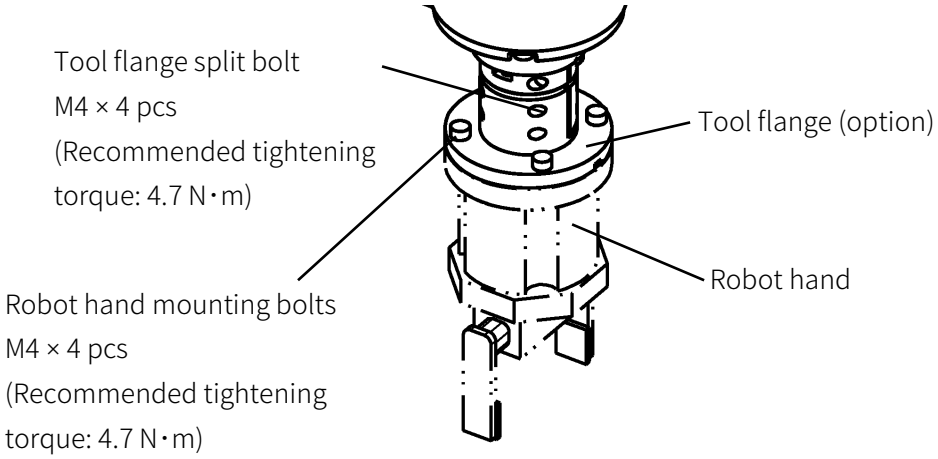


Fig. 13.1 Hand set bolts

13.1.1.2 Check of Robot Installation Bolts

Use the hex key to check that the installation bolts of the robot base are not loose. Tighten any bolts that are loose. (Thread locking adhesive is not needed.)

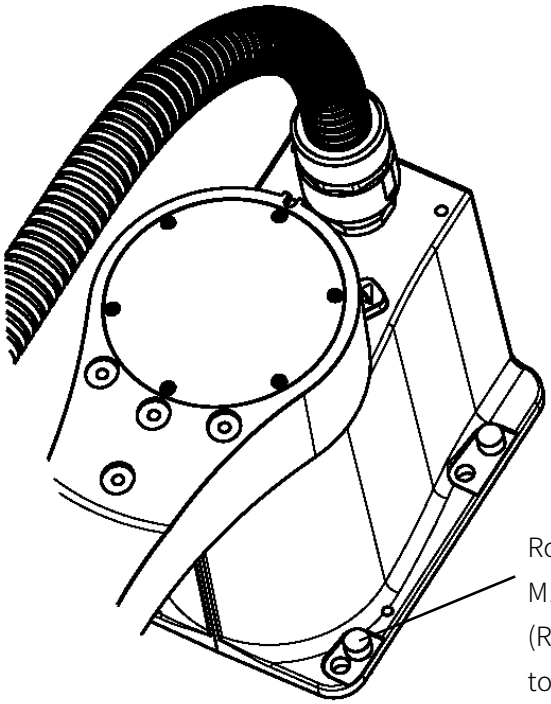


Fig. 13.2 Robot installation bolts

13.1.1.3 Check of Robot Cover Mounting Bolts

Check the looseness of the bolts for mounting the covers of the robot.

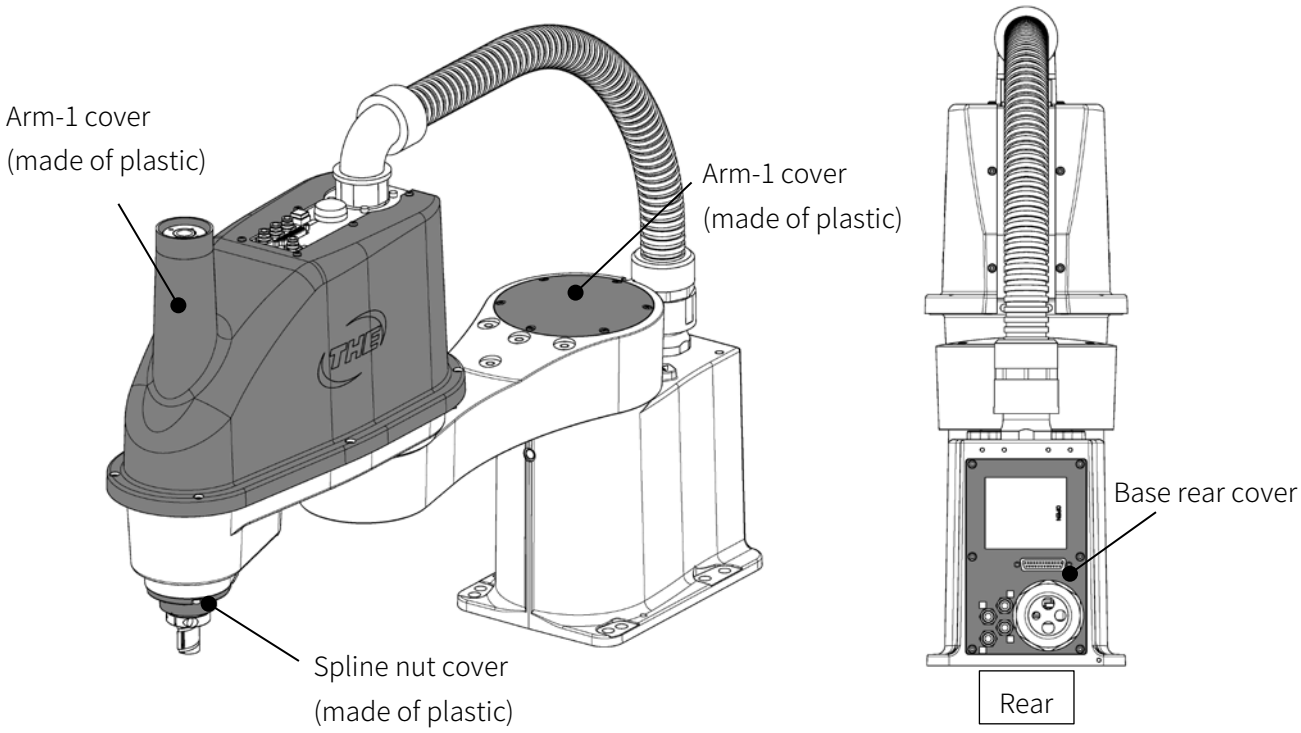


Fig. 13.3 Robot covers

MEMO

When tightening the bolts for attaching the plastic robot covers, if the tightening torque is too high, it could cause the robot cover to crack. Tighten the bolts by referring to the tightening torques shown in the diagrams indicating the mounting bolts for the various covers.

Arm-2 cover

The 2nd arm cover is secured to the 2nd arm and harness guide by 16 hex socket head cap button screws (M4 × 8). Check that the screws are not loose, and tighten them if necessary.
When conducting this check, also check for any looseness in the two hex socket head cap screws (M4 × 8) securing the harness guide.

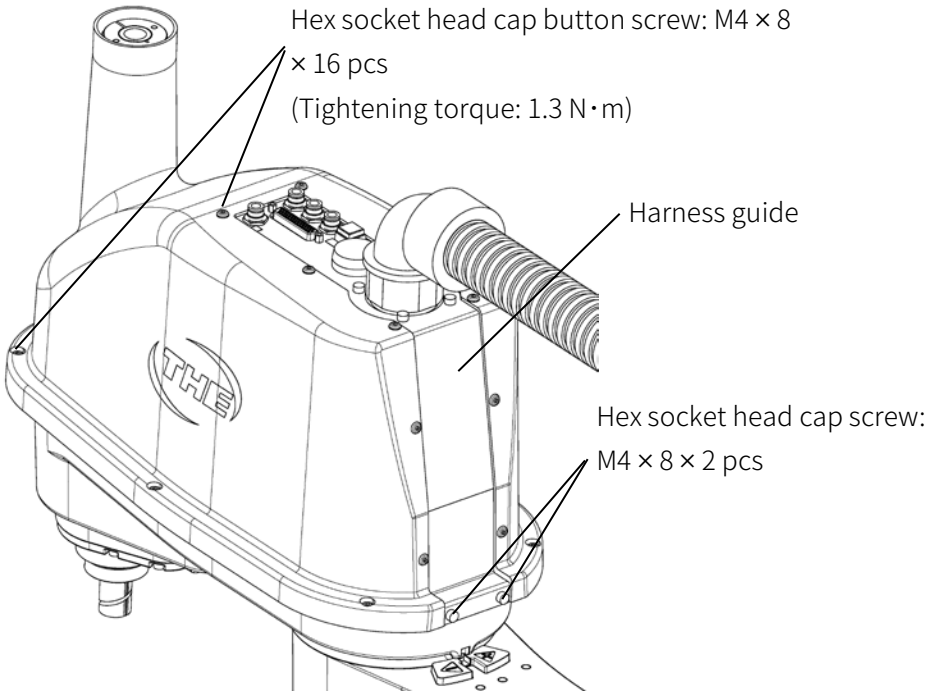


Fig. 13.4 2nd arm cover mounting screws

Arm-1 cover

The 1st arm cover is secured to the 1st arm using six hex socket head cap button screws (M4 × 8). Check that the screws are not loose, and tighten them if necessary.

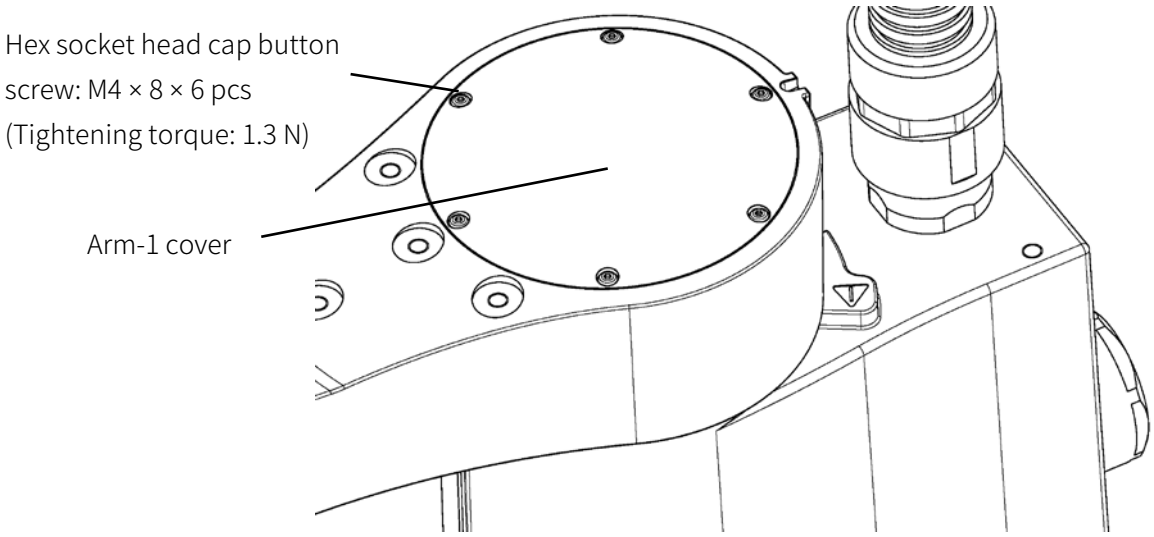


Fig. 13.5 1st arm cover mounting screws

Spline nut cover

The spline nut cover is secured to the 2nd arm using six hex socket head cap screws (M4 × 8). Check that the screws are not loose, and tighten them if necessary.

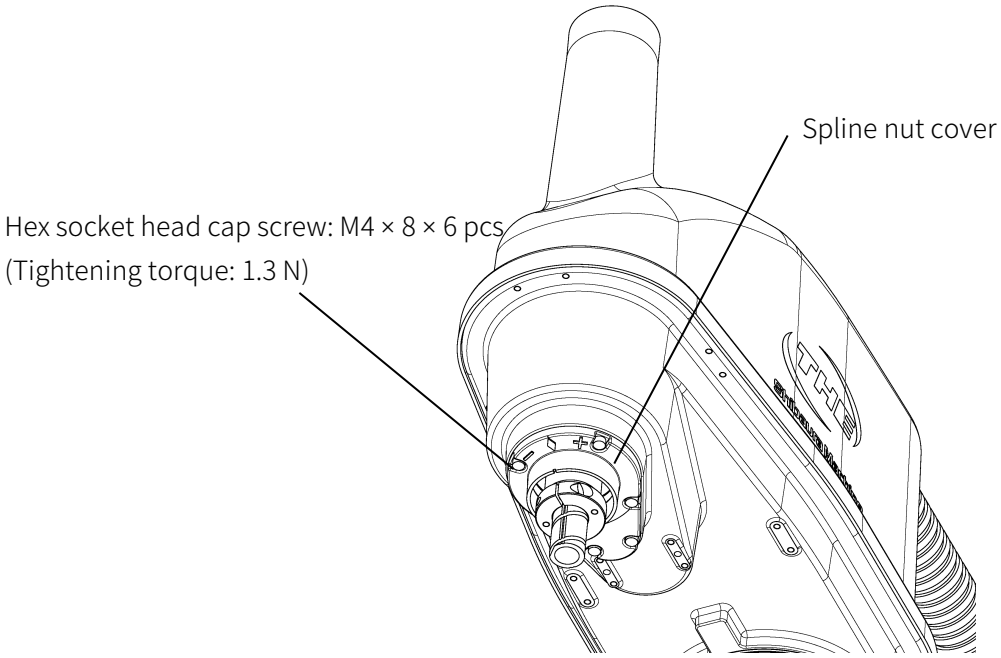


Fig. 13.6 Spline nut cover mounting screws

Base rear cover

The base rear cover is secured to the base using six hex socket head cap button screws (M4 × 8). Check that the screws are not loose, and tighten them if necessary.

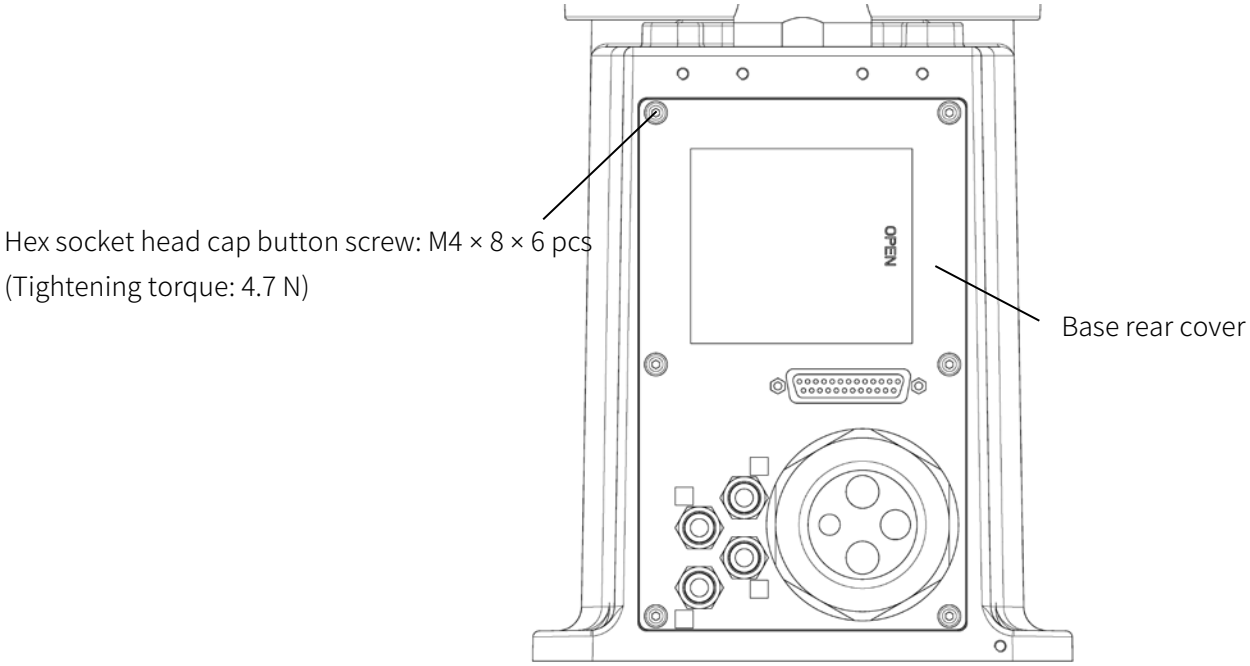


Fig. 13.7 Spline nut cover mounting screws

13.1.2 Check of the Wearing Out of Cables and Air tube

If the air tube passes through the ball screw shaft, check the air tube for any abrasions, breaks, cracks, or other imperfections.

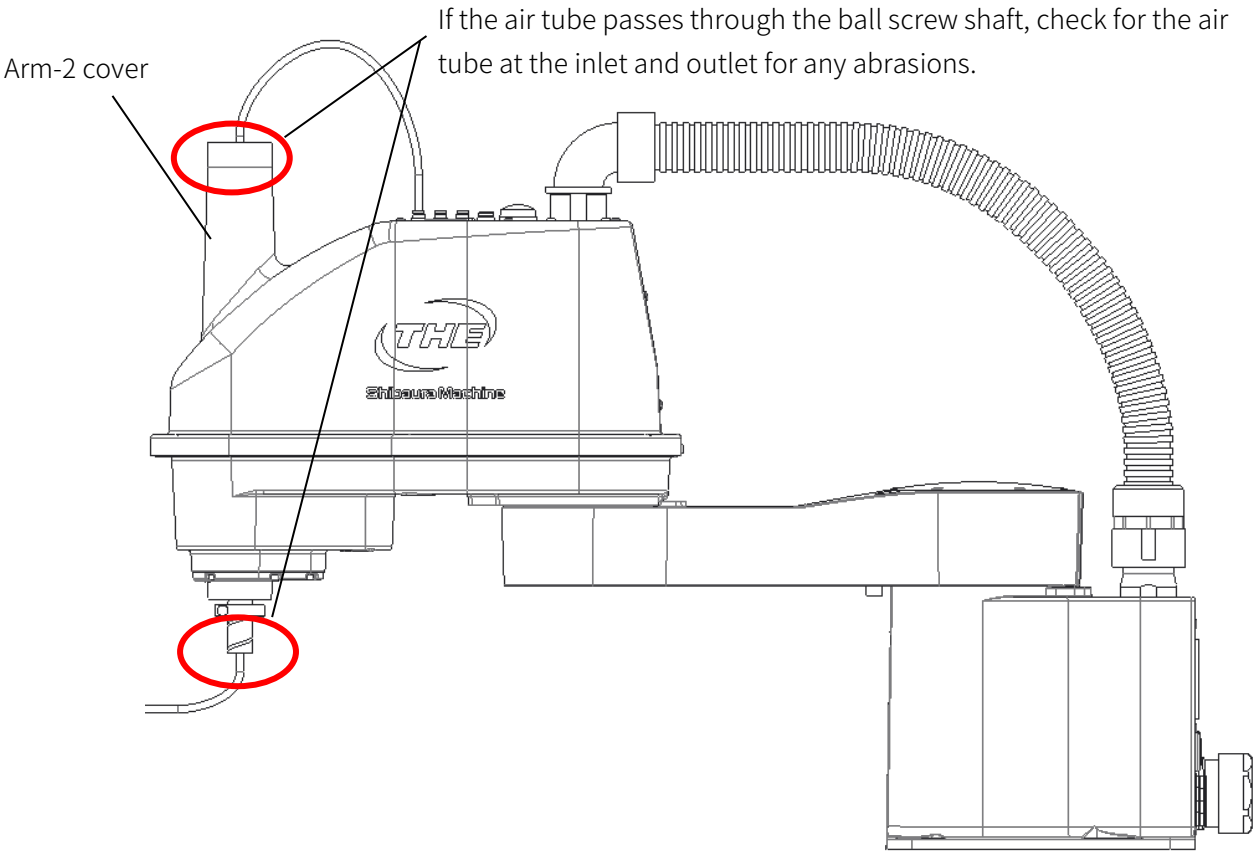


Fig. 13.8 Inspection points of cable

NOTICE

- Be sure to always perform daily inspection for the air tube.
  Replace the air tube immediately if it is damaged.

13.1.3 Check of Tightening of Cable Duct Connectors and Cable Clamps

Use the wrench to check that the clamps securing the cable and cable duct and the cable duct connectors are not loose. The cable duct connectors are found at two locations -- on the 2nd arm side and on the base side. Tighten them if they are loose. Also, tighten the cable duct connectors in the orientation shown in Fig. 13.9. If the cable duct connectors are tightened with the arm bent, the cables cannot be connected correctly.

Bearings are used in the connection part of the 2nd arm-side cable duct connectors. There are screws on the wiring panel on the 2nd arm top surface for securing the bare case, and so check that these screws are also not loose, and tighten them if necessary.
Also, in the same way, check for any looseness and tighten the harness adapters and cable clamps shown in Fig. 13.9.

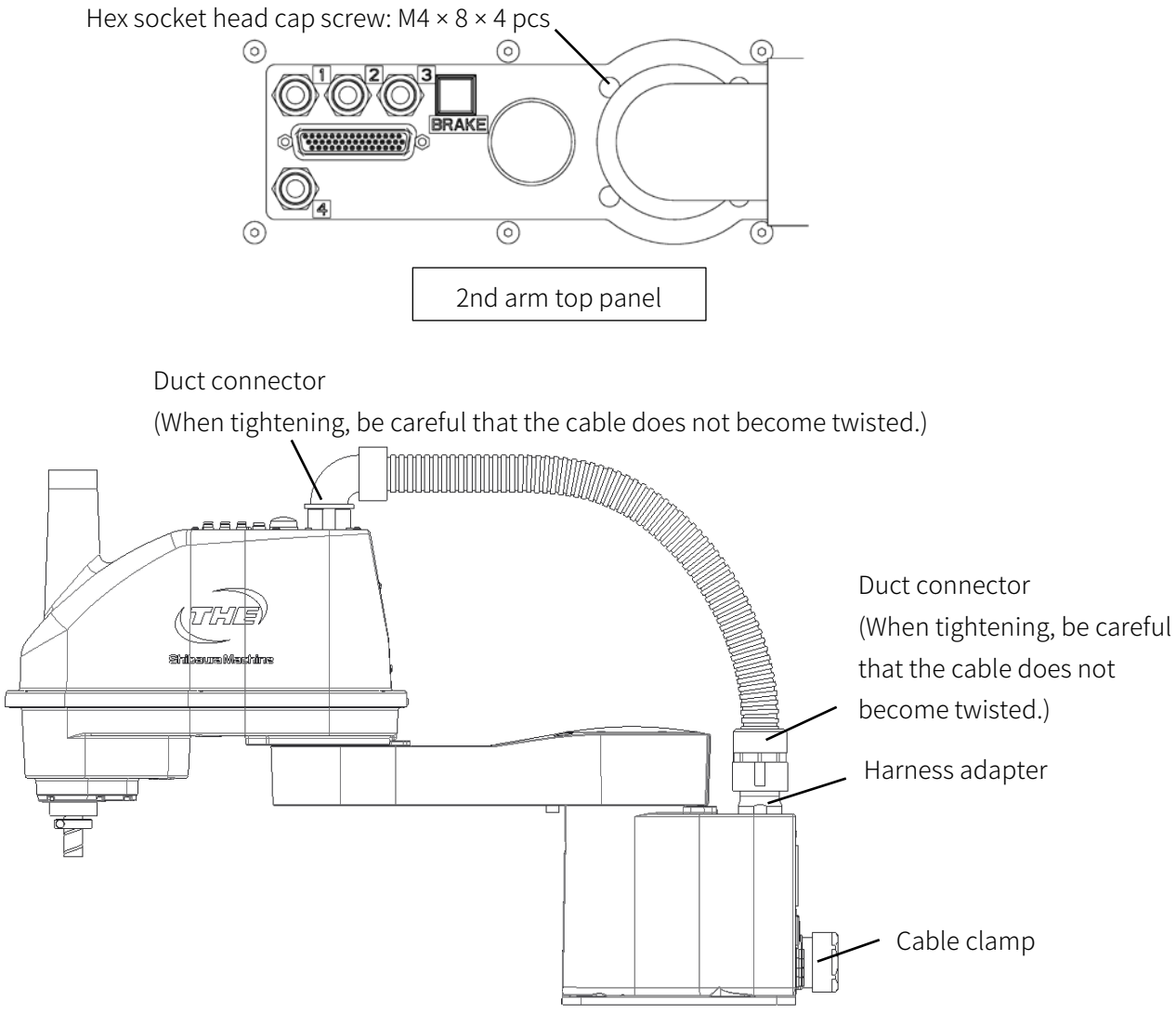


Fig. 13.9 Inspection spot of cable clamp

13.1.4 Check of Each Joint Axis Movement

After connecting the power plug of the controller to the power supply, keep the emergency stop switch pressed. Check that each joint axis moves smoothly when moved by hand. The 3rd axis enters a non-braking state when the brake release switch is pressed. Note that the tool shaft will drop due to the weight of the robot hand.

⚠ WARNING

- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

13.2 Maintenance of ball screw spline unit

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. The SHIBAURA MACHINE Service Department will perform replacement work. A fee is charged for this service.

Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

13.2.1 Location of the Ball Screw Spline Unit

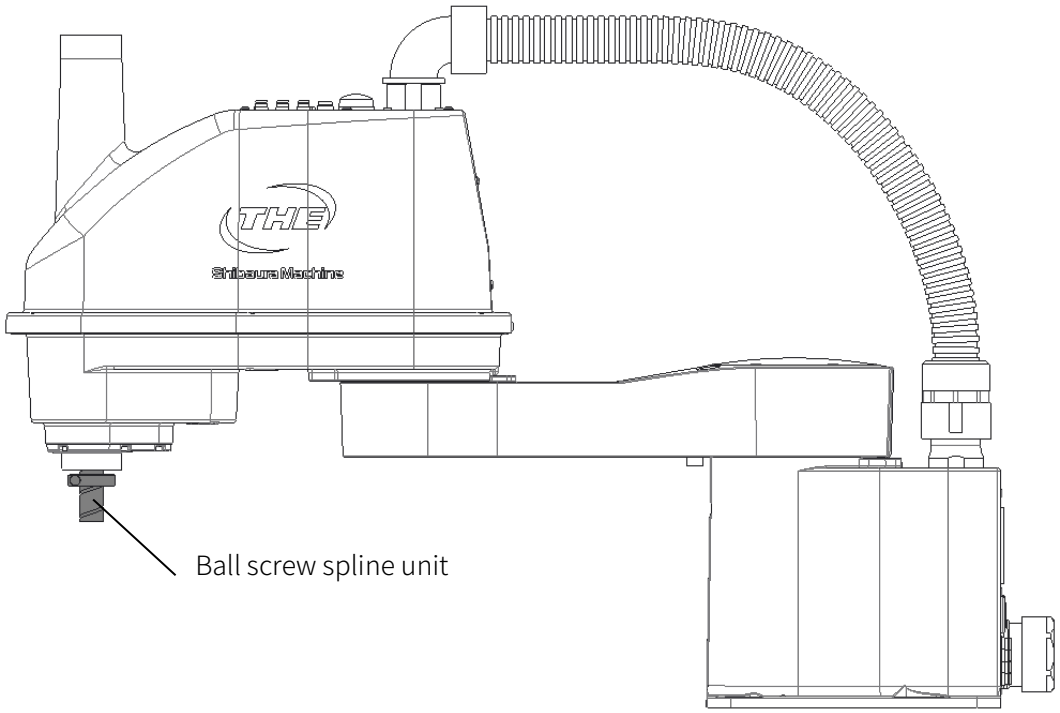


Fig. 13.10 Location of the ball screw spline unit

13.2.2 Greasing the Ball Screw Spline Unit and Applying Corrosion Inhibitor

⚠ WARNING

- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

NOTICE

- Never touch the ball screw spline unit with your bare hands.
  Touching it with bare hands causes the ball screw spline to rust.
  - Be sure to always wear gloves.
- When greasing and applying corrosion inhibitor, be sure to cover the peripheral equipment.
  Grease and corrosion inhibitor could drip on the peripheral equipment.
  - Wipe off any excess grease and corrosion inhibitor with a waste cloth.
- Be sure to check the grease application state daily.
  Insufficient grease or rust formation could cause scratches on the sliding surface, resulting in reduced performance and lifespan of the robot. The robot will malfunction.
- Be sure to use the specified types of grease.
  Otherwise, the performance and lifespan of the robot could be reduced. The robot will malfunction.

MEMO

- *Generally, the ball screw spline unit should be greased every three months, but grease should also be applied whenever it is used up.*
- *Be sure to also check the grease of the ball screw spline unit during daily inspection.*

The state of corrosion inhibitor should also be checked during daily inspection, and if corrosion inhibitor should be applied if it is used up.

Table 13.1 Recommended corrosion inhibitors

Recommended corrosion inhibitor	Maker
Kluber A20	NOK Kluber
WD-40	ST Trading

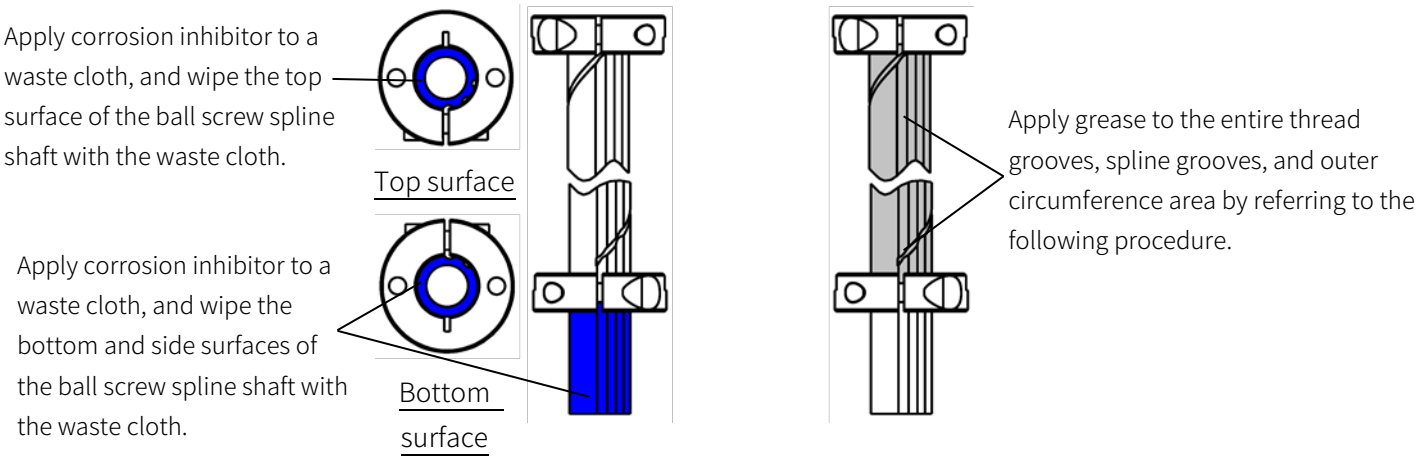


Fig. 13.11 Parts of the ball screw spline unit to which corrosion inhibitor or grease is applied

1 Connect the power plug of the controller, and turn off the servo.

2 Move the robot arm by hand to a position where the 3rd axis can be moved for its full stroke length.

3 Press down on the ball screw spline shaft to the lower limit while pressing the 3rd axis brake release switch.

4 Apply grease to the visible part of the ball screw spline shaft.

Apply grease with a brush or similar implement to the visible part of the shaft. Apply it so that shaft grooves are filled with grease. For the part to apply grease to, see Fig. 13.12.

Table 13.2 Recommended grease

Recommended grease	Maker
AFF grease	THK

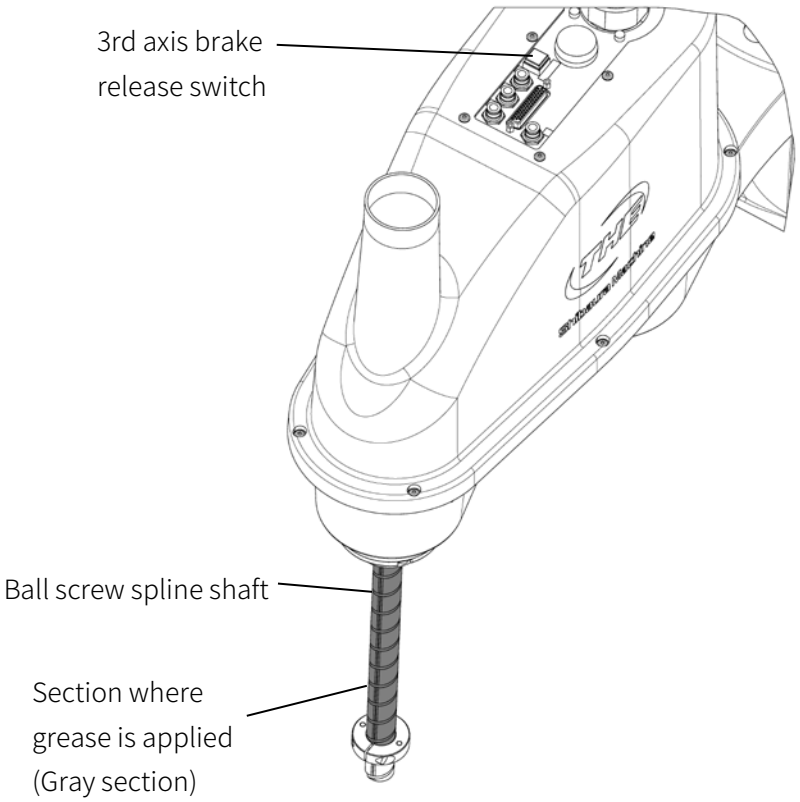


Fig. 13.12 Greasing up the lower side of the ball screw spline unit

⚠ WARNING

- Never touch the grease.
  If it gets on the skin or eyes, it can cause inflammation.
  - Wear protective gloves when applying grease.
  - If it gets on your skin, rinse it off immediately with large amounts of water and soap.
  - If it gets in your eyes, rinse it off immediately with large amounts of water, and then seek immediately medical attention.

5 Smooth out the grease.

Move the shaft up and down several times while pressing the 3rd axis brake release switch to smooth out the grease so that there is no unevenness. Then, wipe off any excess grease to complete the work.

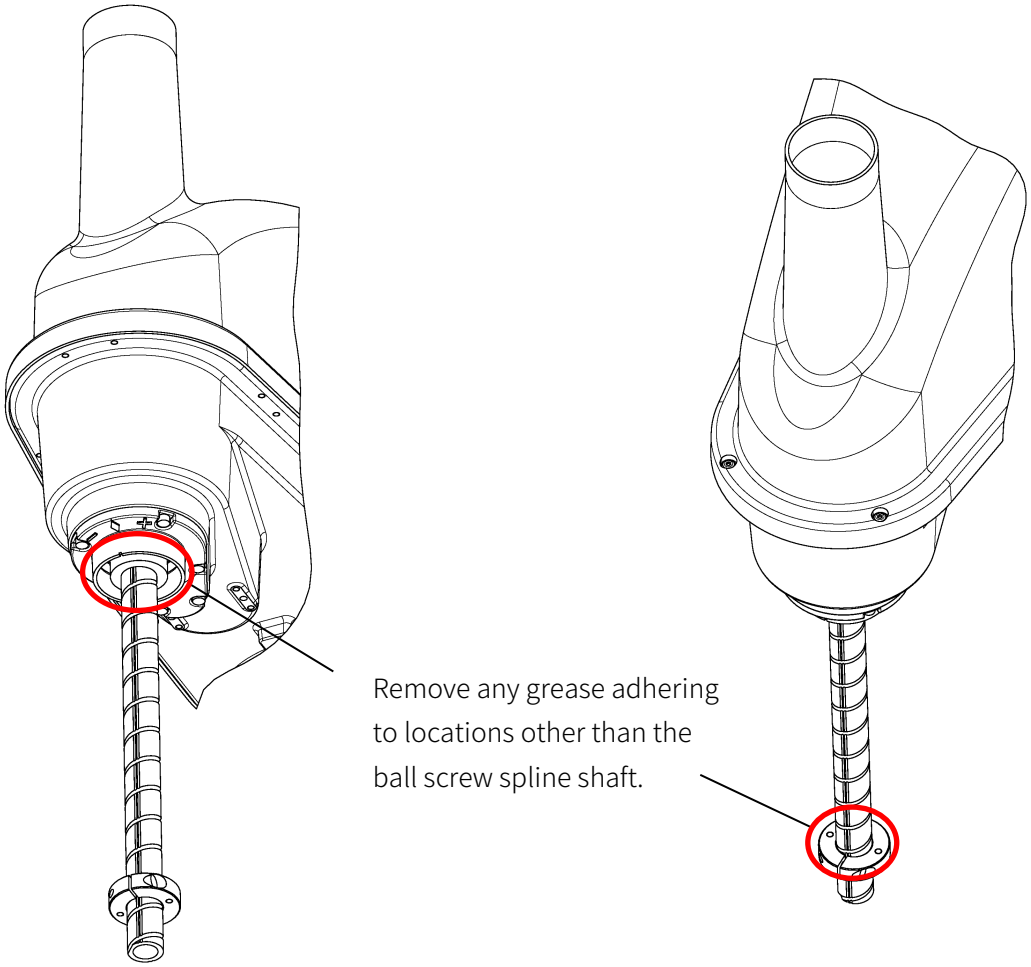


Fig. 13.13 Wiping off grease

6 Perform operation at a low speed.

Perform operation for about four hours at an override of 20% to check that there are no problems with ball screw spline operation.

14. Robot Origin Position and Position Detector Error

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.

Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO

Before starting work, check 10.1 Rescuing a Person Trapped by a Robot, and conduct training.

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.
- Before turning on the servo for the robot, always make sure that there is no one inside the robot's safety protection zone.
  Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, carry a teach pendant set to MANUAL mode, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.
- Be sure to perform maintenance and inspection periodically.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the robot hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

14.1 Setting the Robot Origin Position

At the time of shipment from the factory, the origin position of the robot is set after the arm is fixed in place by the origin position setting jig. Because the position data of the position detector at the time of setting the home position is backed up by the position detector backup batteries, the coordinates of the robot do not need to be set every time the power is turned on.
The position data of the position detector includes two (2) types: servo offset data and multi-turn data.

Servo offset: When the robot origin is set (by the ZEROP operation and REORG operation), the data is overwritten in the robot internal motor.

Multi-turn data: The data is retained in the position detector by backup by the robot-side position detector backup batteries. If the position detector backup battery voltage drops to a certain level, the correct multi-turn data can no longer be guaranteed, and so the position detector itself generates an alarm. This data is set to “0” when the robot origin setting (ZEROP operation and REORG operation) is performed. The data is also set to “0” when the multi-turn data initialization operation is performed.

14.2 Position Detector Error

If a position detector error (encoder alarm) occurs, the alarms shown in the table below appear on the alarm screen of the controller.

Alarm code	Alarm message
8-601	1st joint axis: Multi-turn count error
8-602	2nd joint axis: Multi-turn count error
8-603	3rd joint axis: Multi-turn count error
8-604	4th joint axis: Multi-turn count error

The position detector error is an alarm detected by the position detector itself.
If the machine coordinates were changed without triggering a position detector error, this may have caused a shift in the mechanical connecting position (e.g.: tooth skip in timing belt).
Alarms detected by the position detector itself include a drop in the position detector backup battery voltage, error caused by temperature rise in the position detector, counter overflow, and internal counter data inconsistency. Among these alarms, the alarm which occurs most frequently is a voltage drop in the position detector backup batteries, which is caused by a cable breakage or by the lack of maintenance at specific replacement intervals due to a long-term shutdown. Therefore, to prevent the position detector error, the position detector backup batteries must be replaced on a regular basis. For details on replacing the position detector backup batteries, see "14.3 Replacing the Position Detector Backup Batteries." Also, when the power is turned off, the robot should not be subjected to strong vibrations, and the robot should not be moved suddenly by hand. When transferring and storing the robot, in particular, be sure to secure the robot in an orientation for shipment using the supplied fixing jigs.

Recovering from a position detector error

The following three methods are available to recover from a position detector error. Confirm on the alarm screen that a position detector error has occurred, and check the type of the position detector error and position data on the encoder status screen. Then, perform the recovery operation. For details on the encoder status screen, see "14.5 Encoder Status Screen."

① Multi-turn data initialization operation: See Para. 14.6.
② ZEROP operation using the origin setting jig: See Para. 14.7.
③ Other origin setting operations: See Para. 14.8.

[1] When origin does not need to be set:

- When the position detector backup batteries were replaced in the 1-year inspection or when the battery alarm code is a level 1 alarm (see 14.3.3 Battery Alarm Codes)

The alarm must be reset after replacing the position detector backup batteries. The alarm is reset on the currently occurring alarm screen displayed when the alarm button is tapped. The alarm can be reset by tapping the alarm reset button on the current alarm screen. Then, perform the procedure in "14.9 Check Work."

If the alarm is not reset, contact the SHIBAURA MACHINE Service Department.

[2] When the ① multi-turn data initialization operation is required

- When the position detector backup batteries were replaced when the battery alarm code was a level 8 alarm (see 14.3.3 Battery Alarm Codes)

After the work is completed, perform the ① multi-turn data initialization operation (see 14.6 ① Restoring Origin Position Data by Multi-turn Data Initialization).

[3] When ② ZEROP operation using the origin setting jig or when ③ other origin setting operations are required

- When the motor, reduction gear, timing belt, ball screw spine, or other part was replaced, and the mechanical connection was changed

If the motor or other part needs to be replaced, contact the SHIBAURA MACHINE Service Department.

For reference, the origin setting operation that must be performed after replacement is presented for the ② ZEROP operation using the origin setting jig (see 14.7② Restoring Origin Position Data by ZEROP Function Using the Origin Position Setting Jig) and ③ other origin setting operations (see 14.8③ Other Origin Position Settings).

NOTICE

- If the origin position is set again using the ZEROP function or REORG function, be sure to perform teaching of the teaching points again.
  The origin position shifts slightly due to the error in setting. If the origin position is used without setting it again, the robot could collide with the peripheral equipment.

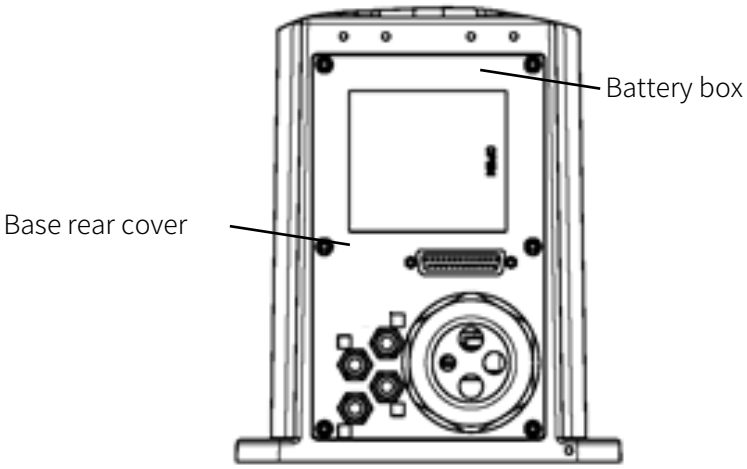
14.3 Replacing the Position Detector Backup Batteries

NOTICE

- Be sure to always replace the position detector batteries during the annual inspection.
  If the position detector data is lost, the robot can no longer be operated.
  - If the position detector backup battery voltage drops to a certain level, a “Battery alarm” will occur. At this stage, data is not lost yet. If the position detector backup batteries are replaced, the alarm is automatically cleared.
  - If the battery voltage drops even further, a “battery alarm” will occur. At this stage, the data is lost. A position detector error alarm will also occur.
  - If the data is lost, the origin position must be set again. For details on how to set the origin position, see “14 Robot Origin Position and Position Detector Error.”
- When the robot is not used while the power turned off, be sure to replace the position detector batteries every two months.
  Because the position detector backup batteries use a large amount of power while the power is turned off, the position detector data will be lost before the periodic inspection is conducted.

14.3.1 Location of Battery Box

The battery box is mounted to the base rear cover on the base rear.



Batteries used: AA cell batteries x 3

Fig. 14.1 Battery box location

14.3.2 Replacing the Batteries

⚠ CAUTION

- Be sure to always check that the polarity of the position detector backup batteries is correct.
  If the polarity is wrong, electrolyte could leak from the position detector backup batteries.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

1 If safety can be ensured, keep the power on and press the emergency stop button (move the emergency stop switch to the ON position) to place the robot in stop state. If safety cannot be ensured, turn off the power.

2 Open the cover of the battery box, and take out the position detector backup batteries.

3 Insert three new position detector backup batteries (AA size alkaline cells) while paying careful attention to the polarity.

MEMO

If you replace the batteries with the power turned off, the replacement process must be completed within 5 minutes.

4 Reset the alarm.

If the alarm "1-40* Joint Axis* Encoder Battery low" shown in Table 14.1 is still displayed, contact the SHIBAURA MACHINE Service Department.

5 Remove the battery box cover.

As shown in the following photo, place the terminals of a tester on the batteries to measure the voltage.

6 Check that the voltage is about 4.5 V.

If the voltage is low, the batteries may not have fit properly.

7 If the voltage is low, remove the batteries, wipe off the metal battery contact of the battery box with a waste cloth, and then insert the batteries again.

Measure the voltage again using a tester, and if the voltage is about 4.5 V, take step 4 again.

Terminals of tester



Fig. 14.2 Battery voltage measurement

14.3.3 Battery Alarm Codes

When a position detector error, including a “battery alarm,” has occurred, the following alarm code is shown on the currently occurring alarm screen.

If the voltage of the position detector backup batteries is low, one of the following alarm codes will appear on the current alarm screen. If one of the following alarm codes is displayed in the alarm history when a position detector error has occurred, it indicates that a “battery alarm” has occurred.

Table 14.1 1-40* Battery alarm codes

Alarm code	Alarm message
1-401	1st joint axis: Encoder battery low
1-402	2nd joint axis: Encoder battery low
1-403	3rd joint axis: Encoder battery low
1-404	4th joint axis: Encoder battery low

If a level 1 alarm shown in Table 14.1 has occurred, replacement of the position detector backup batteries is recommended as soon as possible. If the state in which this alarm appears is left as is, an alarm of level 8 alarm listed in to will occur. Be aware that the information on the origin position of the robot may disappear in some cases.

Table 14.2 8-60* Battery alarm codes

Alarm code	Alarm message
8-601	1st joint axis: Multi-turn count error
8-602	2nd joint axis: Multi-turn count error
8-603	3rd joint axis: Multi-turn count error
8-604	4th joint axis: Multi-turn count error

If an alarm shown in Table 14.2 appears, the counter may skip due to the low power level of the position detector backup batteries. First, replace the position detector backup batteries. If the alarm still appears even after replacing the position detector backup batteries, the position detector may be faulty. In that case, the motor must be replaced. Contact SHIBAURA MACHINE Service Department.

Table 14.3 8-61* Battery alarm codes

Alarm code	Alarm message
8-611	1st joint axis: Encoder battery empty
8-612	2nd joint axis: Encoder battery empty
8-613	3rd joint axis: Encoder battery empty
8-614	4th joint axis: Encoder battery empty

If this alarm is displayed, the voltage of the position detector backup batteries may be below the specified value. Replace the position detector backup batteries. Also, if the voltage of the position detector backup batteries drops below the specified value, the robot origin position information may be lost. If the alarm is not cleared even after changing the position detector backup batteries, set the origin position by referring to "14 Robot Origin Position and Position Detector Error."

Table 14.4 8-40* Battery alarm codes

Alarm code	Alarm message
8-401	1st joint axis: Encoder error
8-402	2nd joint axis: Encoder error
8-403	3rd joint axis: Encoder error
8-404	4th joint axis: Encoder error

Generally, for level 8 battery alarms, the two types of alarms shown in Table 14.2 and Table 14.3 appear, but in certain rare cases, the alarms shown in Table 14.4 may also appear. If any of these alarms appear, replace the position detector backup batteries. Also, if the voltage of the position detector backup batteries drops below the specified value, the robot origin position information may be lost. If the alarm is not cleared even after changing the position detector backup batteries, set the origin position by referring to "14 Robot Origin Position and Position Detector Error."

A battery alarm is a type of position detector error (i.e., encoder alarms), and there are also other causes of the position detector error. The recovery method from the position detector error varies depending on the error that occurred. Consequently, to properly recover from the position detector error, you must carefully read "14 Robot Origin Position and Position Detector Error" and fully understand the position detector errors and recovery operation from the position detector errors.

14.4 Origin Position

This section describes the origin position required for the origin position setting. When moving the 1st, 2nd, and 3rd axes of the robot to the origin position, align the origin position mark for each joint axis by referring to "14.4.1 Locations of Robot Origin Position Marks." The 3rd axis does not have an origin position mark. Move to this position by referring to "14.4.2 3rd Axis Origin Position."
Installation of the origin setting jig (optional) is explained in "14.4.3 Installing the Origin Position Setting Jig (Option)."

14.4.1 Locations of Robot Origin Position Marks

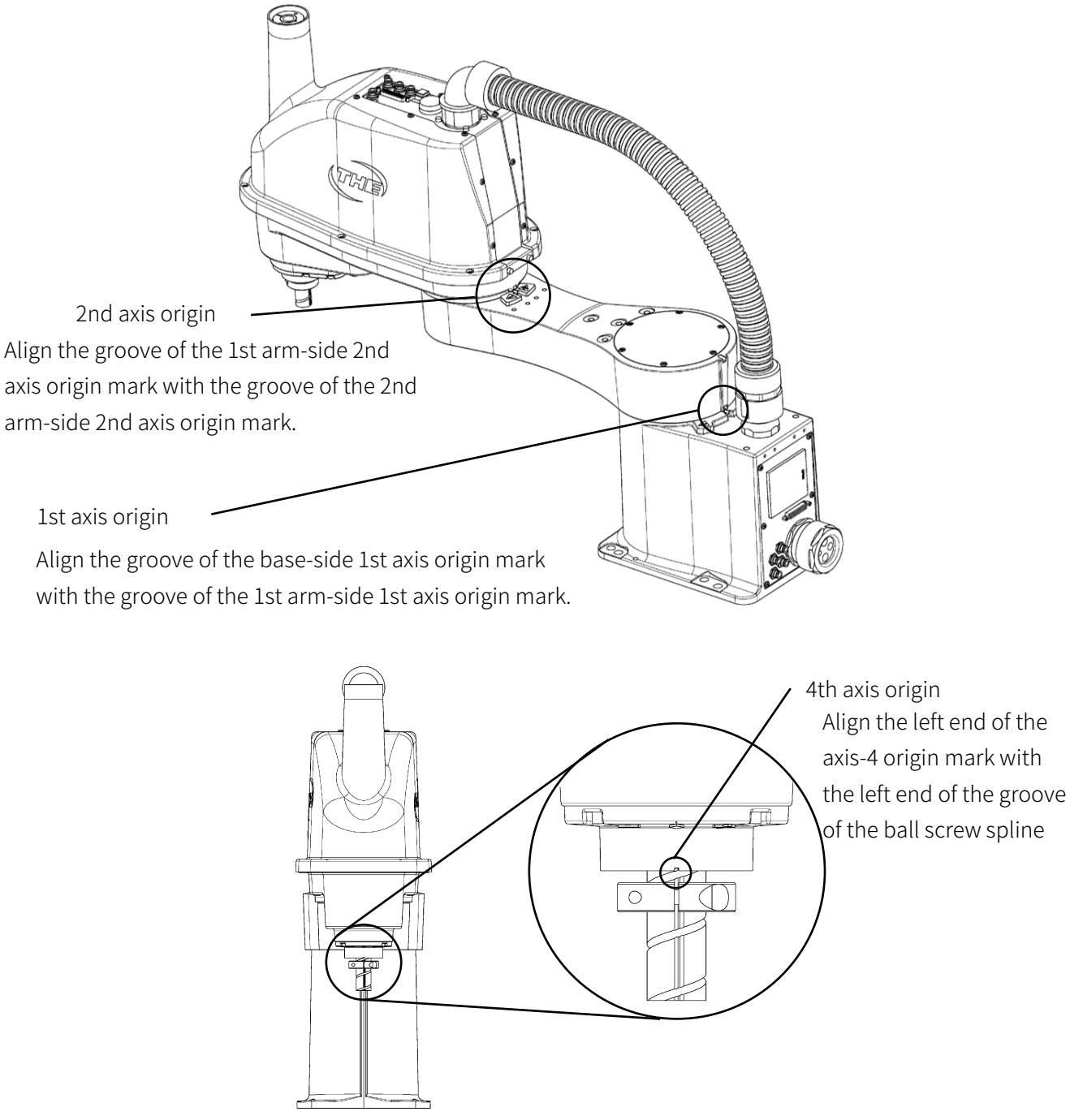


Fig. 14.3 Locations of origin marks

14.4.2 3rd Axis Origin Position

When moving the 3rd axis to the origin position, move the ball screw spine shaft by referring to the dimensions in Fig. 14.4. Also, when measuring the 3rd axis position, set the location shown in Fig. 14.5 as the reference plane.

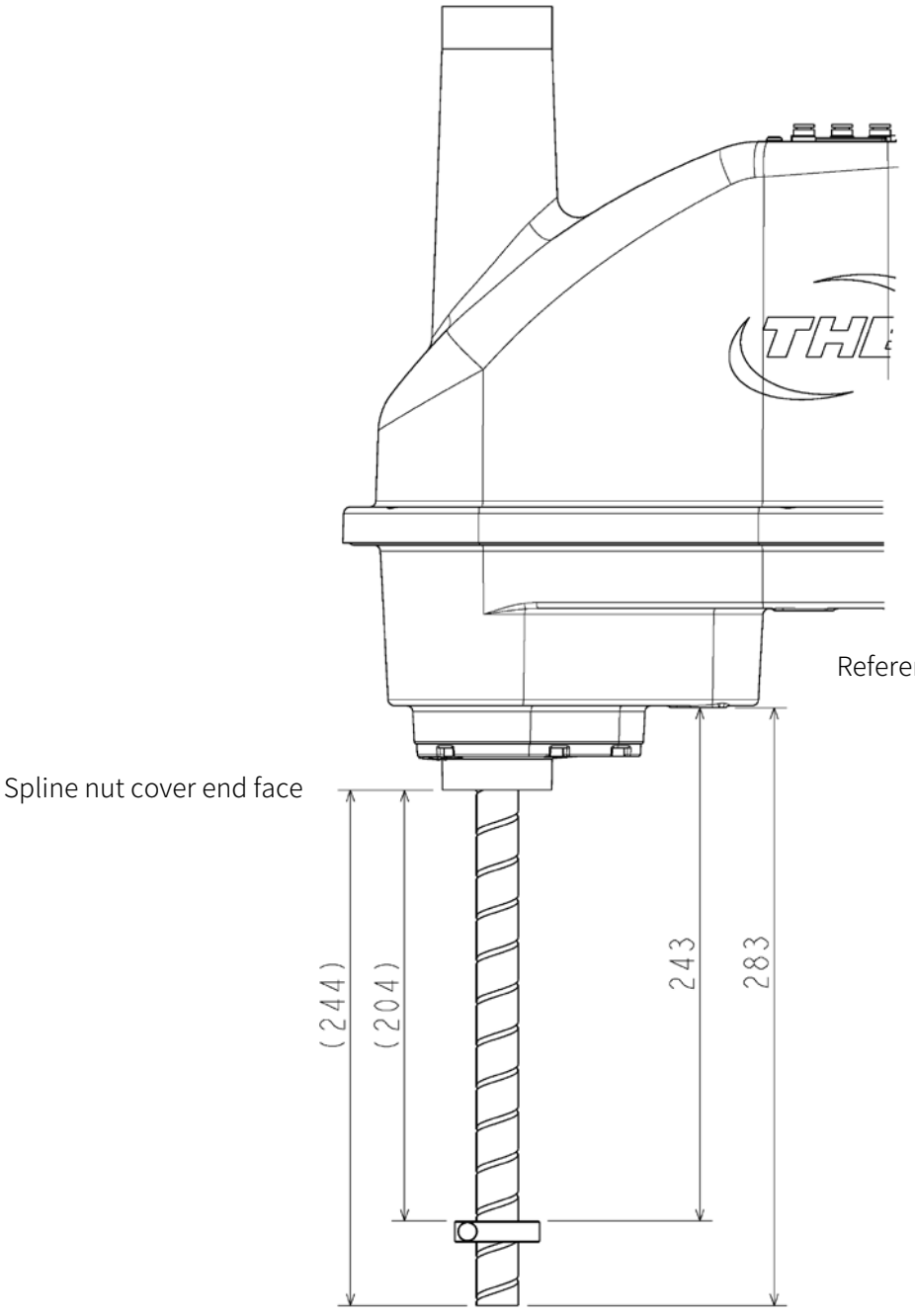


Fig. 14.4 3rd axis origin position dimensions

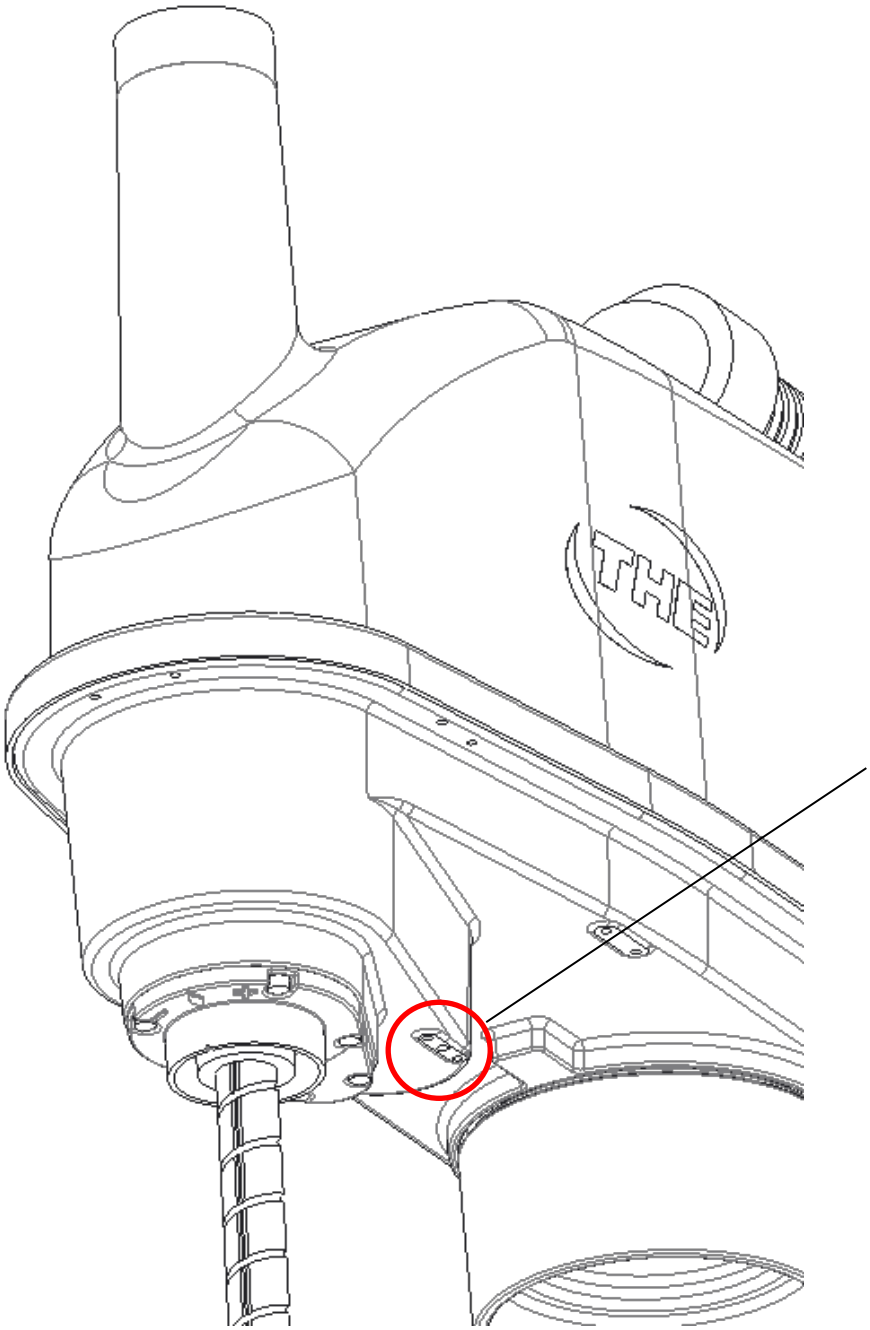


Fig. 14.5 3rd axis origin positioning reference plane

14.4.3 Installing the Origin Position Setting Jig (Option)

This section shows the installation position of the origin position setting jig (option) in "14.7 ② Restoring Origin Position Data by ZEROP Function Using the Origin Position Setting Jig." For details on the procedure for installing and removing the 3rd and 4th axis origin position setting jig, see "14.4.4 Installing and Removing the 3rd and 4th Axis Origin Setting Jig." The origin position setting jig used in this section is shown in Table 14.5. When ordering an origin position setting jig, specify the robot model name, SHIBAURA drawing number, and unit code.

Table 14.5 Types of Origin Position Setting Jigs

Part name	Type	SHIBAURA Dwg. No.
1st and 2nd axis origin position setting jig		M342399
3rd and 4th axis origin position setting jig		L23752G01
Tool flange		L12515G01

The parallel pin and parallel key used by the 3rd and 4th axis origin position setting jig are shown in Table 14.6.

Table 14.6 Types of parallel pin and parallel key

Part name	Type	Size	Material
Parallel pin	Type B	4 × 8	SUS
Parallel key	Round	4 × 4 × 16 (p6)	S45C

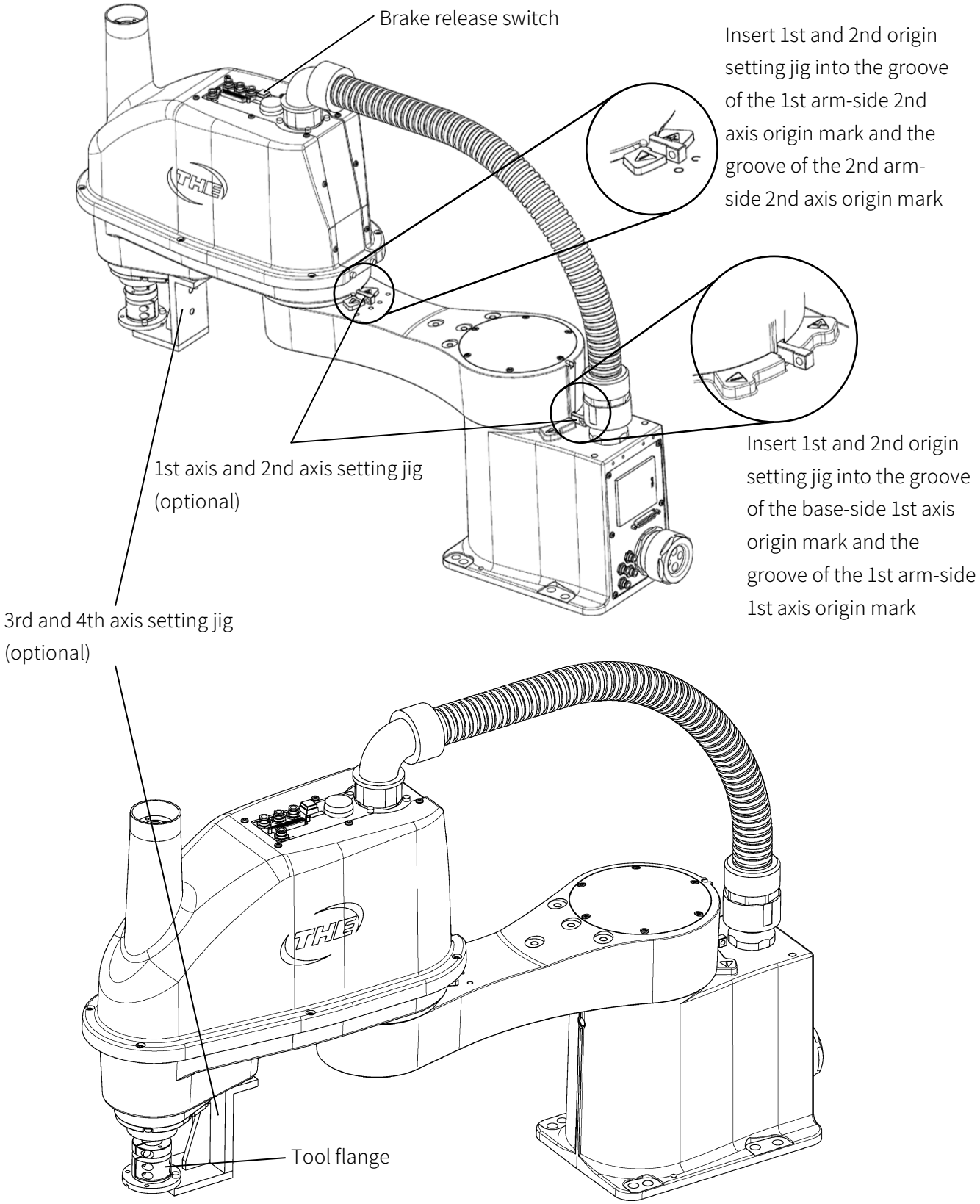


Fig. 14.6 Installing the origin position setting jig (option)

14.4.4 Installing and Removing the 3rd and 4th Axis Origin Setting Jig

1 Mount the tool flange (option).

Fit the tool flange so that the left side of the stopper divider is aligned with the left side of the tool flange divider as shown in Fig. 14.7.

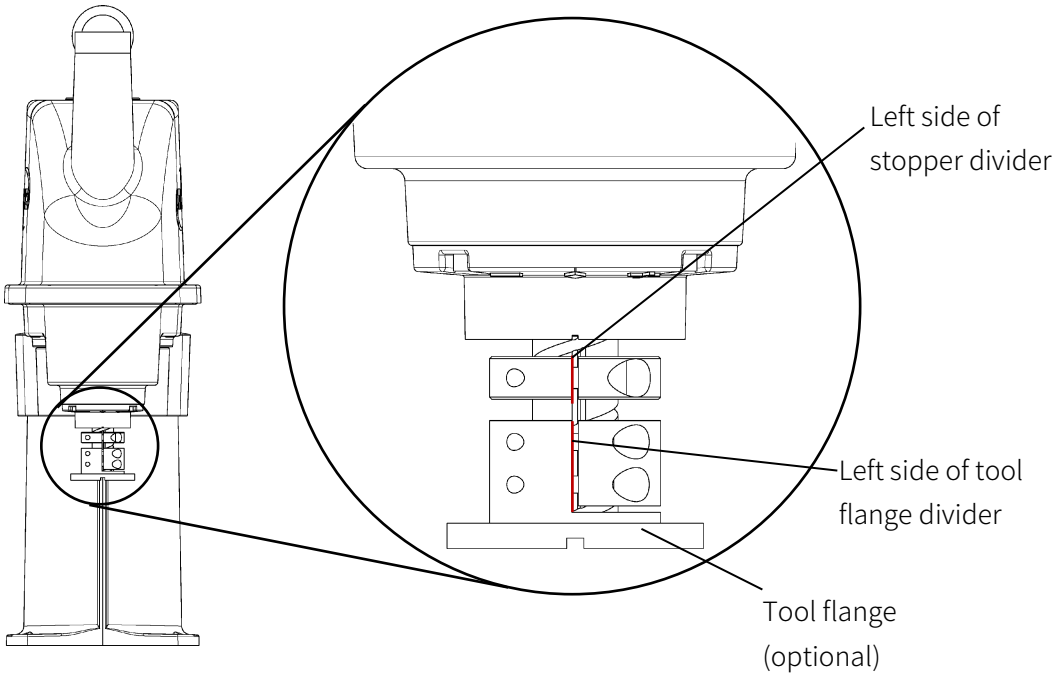


Fig. 14.7 Mounting the tool flange

2 Mount the 3rd axis and 4th axis origin position setting jig into the 2nd arm lower section.

Insert the parallel pins of the 3rd axis and 4th axis origin position setting jig (option) into the pinholes, and secure it in place using the two hex socket head cap screws (M4 × 12).

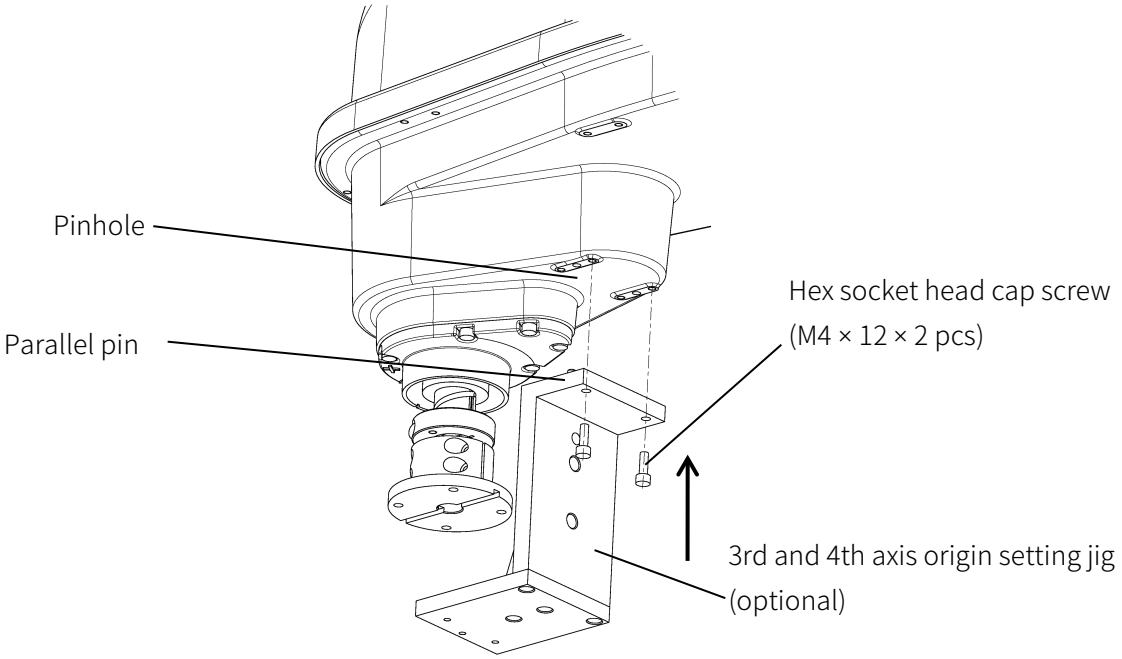


Fig. 14.8 Securing the 3rd and 4th origin position setting jig

3 Lower the tool flange so that the parallel key of the 3rd and 4th axis origin position setting jig fits into the lower key groove of the tool flange.

Press the brake release switch to rotate the ball screw spline and match the orientation.

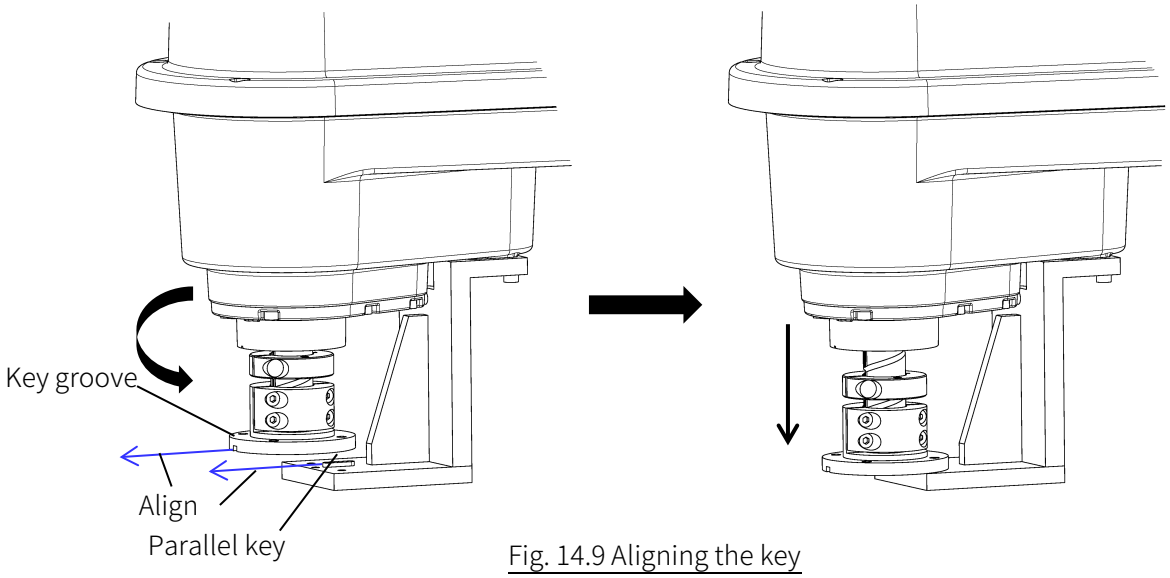


Fig. 14.9 Aligning the key

4 Secure the tool flange to the 3rd axis and 4th axis origin position setting jig.

Use the two hex socket head cap screws (M4 × 12).

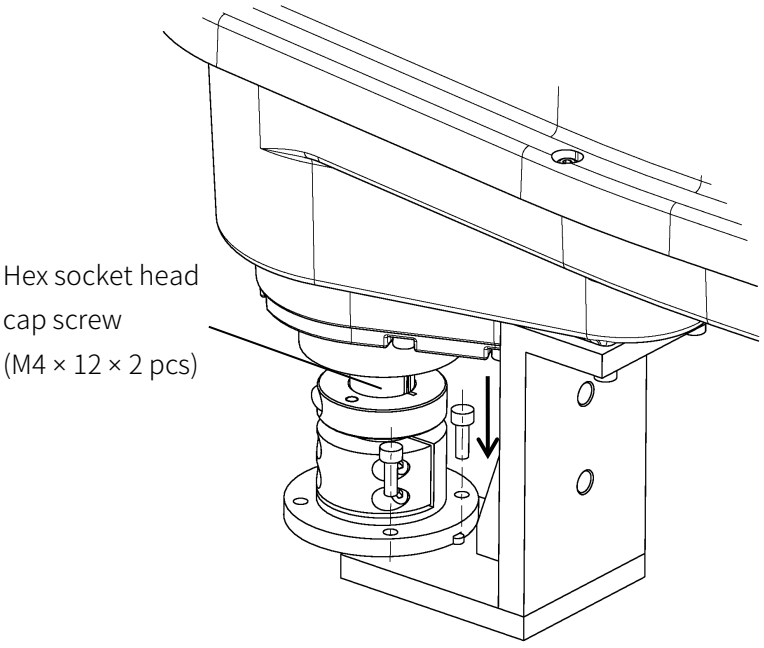


Fig. 14.10 Securing the Tool Flange

5 Set the origin position according to the procedure explained in "14.7 ② Restoring Origin Position Data by ZEROP Function Using the Origin Position Setting Jig ."

6 Remove the 3rd and 4th axis origin position setting jig by performing the above procedure in reverse.

14.5 Encoder Status Screen

Position data and the type of position detector error can be confirmed by changing the teach pendant screen to the encoder information screen. The types of position detector errors are indicated using a bit graph in the status as shown in the following table. The motor multi-turn data, position data in one turn of the motor and the status are displayed under the columns for multi-turn data, position data, and status, respectively.

The procedure for displaying the encoder information screen is shown below.

Table 14.7 Status table

Status	Detail
0000	Normal.
0100	This error occurs when heavy vibrations were applied to the robot or when the robot was moved fast by hand while the power drive cable was disconnected when the power was turned off. The multi-turn data may have shifted.
8000	This error occurs when the voltage of the position detector backup batteries has dropped. This can be recovered to normal by replacing the position detector backup batteries.
4000	The voltage of the position detector backup batteries has dropped even further than the level above. An error may have occurred in the multi-turn data.
C000	The alarm statuses 8000 and 4000 have occurred at the same time.

Opening the encoder information screen

1 Tap the maintenance button on the top screen.

2 Tap the encoder button.

The encoder information screen is opened.

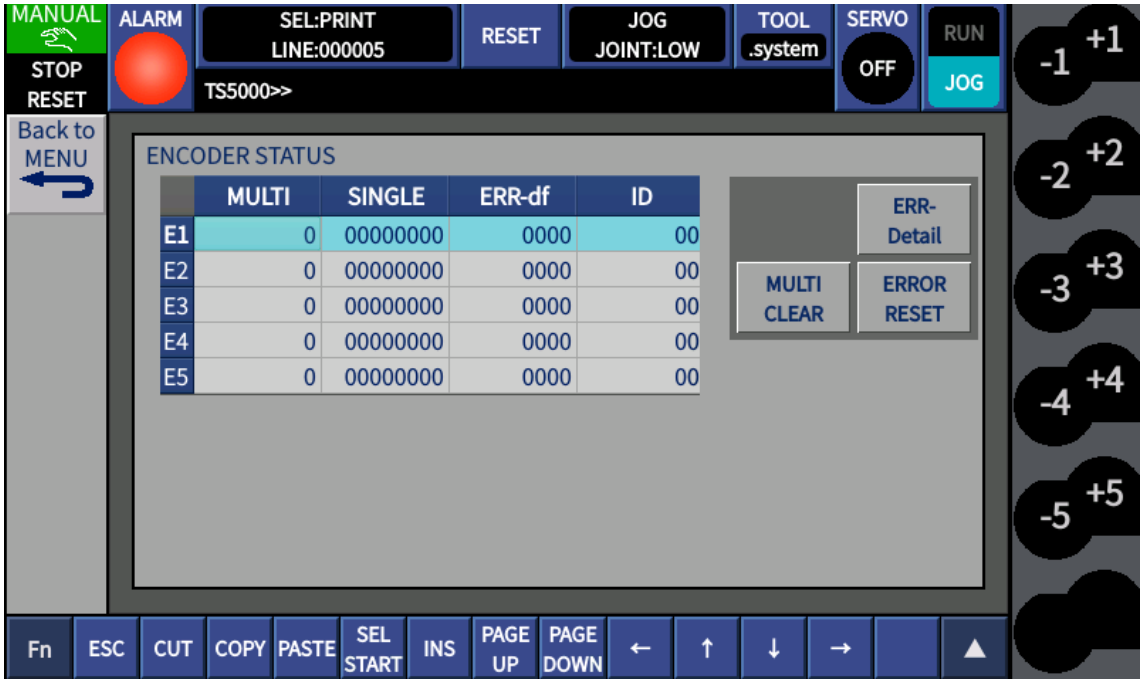


Fig. 14.11 The encoder information screen

If a status listed in the above status table displayed, the system can be restored from the position detector error by the user.

14.6 ① Restoring Origin Position Data by Multi-turn Data Initialization

When the mechanical connection position of the motor and mechanical unit remains unchanged, execute this operation after replacing the position detector backup batteries due to a drop in their voltage or after replacing the cable due to cable breakage, for example.

Multi-turn data initialization procedure

1 See "TP5000 Operation Restrictions," and change the TP5000 user type to "Maintenance."

2 See "14.4 Origin Position," and move the robot to the origin position by hand.

3 Change the master mode to MANUAL.

4 See "14.5 Encoder Status Screen" to display the encoder information screen.

5 Move the cursor to the joint axis where the multi-turn data will be initialized (reset to zero), and tap the multi-turn reset button.

A confirmation message appears.

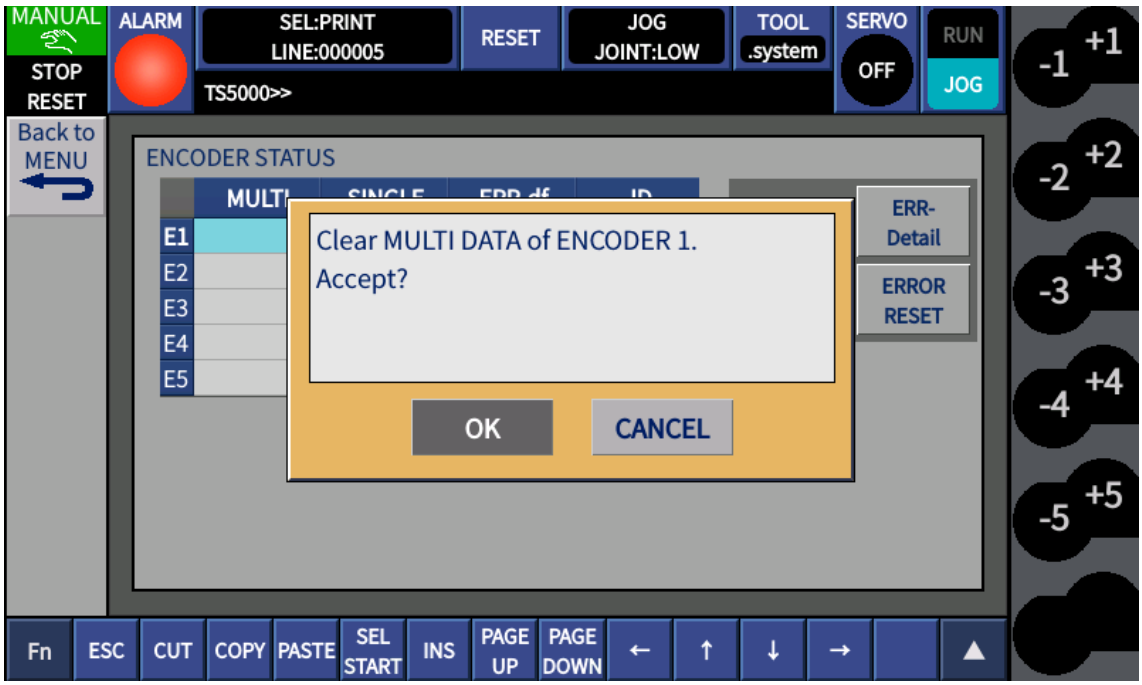


Fig. 14.12 Multi-turn data initialization

6 Check the message, and tap the OK button.

The multi-turn data is initialized (reset to zero).

7 Restart the controller.

8 **Perform "14.9 Check Work ."**

9 **Move the robot to a teaching point or other position to check that the origin position has been set correctly. (In this procedure, the origin position is completely restored to its original state.)**

14.7 ② Restoring Origin Position Data by ZEROP Function Using the Origin Position Setting Jig

If there is a change in the mechanical connection due to motor replacement or other modification that requires the reteaching of teaching points, then perform this procedure.

NOTICE

- If the origin position is set again using the ZEROP function or REORG function, be sure to perform teaching of the teaching points again.
  The origin position shifts slightly due to the error in setting. If the origin position settings in use are used unchanged, the robot will collide with the peripheral equipment or other objects.

14.7.1 Restoring Origin Position Data for All Joint Axes

Procedure for restoring origin position data

1 **See "14.10 TP5000 Operation Restrictions" and change the TP5000 user type to "Maintenance."**

2 **Change master mode to MANUAL mode.**

3 **Mount the origin position setting jig (option) as described in "14.4.3 Installing the Origin Position Setting Jig (Option)."**

4 **Tap the maintenance button on the top screen.**

5 Tap the zero position button.

The zero position screen appears.

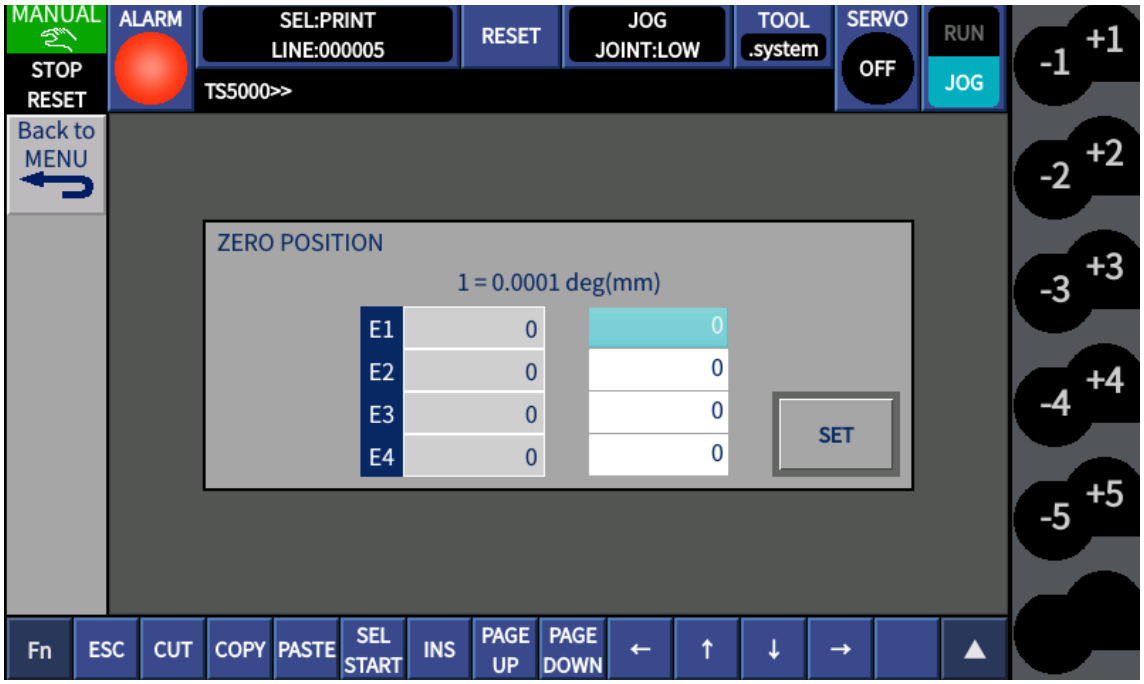


Fig. 14.13 Zero position screen

6 Tap the [↓] key on the keyboard three times or tap the E4 cell.

The selection cursor moves to the E4 (axis 4) cell.

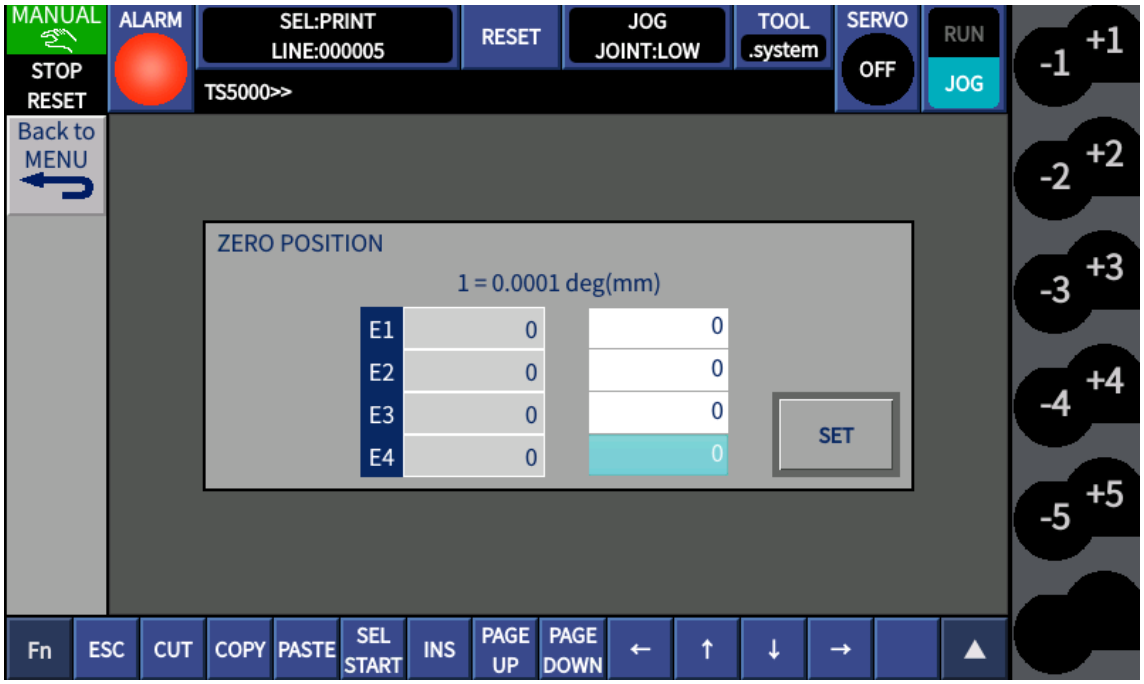


Fig. 14.14 Zero position screen 2

7 Tap the E4 cell.

The cursor is displayed in the cell, indicating that the content can be edited.

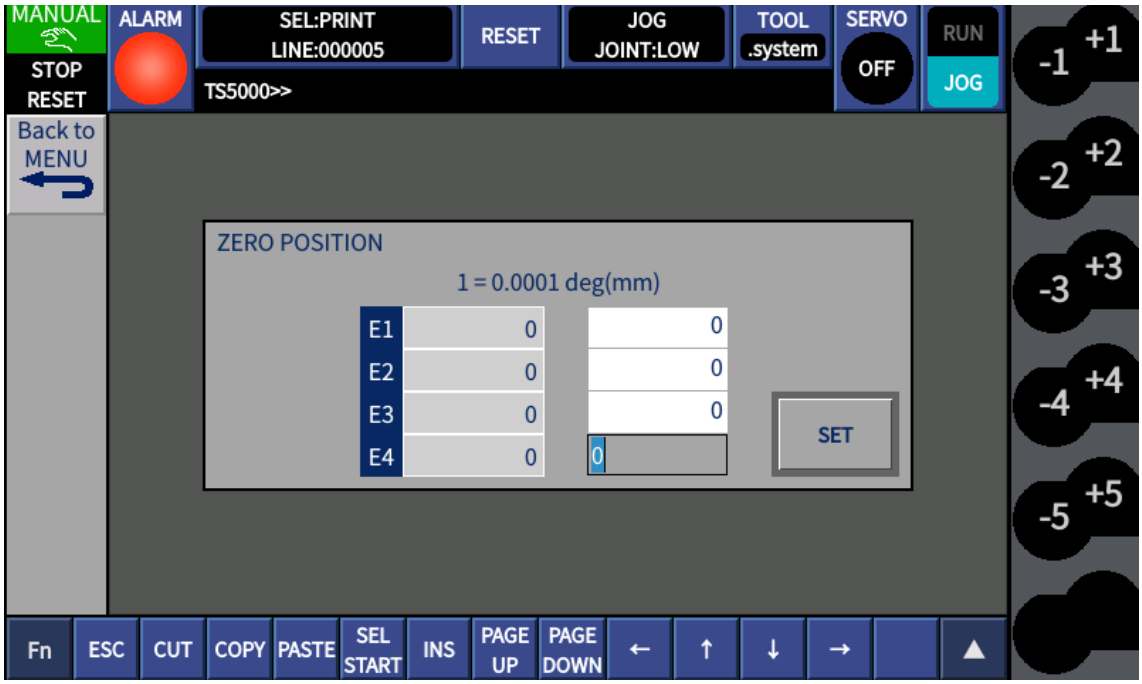


Fig. 14.15 Zero position screen 3

8 Enter “0” and tap the Enter key, or tap a location other than the cell being edited.

9 Tap the setting button.

A confirmation screen appears.

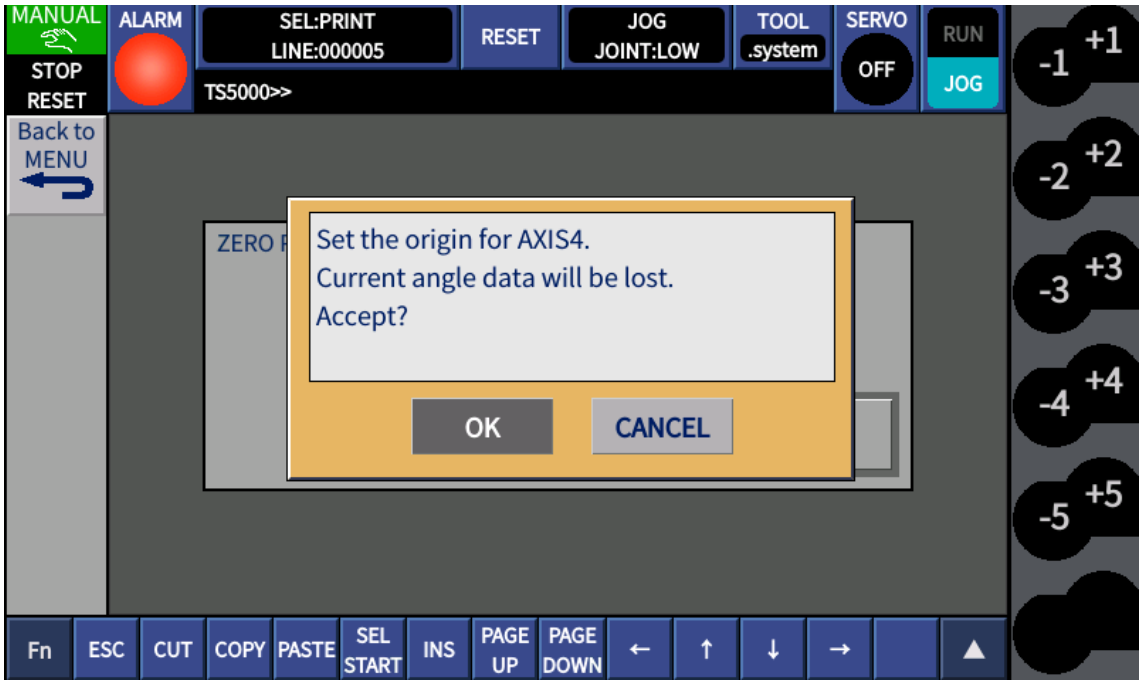


Fig. 14.16 Zero position screen 4

10 Tap the OK button.

The value of E4 becomes 0.

11 Similarly, also perform steps 1 to 9 for E3 to restore the origin position data.

MEMO

For the 3rd axis, set the value in step 7 to 1900000.

12 Similarly, also perform steps 1 to 9 for both E1 and E2 to restore the origin position data.

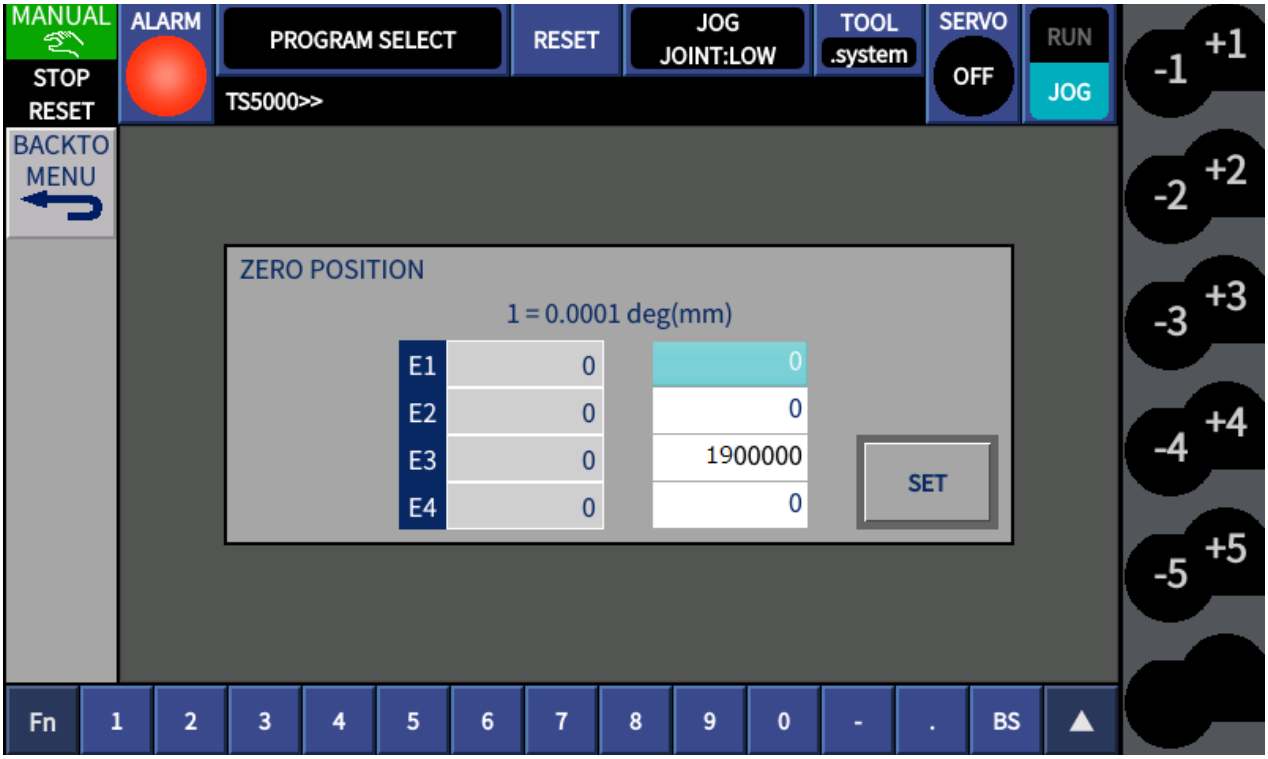


Fig. 14.17 Zero position screen 5

NOTICE

- To set the origin position again using the ZEROP function or REORG function, perform the following procedure.
  Because the 3rd axis operates in tandem with the 4th axis the origin position is not restored if the steps are performed in the wrong order.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter the data, and restore the origin position of the 4th axis.
  3) Move the cursor to E3 on the screen, enter the data, and restore the origin position of the 3rd axis.

- E1 (1st axis) and E2 (2nd axis) do not have any specific order for setting their origin position again.

13 Turn the controller power off and then on again.

14 Remove the origin position setting jig (option) that was mounted in step 3.

15 Perform "14.9 Check Work ."

16 Move the robot to a teaching point or other position to check that the origin position has been set correctly.

14.8 ③ Other Origin Position Settings

14.8.1 Restoring Origin Position Data by HOME Function

14.8.1.1 Overview of HOME Function

This function memorizes a specific coordinate value data in each axis and restores the predetermined origin position data by using the HOME position when the machine origin position data has been destroyed.

The four set points HOME1, HOME2, SYSTEM1 and SYSTEM2 are provided, and the plus (+) side mechanical stopper positions of the 1st axis to 3rd axis are factory-set to SYSTEM1, and the minus (-) side mechanical stopper positions of the 1st axis to 3rd axis are factory-set to SYSTEM2. The 4th axis is set to the 0° position to align with the origin position mark. Because the customer can specify any position for HOME1 and HOME2, it is recommended that a position for securing the robot tip in place be established when building the system. Two patterns can be set: pattern 1 in HOME1 and pattern 2 in HOME2.

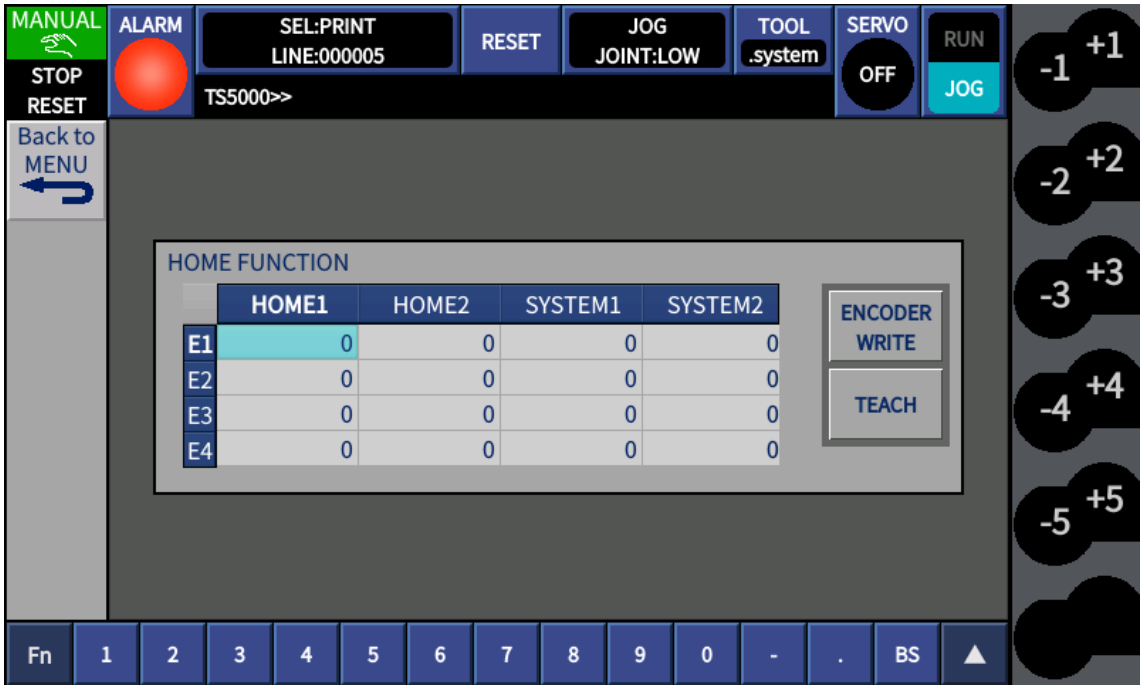


Fig. 14.18 HOME Function screen

MEMO

This screen is used to restore the machine origin position. Never use this screen for other objectives than restoring the origin position.

14.8.1.2 Setting HOME1 and HOME2

NOTICE

- Be sure to enter data to HOME1 and HOME2 for the REORG function when installing the robot.
  Even if the data is entered after the origin position has shifted, it cannot be used as data for restoring the origin position.

1 See "14.10 TP5000 Operation Restrictions," and change the TP5000 user type to "Maintenance."

2 Guide the robot to secure the robot hand at any position.

Fig. 14.19 shows an example of securing the hand using the tool flange (optional).
Fig. 14.20 shows an example of the shape that should be prepared by the customer when the tool flange (option) is secured in place.

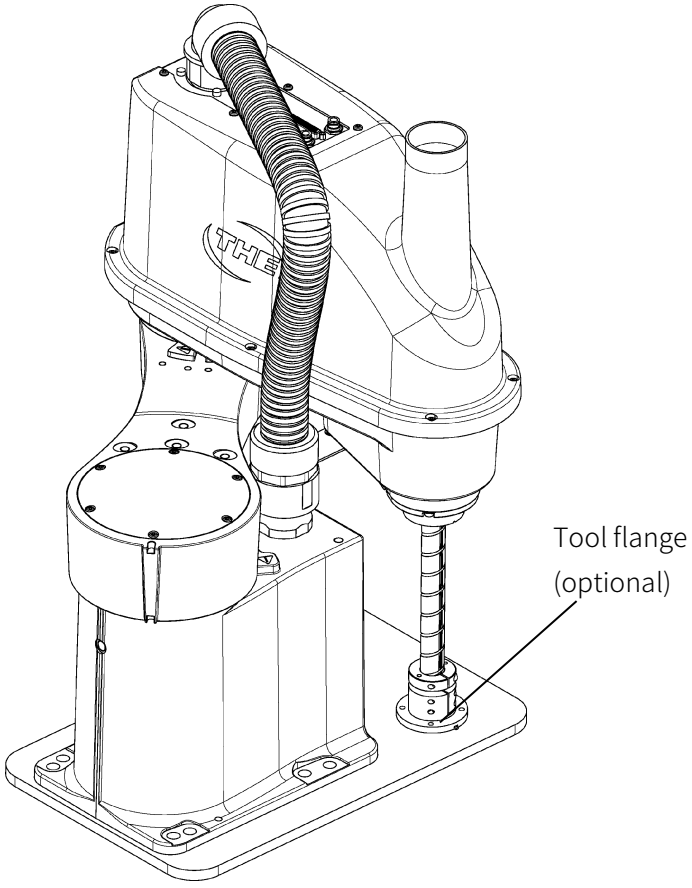


Fig. 14.19 Example of securing the hand

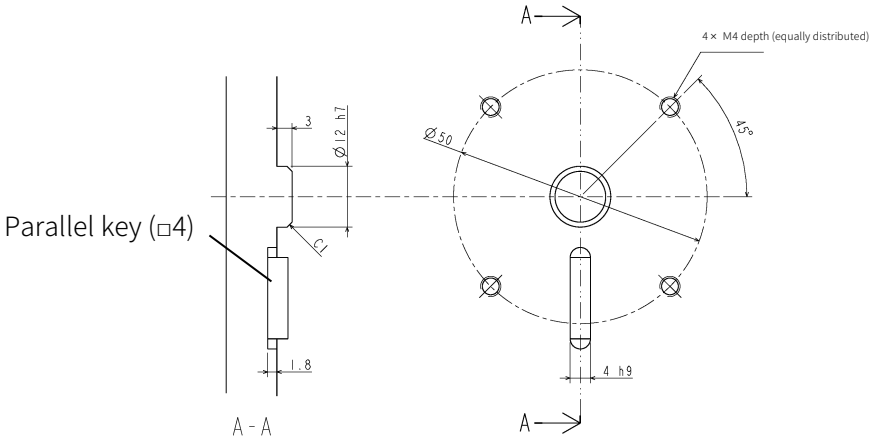


Fig. 14.20 Example of Shape of Securing the Tool Flange (Optional)

3 Turn off the servo power.

4 Tap the maintenance button on the top screen.

5 Tap the origin position restore button.

The origin position restore screen appears.

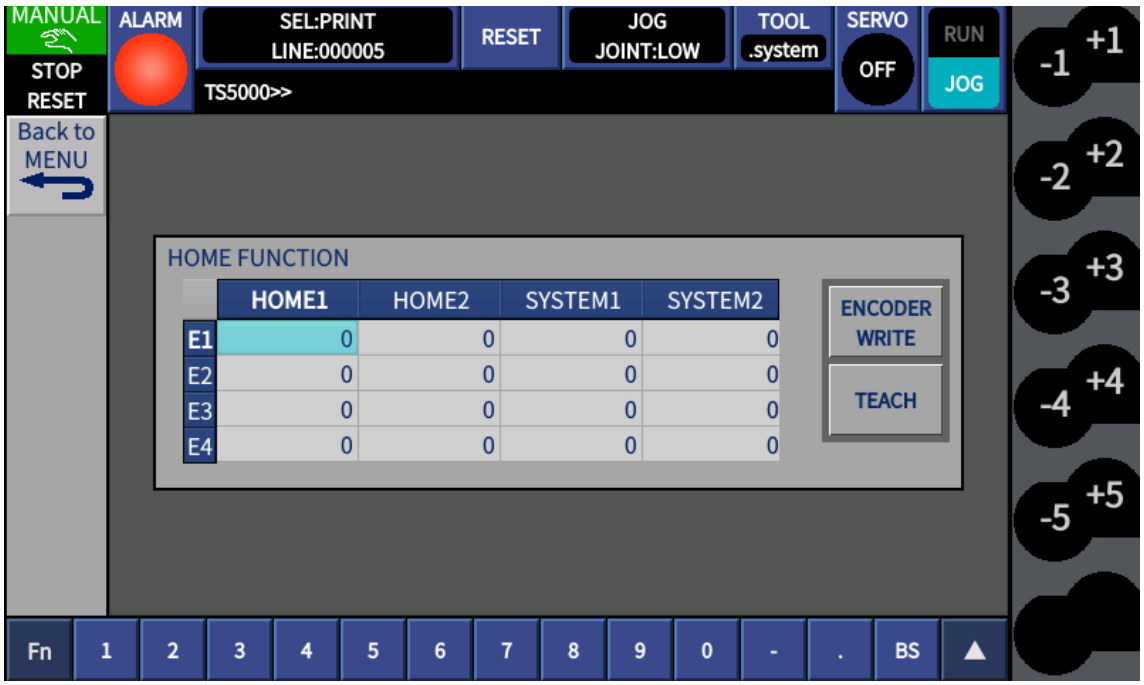


Fig. 14.21 Origin position restore screen

6 Tap the cell corresponding to the joint axis where the origin position will be restored.

The selection cursor moves to the tapped cell.

7 Tap the setting button.

The origin position setting confirmation screen appears.

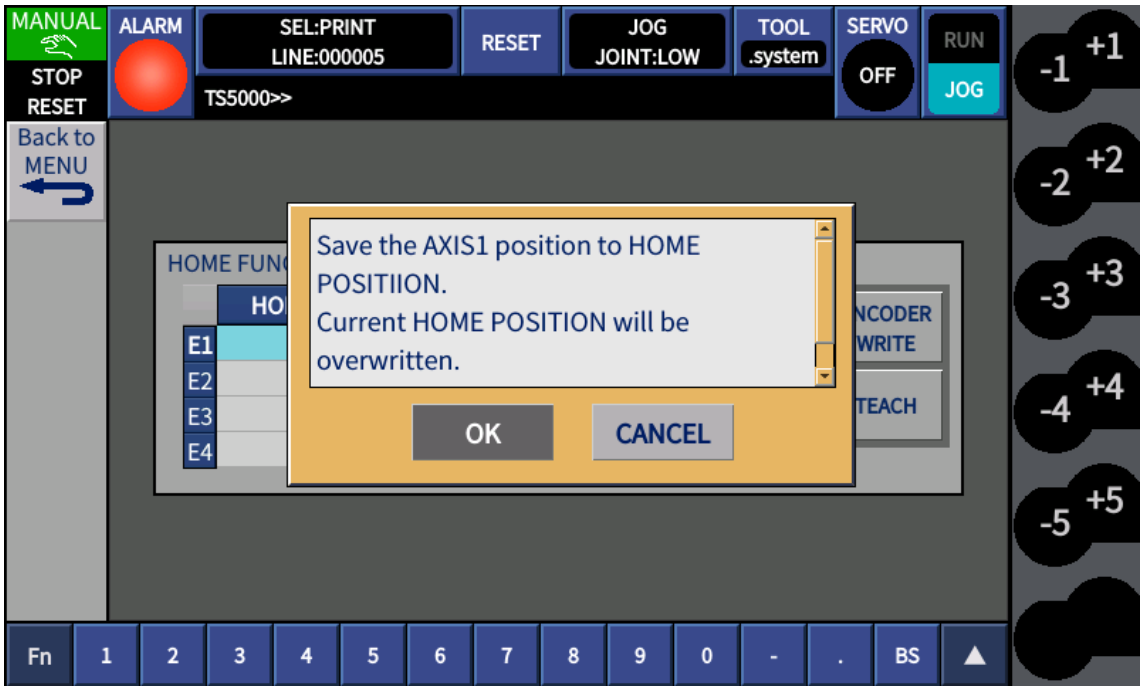


Fig. 14.22 HOME Function screen(2)

8 Tap the OK button.

The value of the cell at the selection cursor is rewritten to the value of the current joint axis.

14.8.1.3 Restoring Data by HOME1 or HOME2

If there is a change in the mechanical connection due to motor replacement or other modification that requires the reteaching of teaching points, then perform this procedure. This section shows the procedure when using HOME1. If HOME2 is used, data is also restored in the same way.

NOTICE

- If the origin position is set again using the ZEROP function or REORG function, be sure to perform teaching of the teaching points again.
  The origin shifts slightly due to the error in setting. If the origin position settings in use are used unchanged, the robot will collide with the peripheral equipment or other objects.

1 See "14.10 TP5000 Operation Restrictions," and change the TP5000 user type to "Maintenance."

2 Guide the robot by hand to the position set in HOME1, and secure it at the position. (See Fig. 14.19)

3 Turn off the servo power.

4 Tap the maintenance button on the top screen.

5 Tap the origin position restore button.

The origin position restore screen appears.

6 Tap the cell corresponding to the joint axis where the origin position will be restored in the HOME1 column. The selection cursor moves to the tapped cell.

7 Tap the origin position restore button.

The origin position restore confirmation screen appears.

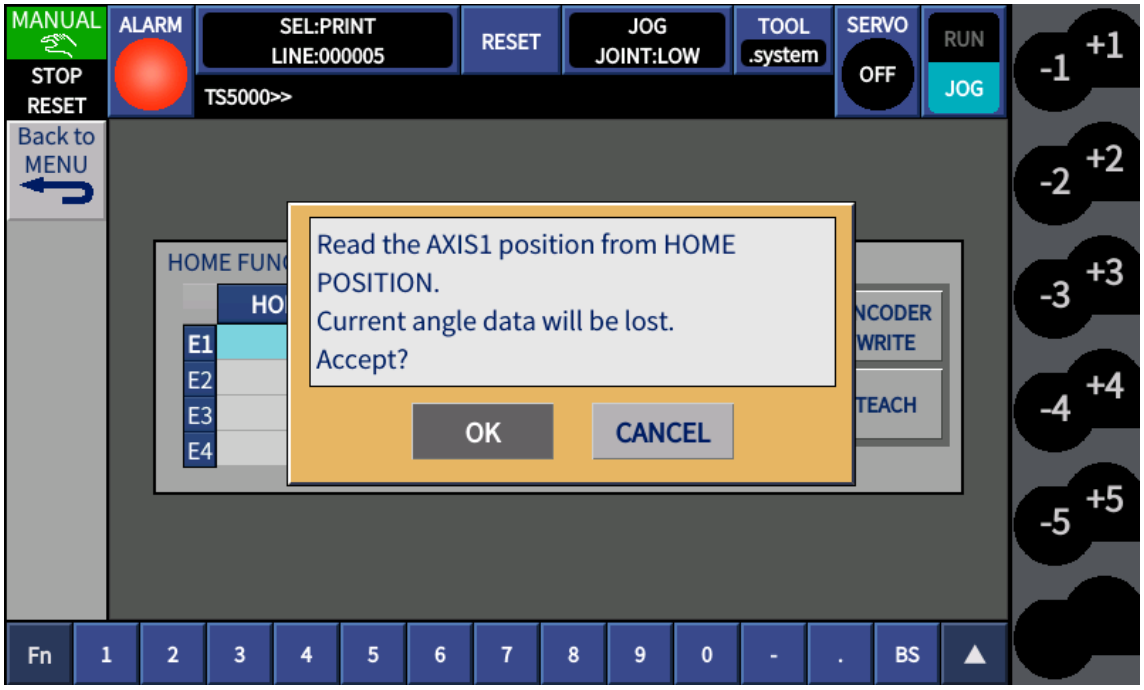


Fig. 14.23 Origin position restore confirmation screen

8 Tap the OK button.

The current position data for the joint axis in the cell at the selection cursor is rewritten to the value of the cell.

MEMO

Even after HOME1 is performed, the robot can be used, but the data format is different from the factory default setting. If you continue to use the robot in this state, the origin position will not be able to be set correctly even by initializing the multi-turn data after the position detector backup batteries run out of power. See "14.8.1.5 Rewriting the Origin Position Data" to rewrite the origin position data.

9 Perform "14.9 Check Work."

10 Move the robot to a teaching point or other position to check that the origin position has been set correctly.

NOTICE

- If re-setting the origin position using the ZEROP function or REORG function, be sure to observe the operation procedure.
  Because the 3rd axis operates in tandem with the 4th axis the origin position is not restored if the steps are performed in the wrong order.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter the data, and restore the origin position of the 4th axis.
  3) Move the cursor to E3 on the screen, enter the data, and restore the origin position of the 3rd axis.
  - E1 (1st axis) and E2 (2nd axis) do not have any specific order for setting their origin position again.

14.8.1.4 Restoring Data by SYSTEM1 or SYSTEM2

If there is a change in the mechanical connection due to motor replacement or other modification that requires the reteaching of teaching points, then perform this procedure. The plus (+) side mechanical stopper positions for the 1st axis to 3rd axis are factory-set to SYSTEM1, and the minus (-) side mechanical stopper positions for the 1st axis to 3rd axis are factory-set to SYSTEM2. The 4th axis at the position of the origin position mark. (Because the 4th axis is set based on the origin position mark, it may be unable to be completely restored to original state.)
This section describes how to restore data using SYSTEM1.

NOTICE

- If the origin position is set again using the ZEROP function or REORG function, be sure to perform teaching of the teaching points again.
  The origin position shifts slightly due to the error in setting. If the origin position settings in use are used unchanged, the robot could interfere with the peripheral equipment or other objects.

1 Turn off the servo power.

2 Tap the maintenance button on the top screen.

3 Tap the origin position restore button.

The origin position restore screen appears.

4 Manually rotate the 1st arm in the plus (+) direction to bring the mechanical stopper into contact with the 1st axis.

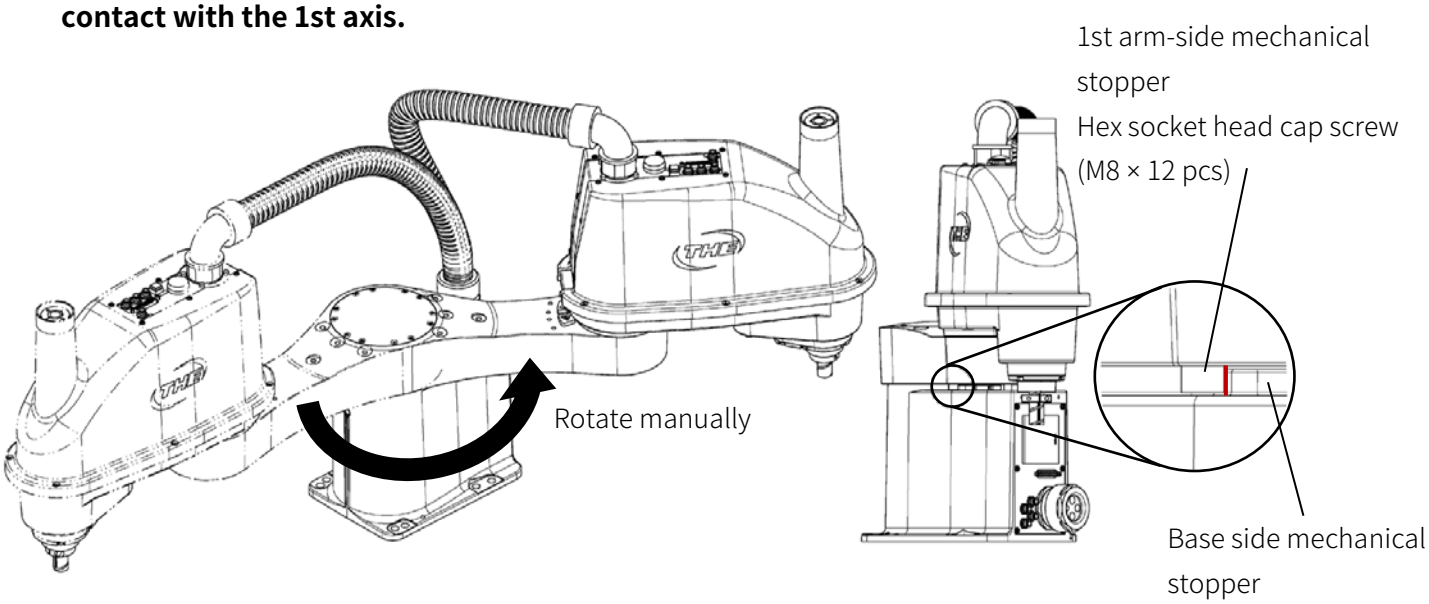


Fig. 14.24 Work to Bring the Mechanical Stopper into Contact with the Axis 1 (SYSTEM1)

5 Tap the (E1) cell of SYSTEM1.

6 Tap the origin position restore button.

The machine origin position of the 1st axis is restored.

7 Similarly, manually rotate the 2nd arm in the plus (+) direction to bring the mechanical stopper into contact with the 2nd axis.

8 Tap the (E2) cell of SYSTEM1.

9 Tap the origin position restore button.

The machine origin position of the 2nd axis is restored.

10 Align the 4th axis with the 4th axis origin position mark while pressing the brake release switch.

(See "14.4.1 Locations of Robot Origin Position Marks.")

11 Tap the (E4) cell of SYSTEM1.

12 Tap the origin position restore button.

The 4th axis is restored.

13 For the 3rd axis, bring the mechanical stopper into contact with the spline nut while pressing the brake release switch.

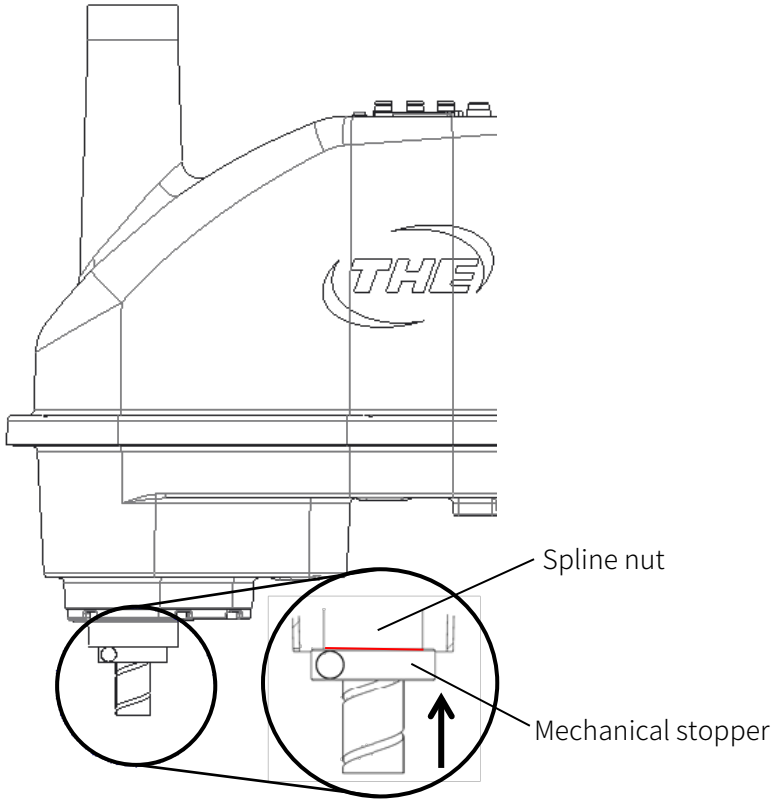


Fig. 14.25 Work to Bring the Mechanical Stopper into Contact with the Axis 3 (SYSTEM1)

14 Tap the (E3) cell of SYSTEM1.

15 Tap the origin position restore button.

The 3rd axis is restored.

16 Return to the maintenance screen.

17 Perform "14.8.1.5 Rewriting the Origin Position Data ."

Even after SYSTEM1 or SYSTEM2 is performed, the robot can be used, but the data format is different from the factory default setting. If you continue to use the robot in this state, the origin position will not be able to be set correctly even by erasing the multi-turn data after the position detector backup batteries run out of power.

18 Perform "14.9 Check Work ."

19 Move the robot to a teaching point or other position to check that the origin position has been set correctly.

14.8.1.5 Rewriting the Origin Position Data

After performing the origin position restore operation, the servo offset and multi-turn data are in formats different from the factory default setting. The robot operates normally even if the data formats are used without performing this operation. However, the origin position will be unable to be set normally by initializing multi-turn data when the position detector backup batteries run out of power. After HOME operation, perform this operation to return origin position data to the factory default setting.

1 Change the master mode to MANUAL mode.

2 Directly execute the MOVEA command to move each joint axis to the “0” position.

During this operation, perform OVRD at a low speed while checking for safety.

For detail on how to execute commands directly, refer to “STEA1162 TP5000 Operation Manual.”

3 Tap the maintenance button on the top screen.

4 Turn off the servo power.

5 Tap the zero position button.

The zero position screen appears.

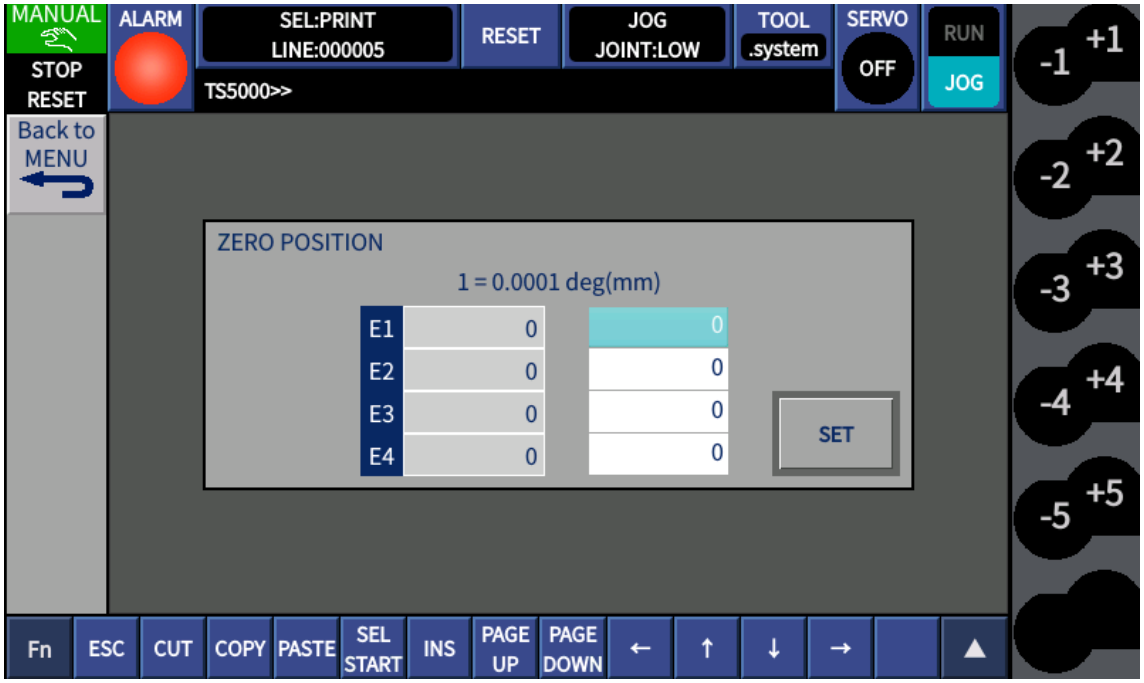


Fig. 14.26 Zero position screen 1

6 Tap the [↓] key on the keyboard three times or tap the E4 cell.

Move the selection cursor to the E4 (axis 4) cell.

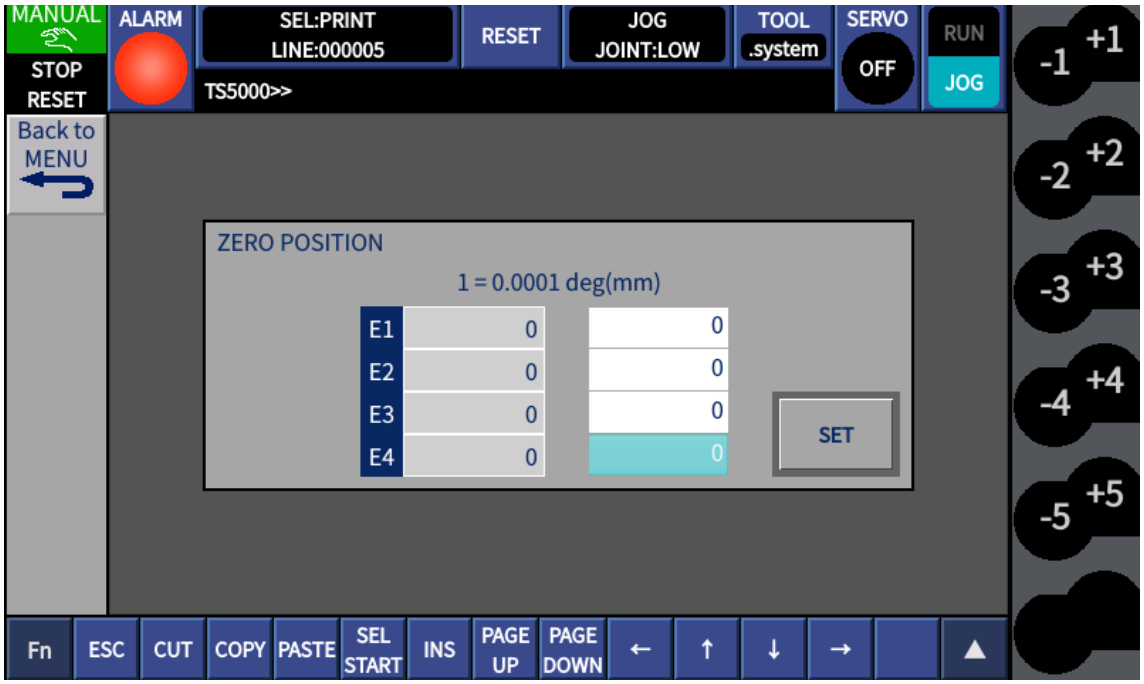


Fig. 14.27 Zero position screen 2

7 Tap the E4 cell.

The cursor is displayed in the cell, indicating that the content can be edited.

8 Enter “0” and tap the [Enter] key or tap the place other than the call being edited.

9 Tap the setting button.

A confirmation screen appears.

10 Tap the OK button.

The value of E4 becomes 0.

11 Similarly, also perform steps 1 to 8 for E3 to rewrite origin position data.

12 Similarly, also perform steps 1 to 8 for E1 and E2 to rewrite origin position data.

NOTICE

- To set the origin position again using the ZEROP function or REORG function, perform the following procedure.
  Because the 3rd axis operates in tandem with the 4th axis the origin position is not restored if the steps are performed in the wrong order.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter the data, and restore the origin position of the 4th axis.
  3) Move the cursor to E3 on the screen, enter the data, and restore the origin position of the 3rd axis.
  - E1 (1st axis) and E2 (2nd axis) can be performed in any order for setting the origin again.

13 Turn the controller power off and then on again.

14 Perform "14.9 Check Work."

15 Move the robot to a teaching point or other position to check that the origin position has been set correctly.

14.8.2 Restoring Origin Position Data Using Origin Position Mark

NOTICE

- If the origin position is set again using the ZEROP function or REORG function, be sure to perform teaching of the teaching points again.
  The origin position shifts slightly due to the error in setting. If the origin position settings in use are used unchanged, the robot will collide with the peripheral equipment or other objects.

1 Change the master mode to MANUAL mode.

2 See "14.4.1 Locations of Robot Origin Position Marks" and "14.4.2 3rd Axis Origin Position," and move the robot to the origin position.

3 Tap the maintenance button on the top screen.

4 Turn off the servo power.

5 Tap the zero position button.

The zero position screen appears.

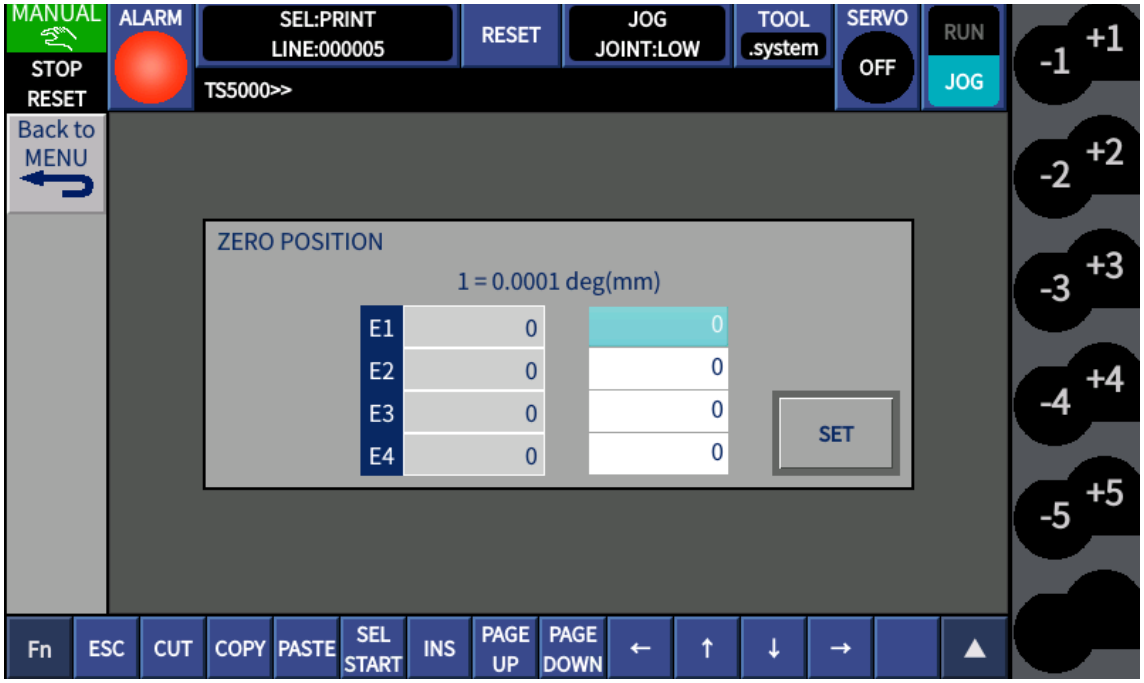


Fig. 14.28 Zero position screen 1

6 Tap the [↓] key on the keyboard three times, or tap the E4 cell.

Move the selection cursor to the E4 (axis 4) cell.

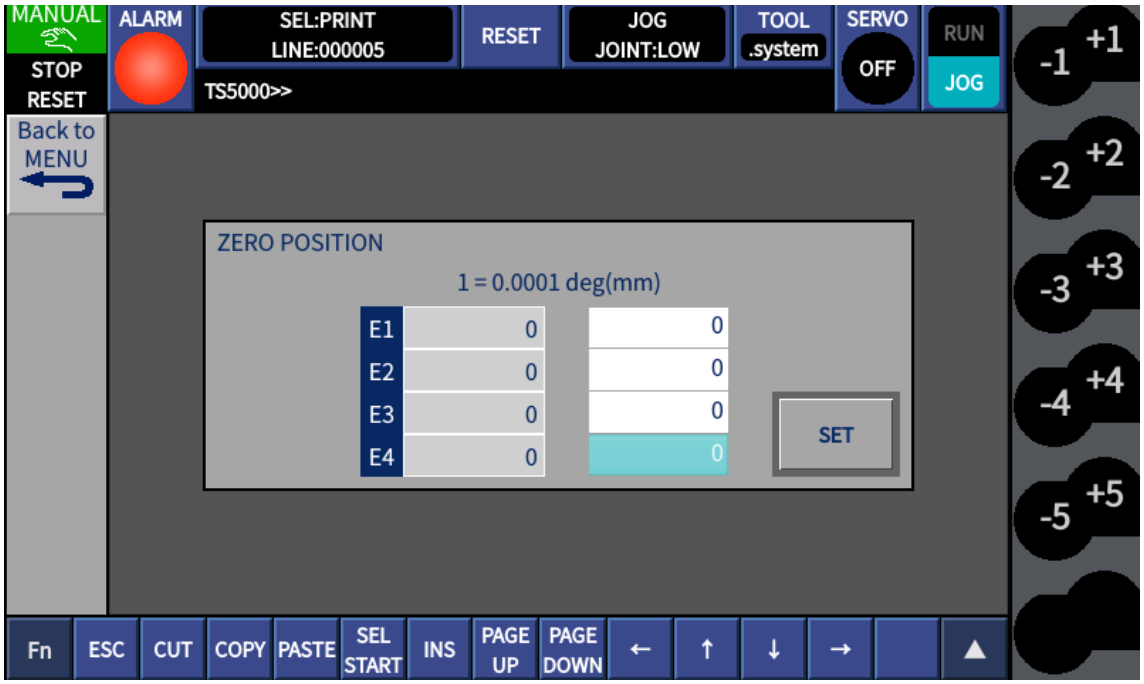


Fig. 14.29 Zero position screen 2

7 Tap the E4 cell.

The cursor is displayed in the cell, indicating that the content can be edited.

8 Enter “0” and tap the [Enter] key or tap the place other than the call being edited.

9 Tap the setting button.

A confirmation screen appears.

10 Tap the OK button.

The value of E4 becomes 0.

11 Similarly, also perform steps 1 to 8 for E3 to restore the origin position data.

12 Similarly, also perform steps 1 to 8 for both E1 and E2 to restore the origin position data.

NOTICE

- To set the origin position again using the ZEROP function or REORG function, perform the following procedure.
  Because the 3rd axis operates in tandem with the 4th axis the origin position is not restored if the steps are performed in the wrong order.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter the data, and restore the origin position of the 4th axis.
  3) Move the cursor to E3 on the screen, enter the data, and restore the origin position of the 3rd axis.
  - E1 (1st axis) and E2 (2nd axis) do not have any specific order for setting their origin position again.

13 Turn the controller power off and then on again.

14 Perform the procedure in "14.9 Check Work. "

15 Move the robot to a teaching point or other position to check that the origin position has been set correctly.

14.9 Check Work

Perform check work in order of the 1st axis, 2nd axis, 3rd axis, and 4th axis. After all axes are checked, also check for any deviation in the teaching points.

NOTICE

- After the origin point has been set again, be sure to always perform the check work.
  Unless the origin position is restored correctly, the robot will collide with the mechanical stopper or peripheral equipment when it operates.

14.9.1 Checking the Axis 1 and Axis 2

Manually move the 1st and 2nd axes to the travel limit in an emergency stop state, and check that a software limit alarm occurs. Also, when manually bringing an arm into contact with the mechanical stopper, check on the POS screen of the utility that the position is not largely deviated from the values of axis 1 and axis 2 in SYSTEM1 or SYSTEM2 used in 14.8.1.4 Restoring Data by SYSTEM1 or SYSTEM2 ." Fig. 14.30 shows check work that rotates the arm 1 in the plus (+) direction to bring the mechanical stopper into contact with the arm 1. Rotate the 1st axis in the minus (-) direction, and check the 2nd axis similarly.

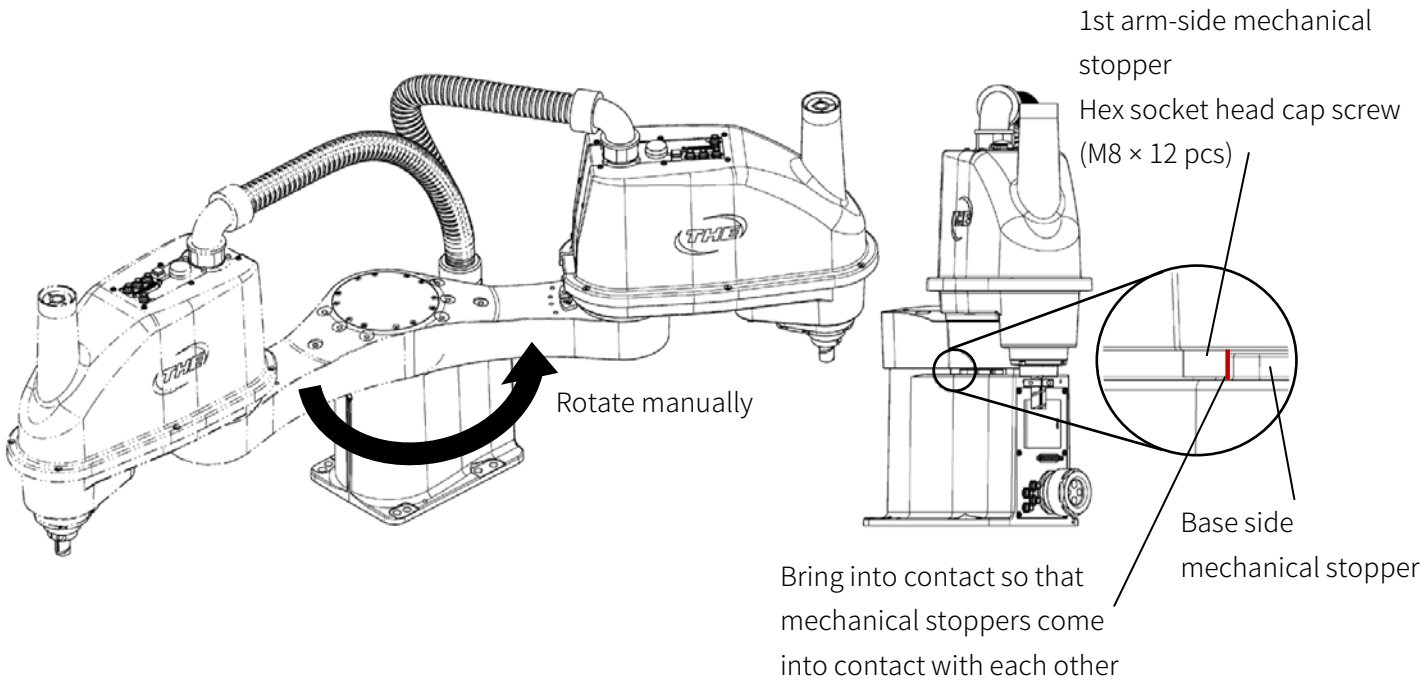
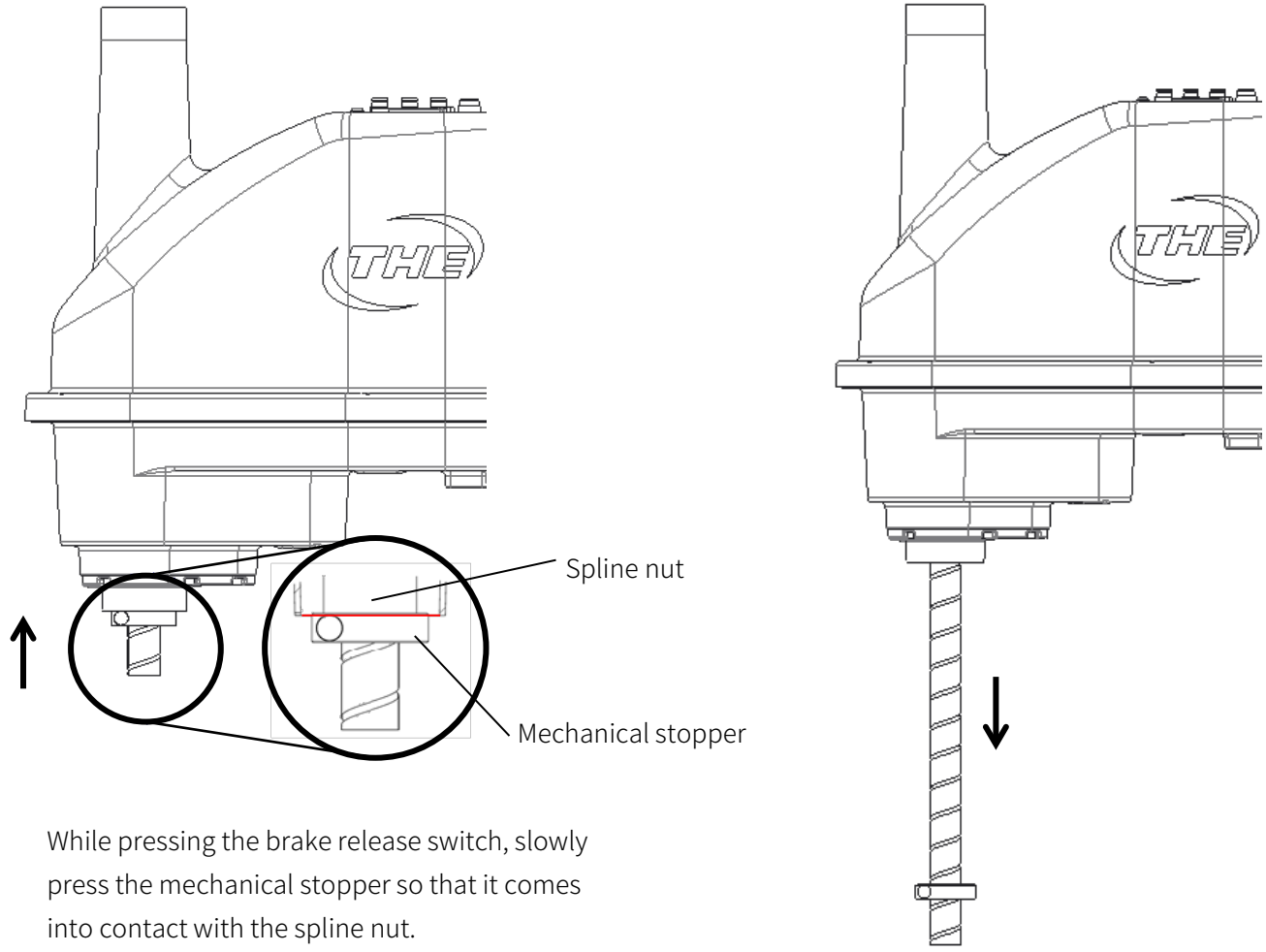


Fig. 14.30 Check Work (Axis 1 in Plus (+) Rotation Direction)

14.9.2 Checking the 3rd Axis

While pressing the brake release switch in an emergency stop state, manually move the 3rd axis to the travel limit to check that a software limit alarm occurs. Also, when manually bringing an arm into contact with the mechanical stopper, check on the POS screen of the utility that the value of the 3rd axis has not deviated significantly from the value of the 3rd axis in SYSTEM1 or SYSTEM2 used in 14.8.1.4 Restoring Data by SYSTEM1 or SYSTEM2 in "14.8.1 Restoring Origin Position Data by HOME Function." Fig. 14.31 shows check work for the axis 3.



While pressing the brake release switch, slowly press the mechanical stopper so that it comes into contact with the spline nut.

While pressing the brake release switch, slowly press the mechanical stopper so that it comes into contact with the internal section.

Fig. 14.31 Check Work (Axis 3)

MEMO

When lowering the 3rd axis and pushing it against the mechanical stopper, perform work while checking the position of the 3rd axis using the teaching pendant, and slowly lower the 3rd axis when the value is close to zero to ensure safety.

14.9.3 Checking the 4th Axis

While pressing the brake release switch in an emergency stop state, manually align the axis 4 with the origin position as shown in Fig14.32. During this operation, check on the POS screen of the utility that the value of the 4th axis has not shifted from 0.

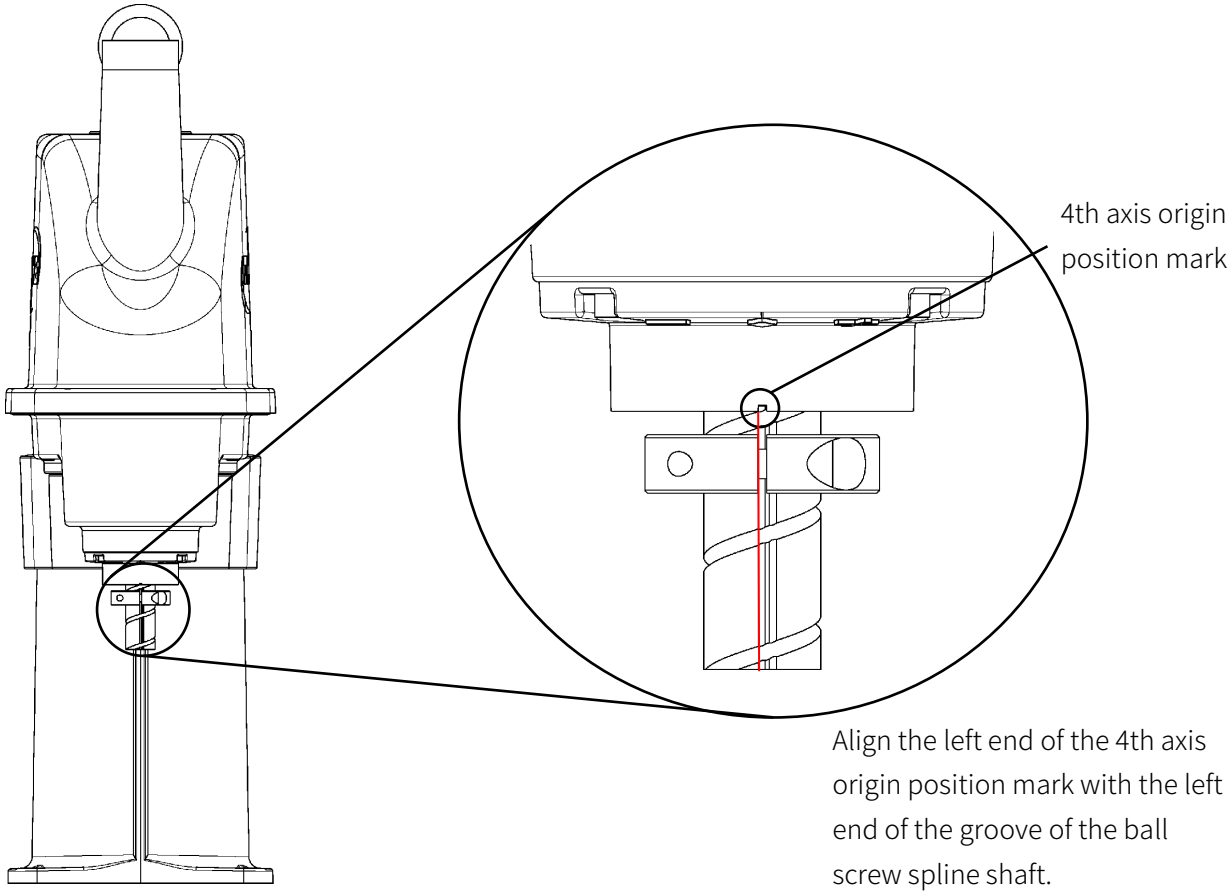


Fig14.32 Check Work (Axis 4)

14.10 TP5000 Operation Restrictions

Screen operation restrictions based on the user type are provided on the teach pendant TP5000 to prevent data (origin position data, etc.) from being lost due to mistaken operation by a third party. Screen operation restrictions are applied based on the user type.

To perform maintenance operations on the maintenance screen, the user type must be changed to "Maintenance".

Changing the user type

1 Tap the maintenance button on the top screen.

The maintenance menu screen appears.

2 Tap the authority setting button.

A password input window appears.

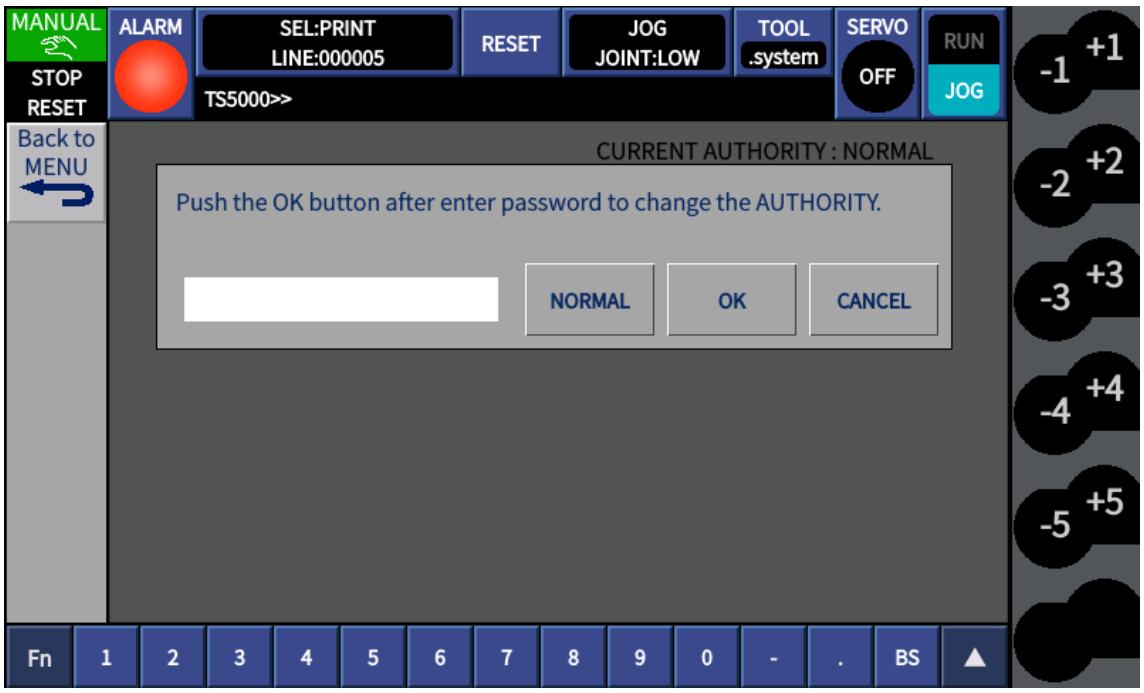


Fig. 14.33 Password input window 1

3 Tap the text box.

The cursor is displayed in the text box, indicating that it can be edited.

4 Use the keyboard to enter “0,” and tap the OK button.

The user type is changed to “Maintenance.”

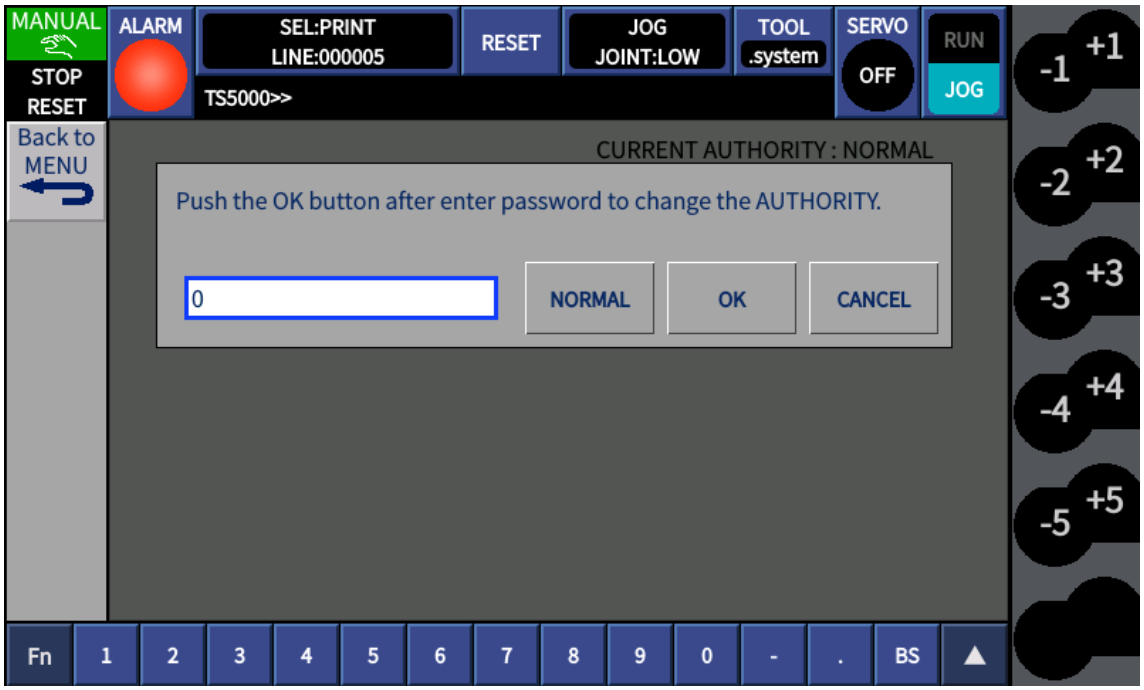


Fig. 14.34 Password input window 2

MEMO

- The changed user type remains activated until it is changed to another user type or the power is turned off.

- When the power to TP5000 is turned on, the user type is “General.”

15. Service Parts

This section describes the robot consumable parts that can be replaced by the SHIBAURA MACHINE Service Department.

MEMO

Before starting work, check the information in *10.1 Rescuing a Person Trapped by a Robot, and* conduct training.

15.1 Notes on Service Parts

⚠DANGER

- Be sure to replace parts with the SHIBAURA MACHINE-specified service replacement parts.
  Use of unauthorized parts could lead to an unpredictable accident.
  - All items other than the position detector batteries are special-specification products. To order them, please contact SHIBAURA MACHINE.
  - Be sure to contact the SHIBAURA MACHINE service company to request any replacement work other than replacement of the position detector batteries. A fee will be charged for any replacement work.
  - SHIBAURA MACHINE does not cover any accidents and damage resulting from replacement work performed by the customer and its results.

NOTICE

- Be sure to replace parts with the SHIBAURA MACHINE-specified maintenance replacement parts.
  Use of unauthorized parts could lead to reduced robot performance and lifespan. The robot will malfunction.

15.2 Service Parts List

The service parts list is shown below. When replacing consumable parts, contact the SHIBAURA MACHINE Service Department by referring to Service Network.

No.	Part name	Model	Drawing No.	Maker	Q'ty	Remarks
1	Arm-2 cover		F127717	SHIBAURA MACHINE	1	
2	Arm-1 cover		M342436		1	
3	AC servomotor		S968416		1	1st axis
4			S968417		1	2nd axis and 4th axis
5			S968418		1	3rd axis
6	Reduction gear		S968522		1	1st axis
7			S981478		1	2nd axis
8	Timing pulley		S968379		1	3rd axis
9			S968380		1	3rd axis
10			S968381		1	4th axis
11			M342404		1	4th axis
12			M342405		1	4th axis
13			M342406		1	4th axis
14	Timing belt		S968393		1	3rd axis
15			S968395		1	4th axis
16			S968397		1	4th axis
17	Ball screw spline unit		H880457		1	
18	Body harness		F127899		1	
19	Brake release switch		M355316		1	
20	LED Lamp		M355317		1	
21	Cable between robot and controller		2M1677		1	
22	O-ring	CO0540A		NOK Corp.	1	
23		CO0543A			1	
24	Grease	Multemp FZ No. 00		Kyodo Yushi Co., Ltd.		1st axis
25		SFB No.1 (For reduction gear)		Nidec-Shimpo Corp.		2nd axis
26		LGAF 3E (For key)*1		Japan SKF Co., Ltd.		All axes
27	Tension adjustment jig		L27611G01	SHIBAURA MACHINE		4th axis
28			L27612G01			3rd axis and 4th axis

16. Disposal

The worker should receive safety training defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures or accidents resulting from the work done by the customer.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

MEMO

- *This describes the key points for disposal. For details on the repacking work, refer to the following chapters.*
  - Installation
  - Transfer
- *Before starting work, check the information in 10.1 Rescuing a Person Trapped by a Robot, and conduct training.*

16.1 Safety Information When Removing the Robot

This section provides safety information when removing the robot from the installation site.

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a malfunction due to noise could cause an expected accident.
- Do not remove the ground wire until the power is turned off.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.
- Before turning on the servo for the robot, always make sure that there is no one inside the robot's safety protection zone.
  Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, carry a teach pendant set to MANUAL mode, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the robot hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - A worker should support the shaft to prevent hand or finger from getting caught.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Do not remove the robot until the power is turned off and the cable is removed from the controller.
  Otherwise, a worker could receive an electric shock due to touching a cable that was broken if the robot is dropped.
- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

NOTICE

- The installed robot should never be removed by only one worker.
  The worker may be unable to support the robot and it could fall over.

16.2 Safety Information When Transferring the Robot

This section provides safety information when transferring the robot to the disposal site.

⚠ WARNING

- Gather the cables together before transferring the robot.
  Otherwise, a worker could trip and fall over a cable on the floor.
  - Workers should wear a helmet to protect their head.

⚠ CAUTION

- When transferring the robot, be sure to always secure in place using the supplied fixing jigs.
  Otherwise, a worker could get caught in an arm that moved.
  - The weight of the arm section is approximately 17 kg.
  - Wear gloves to prevent slip and protect hands.

16.2.1 Transferring the Robot by Crane

⚠ WARNING

- When hoisting the robot, be sure to always follow the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the attached eyebolts.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
  - Be sure to always hoist and lower the robot slowly.
  - Clear a transfer route, and make sure that no one goes under the suspended load.

NOTICE

- Never touch the ball screw spline shaft with your bare hands.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to hoist and lower the robot gently.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- Be sure to secure the robot to be shipped in a packing box as it was delivered, and then remove the eyebolts.
  If the robot is moved while the eyebolts are mounted, the robot could be damaged or fail.
  - This is a mechanism indicating that a robot with the eyebolts mounted cannot be operated.
  - Keep the removed eyebolts with the robot.
- When hoisting and transferring the robot, use the eyebolts mounted into the tapped holes indicated by the arrow marks. Otherwise, the robot could lose its balance and fall over.

16.2.2 Transferring the Robot by Workers

⚠ WARNING

- The robot should never be transferred by only one worker.
  Otherwise, the robot cannot be fully supported and could drop.
  - The robot weight is 31 kg.
  - Lift the robot using more than one person.
  - Use a cart or similar device to the transfer location.
  - Clear a transfer route, and make sure that a dropped robot will not hit anyone.
  - Wear protective gloves and safety shoes.
- Never use an awkward posture when inserting the product into the packing box.
  Overexertion could injure the worker's back.
  - Provide space where work can be performed easily.

NOTICE

- When lifting up the robot by workers, be sure to hold the gray portions shown in Fig. 5.3 by hands.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.
- The robot should never be transferred by only one worker.
  Otherwise, the worker may be unable to support the robot and it could fall over.

16.3 Notes on Disposal

When disposing of the pallets and packing materials that are no longer needed after unpacking the robot and other product, unneeded parts after replacement, and waste cloths used to wipe up grease, these items should be treated as industrial waste.
Dispose of them in accordance with your company's rules. SHIBAURA MACHINE will not collect and dispose of these items.

MEMO

For details on removing the position detector backup batteries, check the information in "*14.3 Replacing the Position Detector Backup Batteries.*"

⚠ CAUTION

- Dispose or recycle the position detector backup batteries in accordance with the rules for your company or region.
  If a battery is incinerated, disassembled, recharged, or heated, leaked fluid could get on your skin or in your eyes, resulting in inflammation.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Use tape or other insulator to isolate the terminals of used batteries.
- Place the robot at the disposal site in a stable orientation.
  Otherwise, the robot will fall over.
  - If possible, place the robot on its side to ensure safety.

NOTICE

- Be sure to dispose of any unneeded packing materials in accordance with your company's rules.